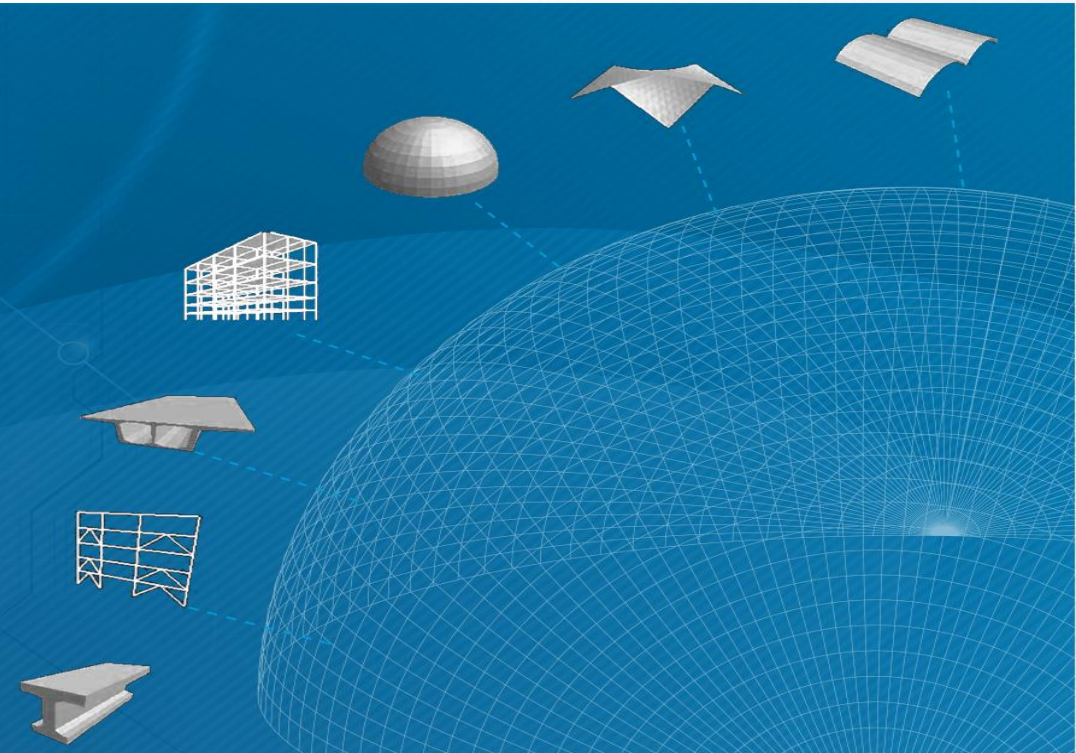
CE – 412 Design of Concrete Structures

## Lecture 3:

# Cross-sections –
# Behavior in Flexure (Beams)

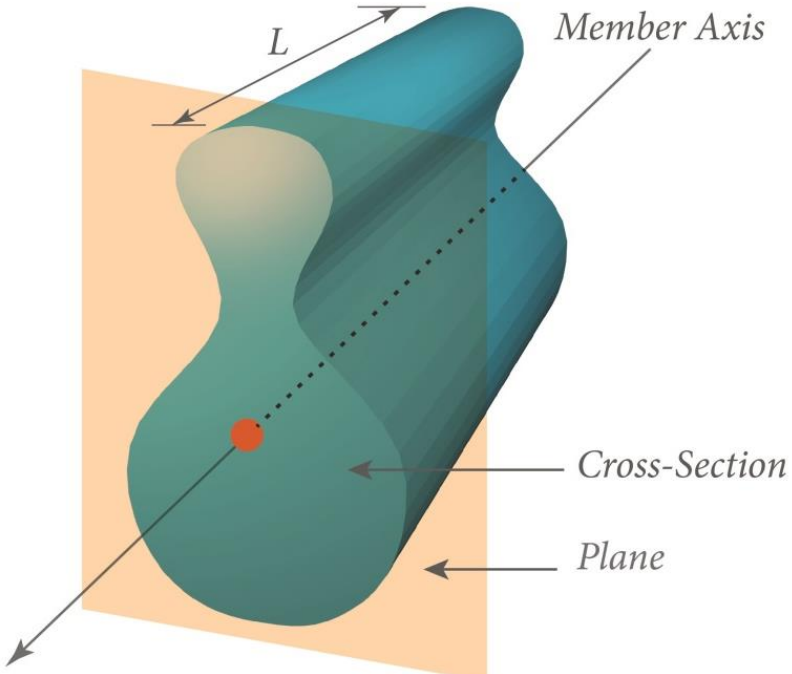


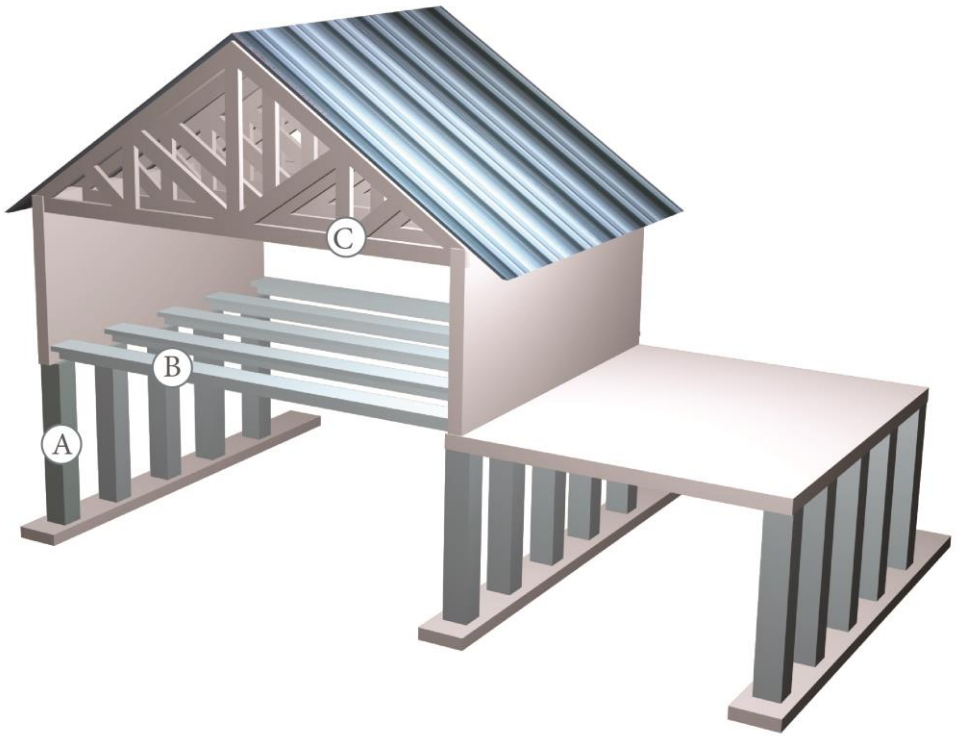
October - 2020

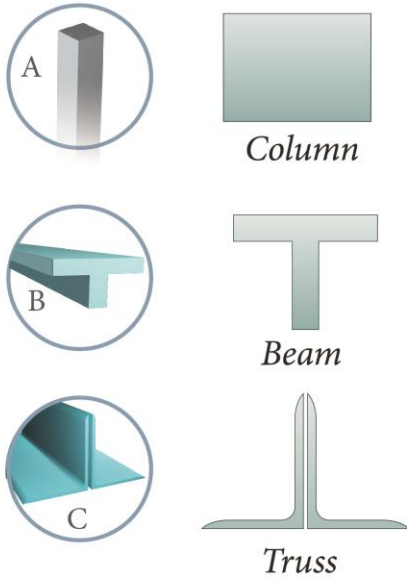

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Asian Institute of Technology (AIT), Thailand

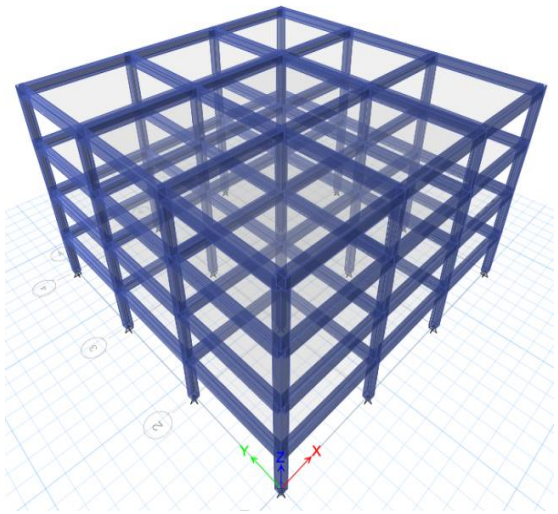
# Cross-sections

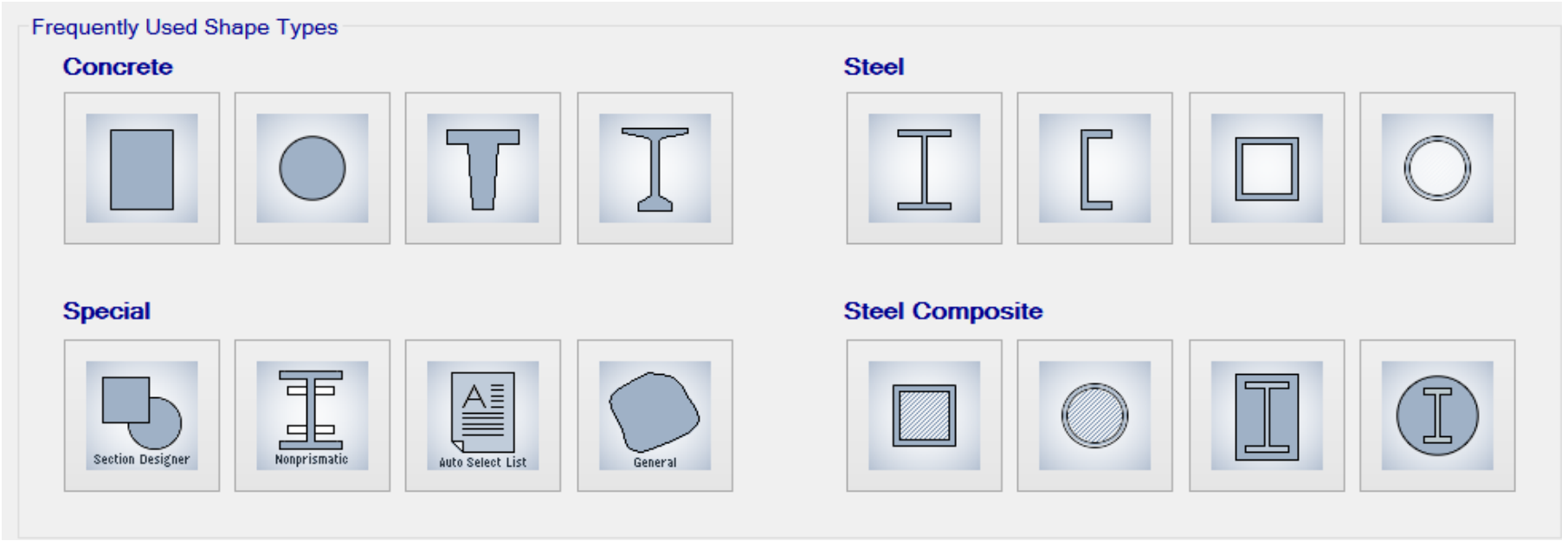


# Frame Members and Sections





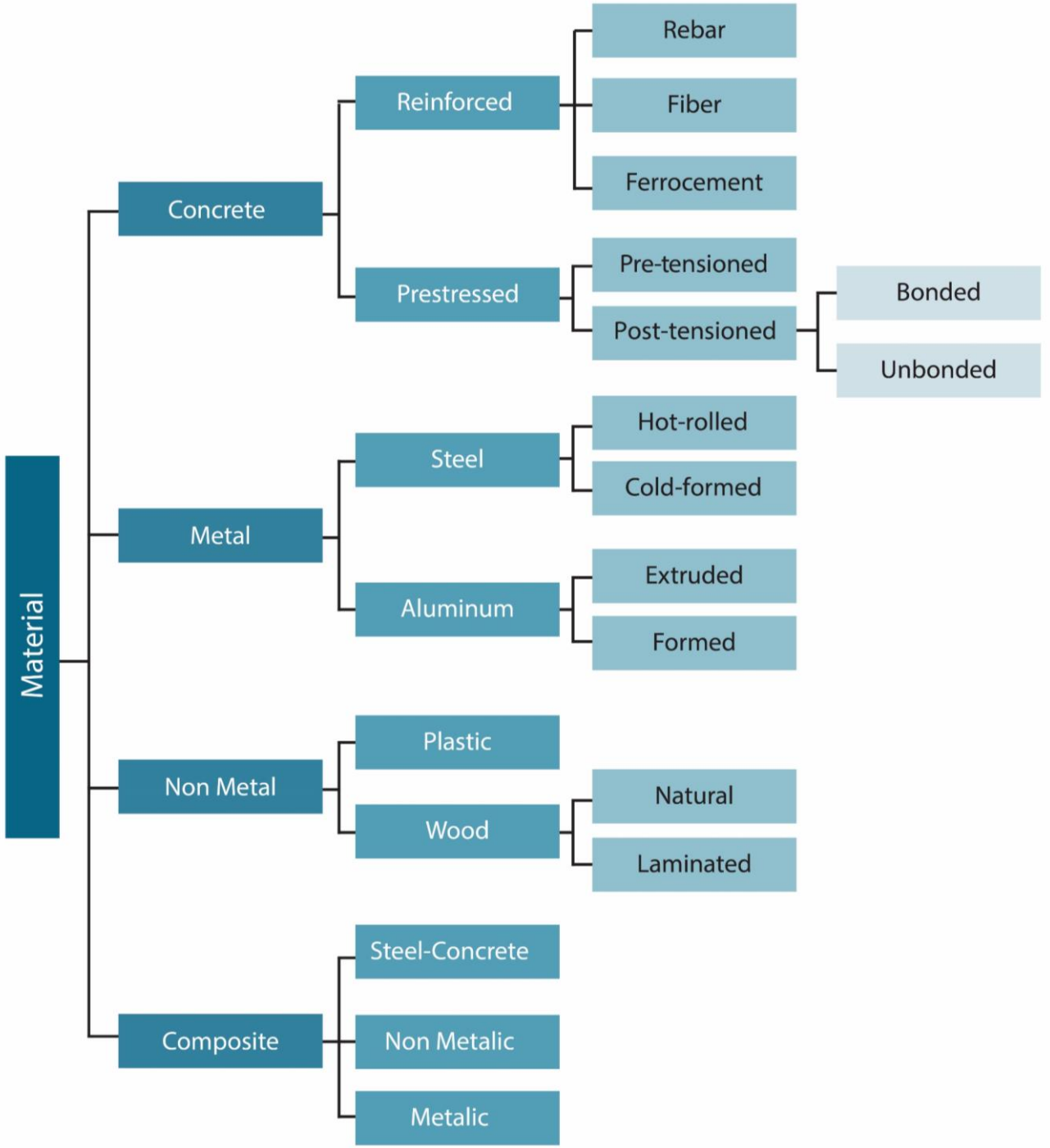
# Basic Section Types - Proportions

- Slender
    - Buckling of section parts before reaching material yielding
    - Cold formed, thin walled metal sections
- Compact
    - Material yielding first, followed by bucking of section parts
    - Most hot rolled and built-up metal sections
    - Some thin concrete sections
- Plastic
    - Material failure (yielding, rupture, but no buckling)
    - Most concrete sections

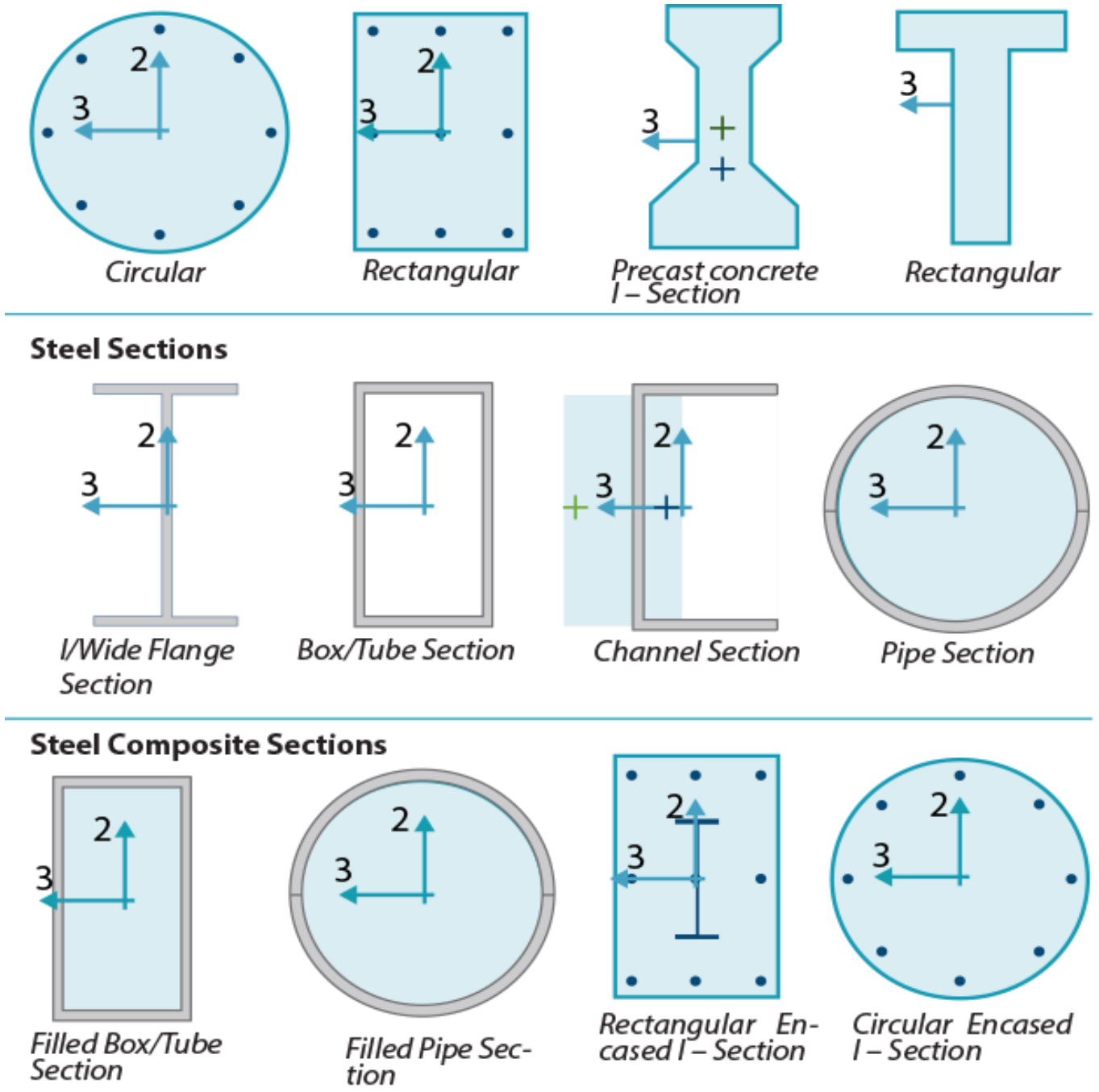
# Section Types – Member Usage

- Beams
    - Primarily bending, shear and torsion
- Trusses
    - Primarily tension and compression
- Columns
    - Primarily compression, bending
    - Shear and torsion also important

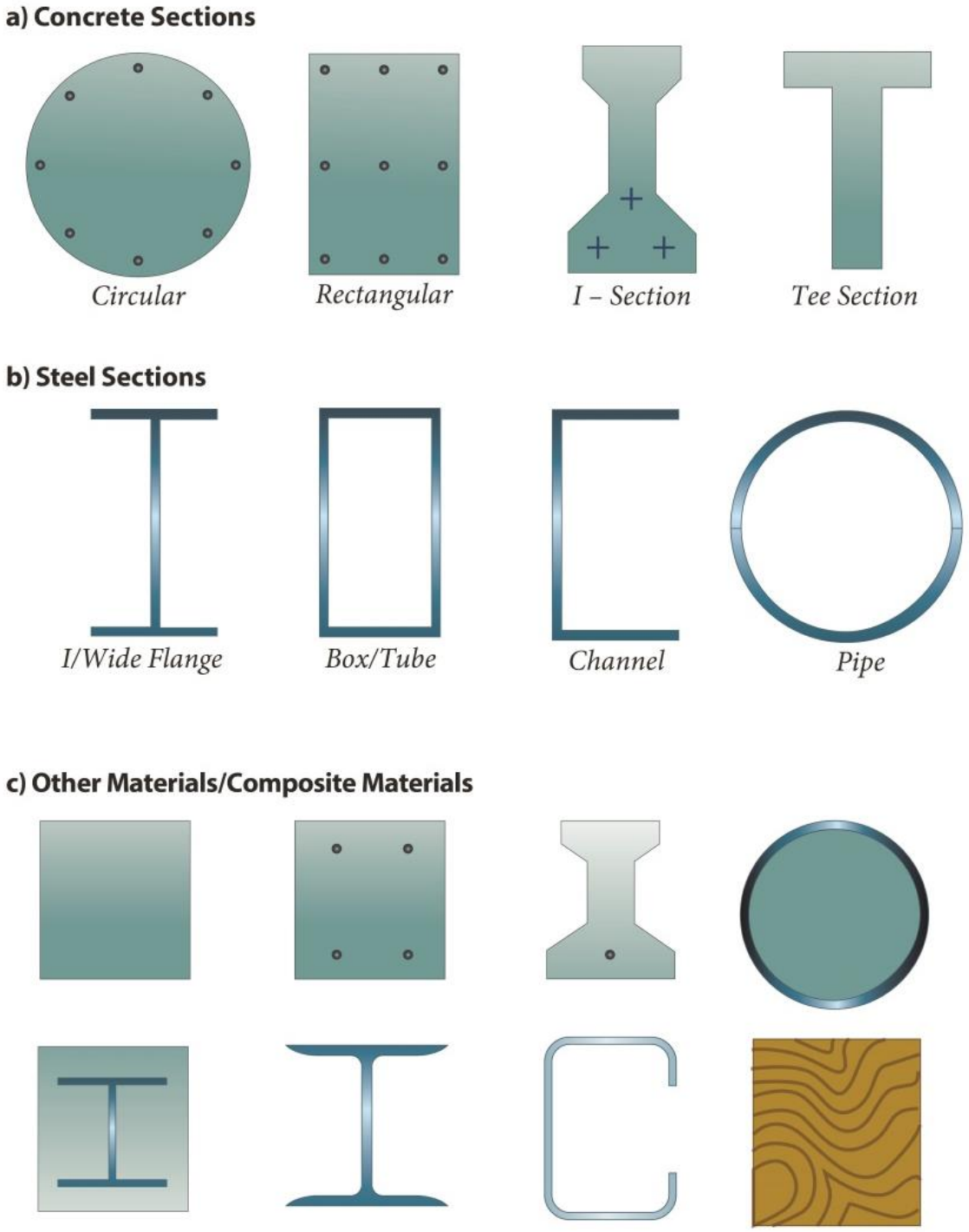
## Cross-section classification based on primary material composition



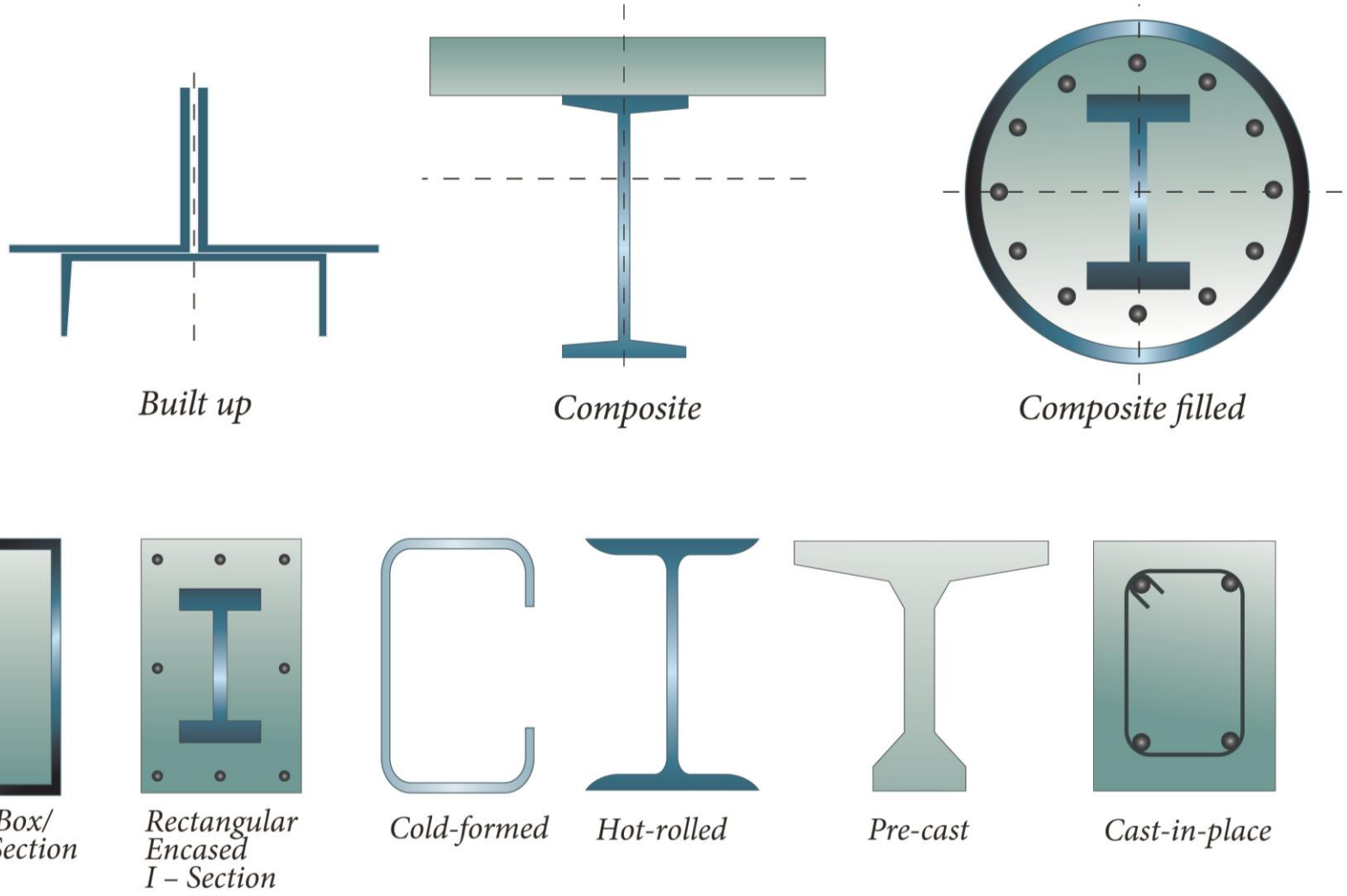
**Some of the shapes used for Reinforced and Pre-stressed concrete sections defined in CSI ETABS Section Designer.**

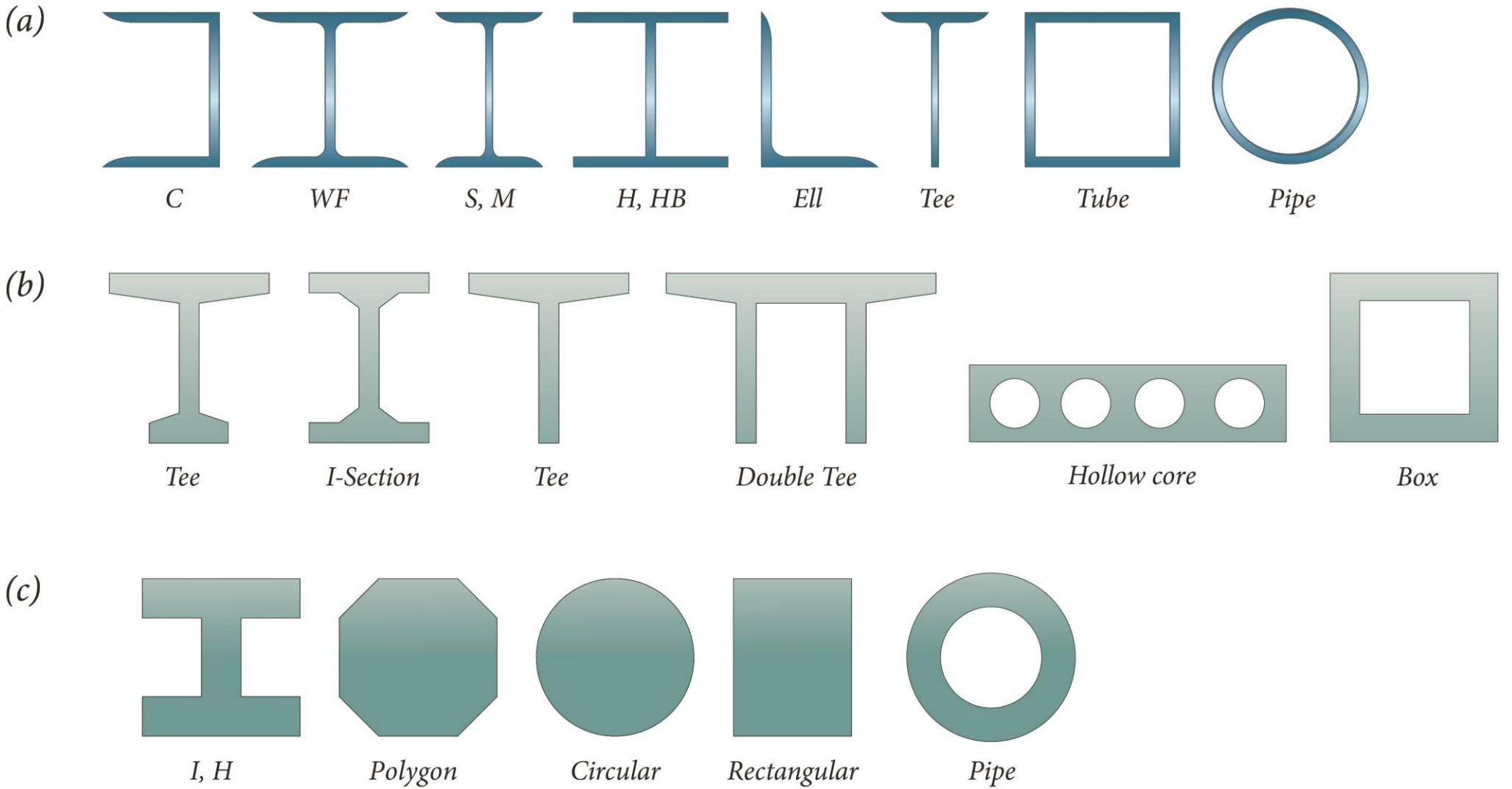


**Some common cross-sectional types based on materials and geometry**



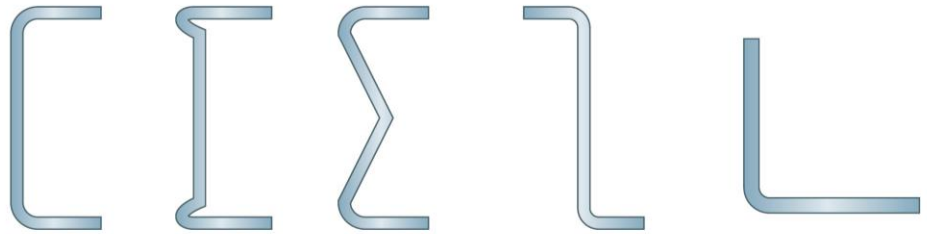
# Some common cross-sectional types based on materials and geometry



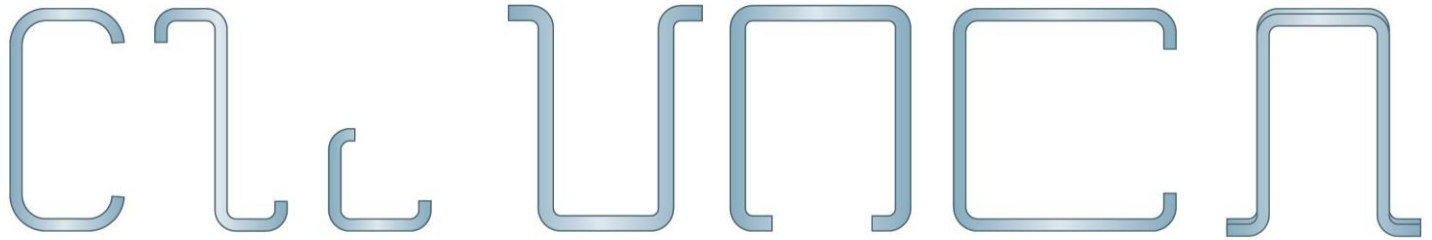



Some typical standard cross-section shapes used (a) in AISC database, (b) in BS Database, (c) in pre-cast, pre-stressed girders and slabs, (d) in pre-cast concrete piles

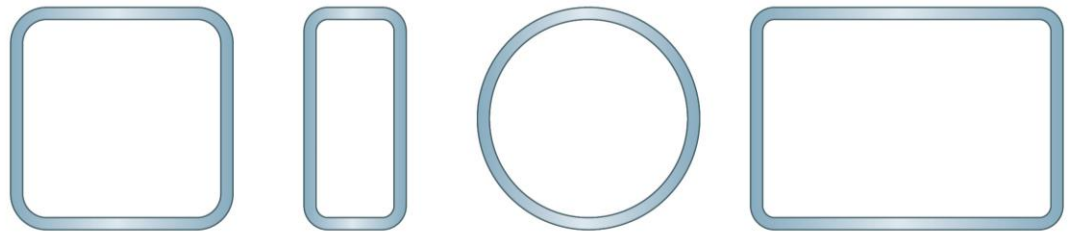
(d) *Simple Sections*



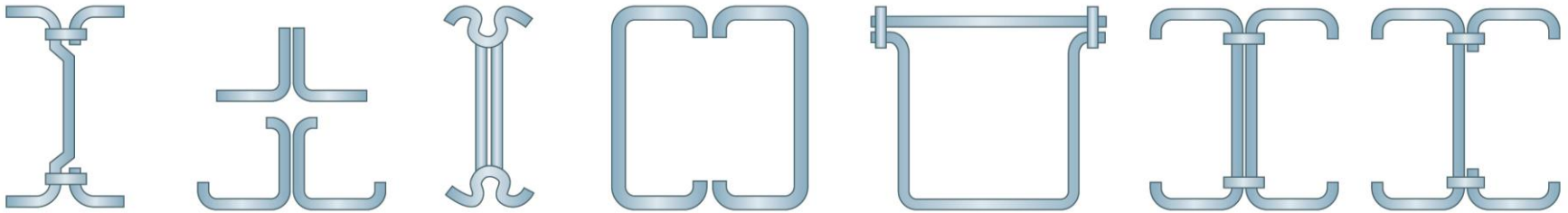
*Lipped Sections*



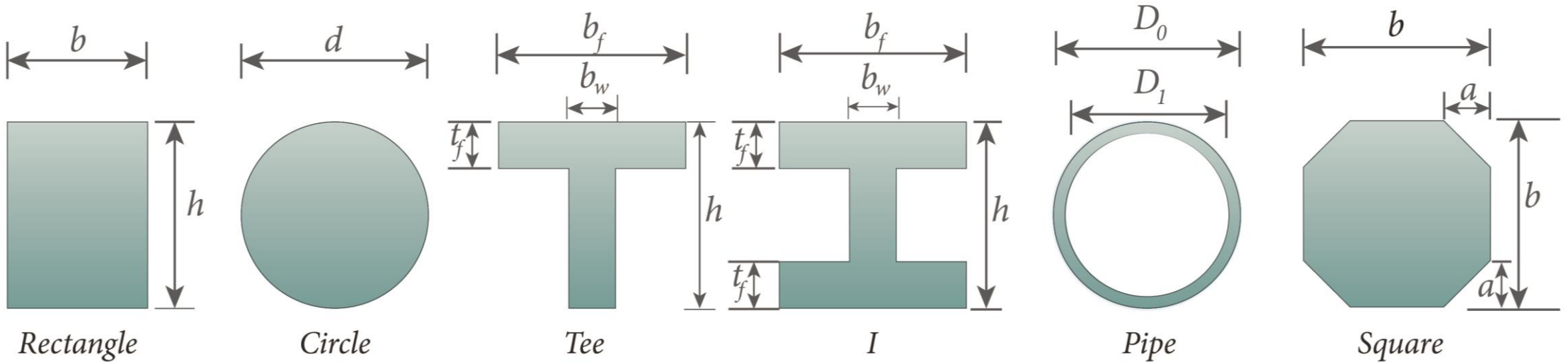
*Tubular Sections*

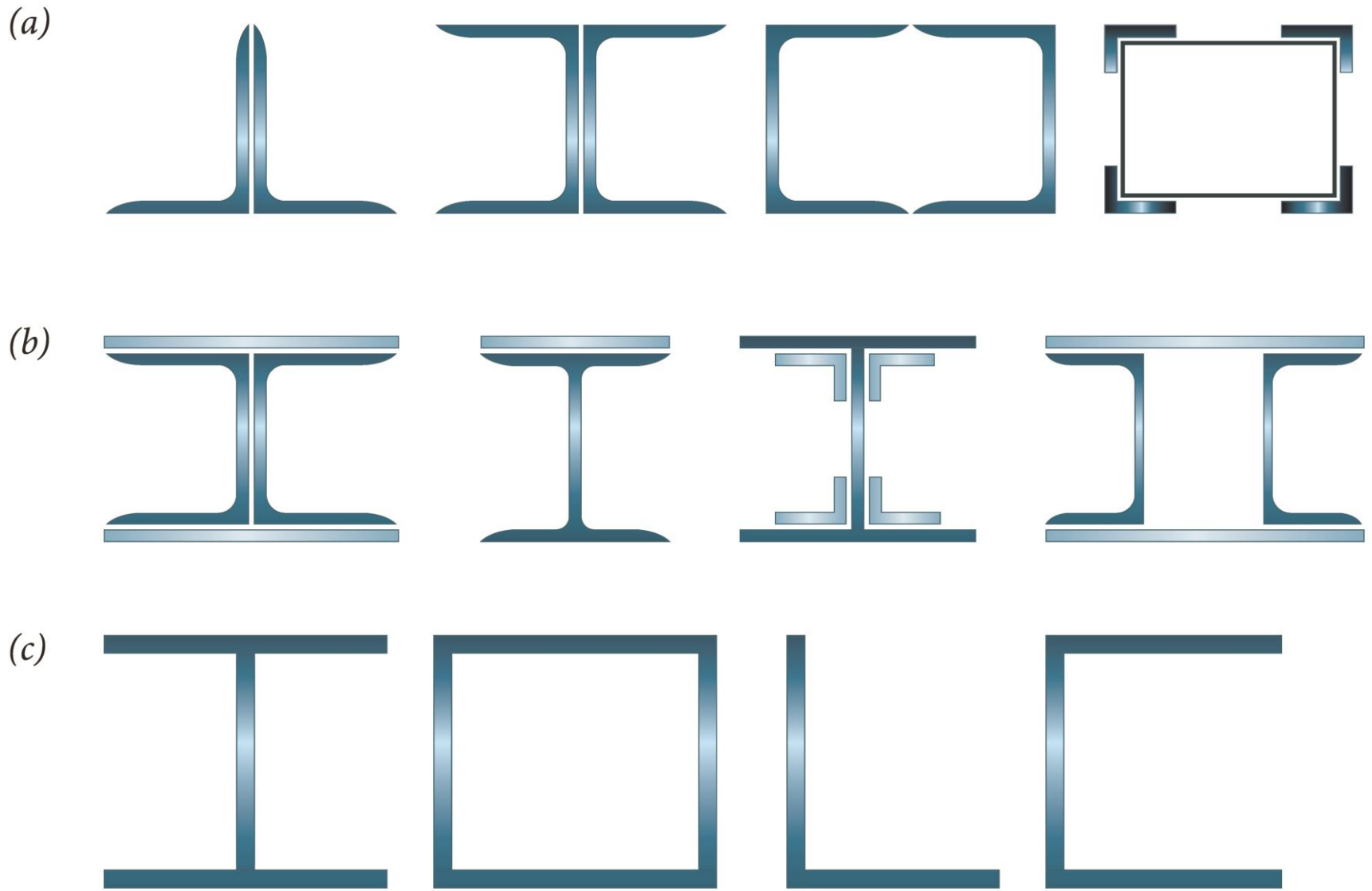


*Build-up Sections*

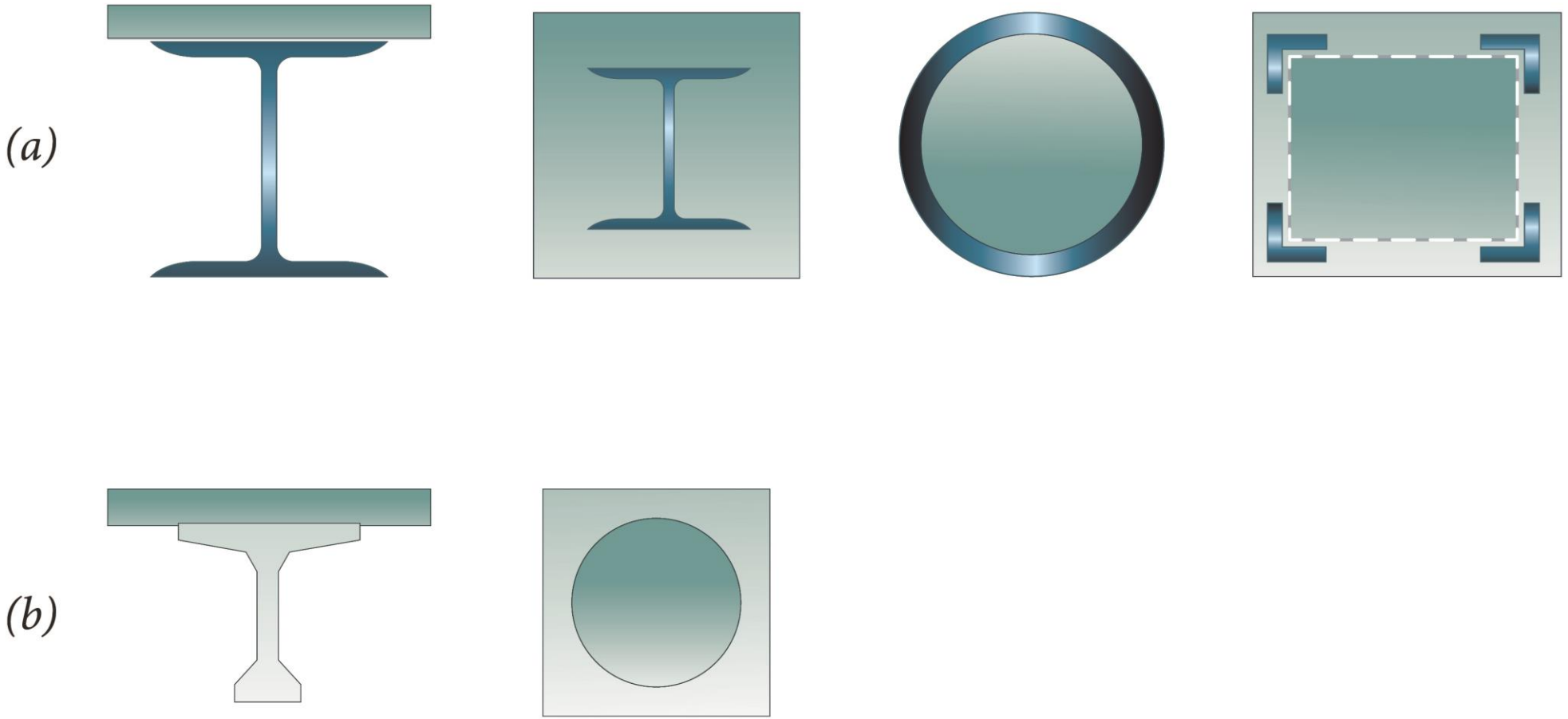


# Some typical parametrically defined cross-section shapes

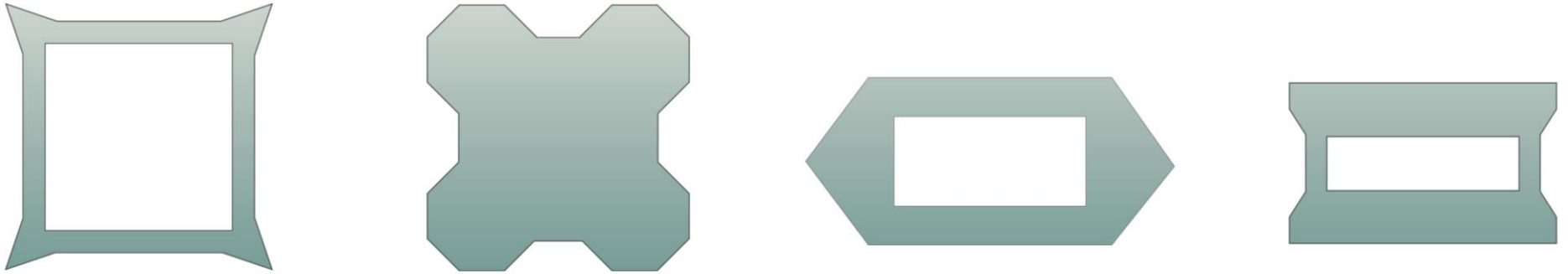





Some typical built-up shapes and sections (a) made from standard shapes, (b) made from standard shapes and plates, (c)made from plates

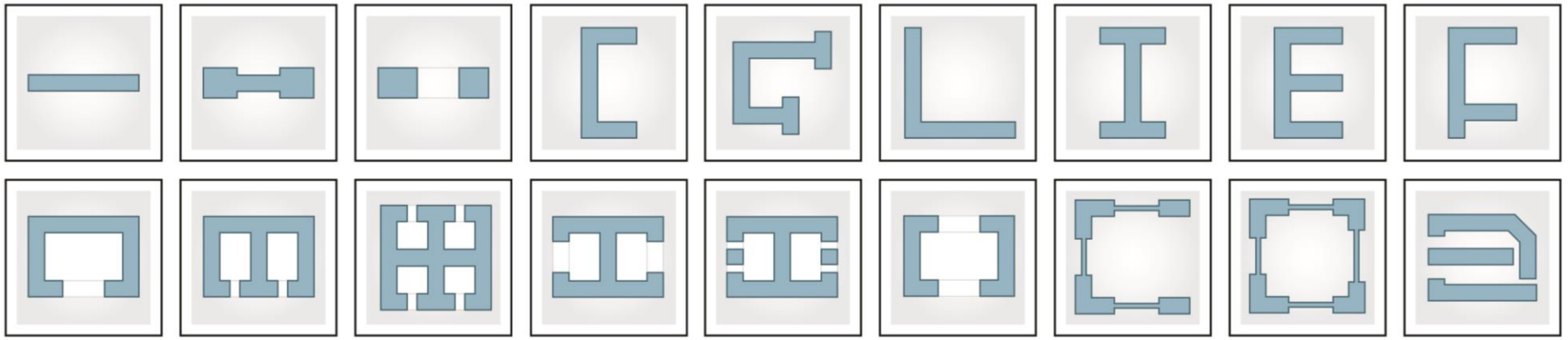



Some typical composite sections. (a) Concrete-Steel composite, (b) Concrete-Concrete composite

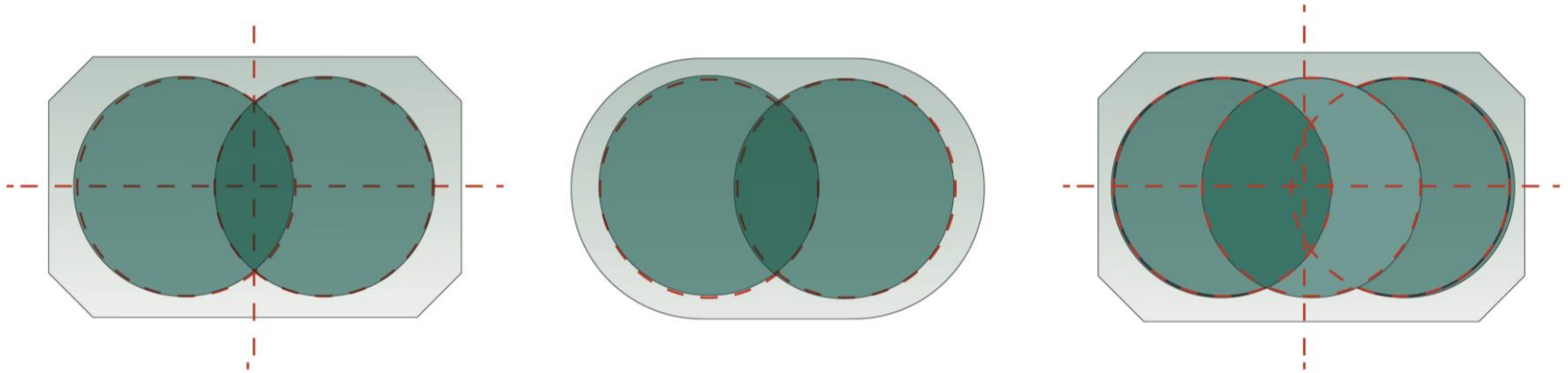


Star Pier | Flower Style Pier | Hollow Hexagon Pier | Hollow Pier

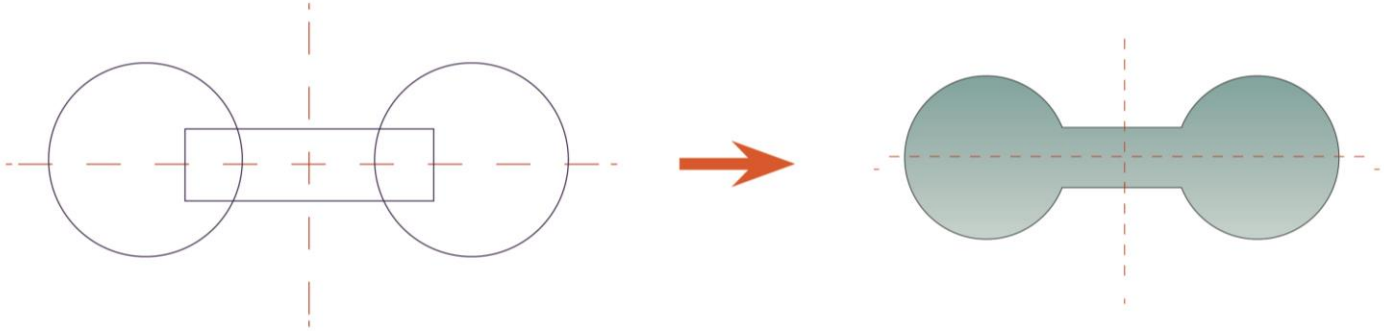
a) Bridge Piers (CSI Section Builder)



b) Shear Wall Stacks (ETABS 2015)

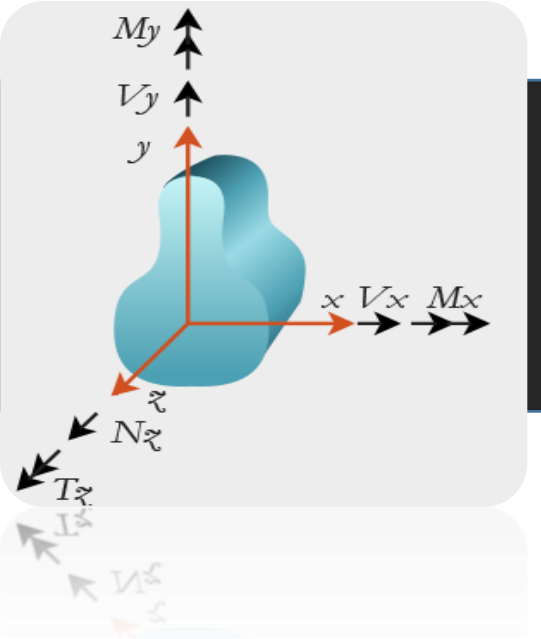


*c) CALTRANS Bridge Piers (CSI Section Builder)*

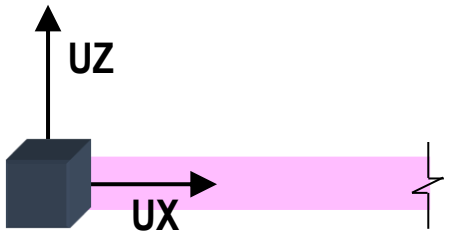


*d) Merging of Shape (CSI Section Builder)*

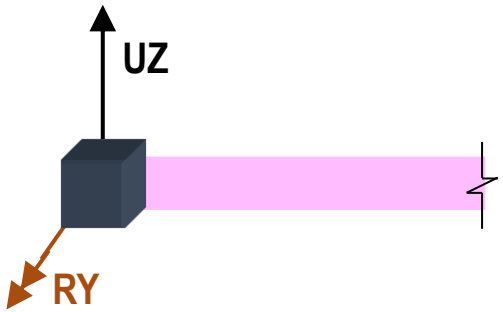
# DOFs and Cross-Sectional Properties



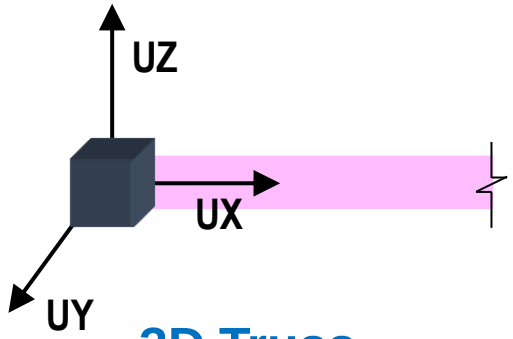
# DOF for 1D Elements



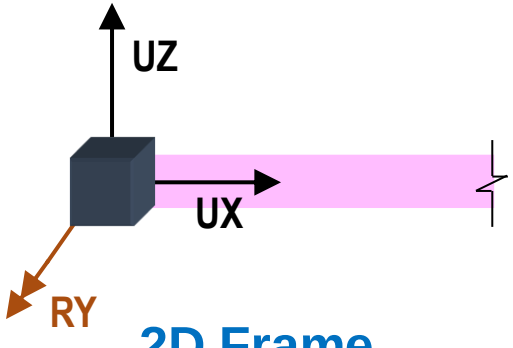

2D Truss



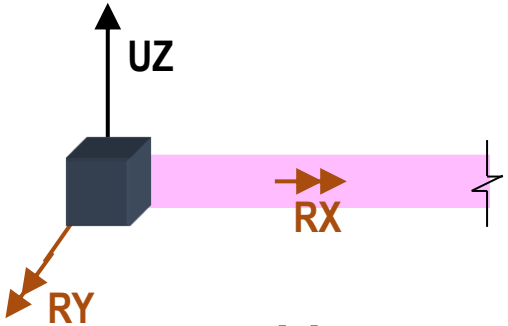

2D Beam



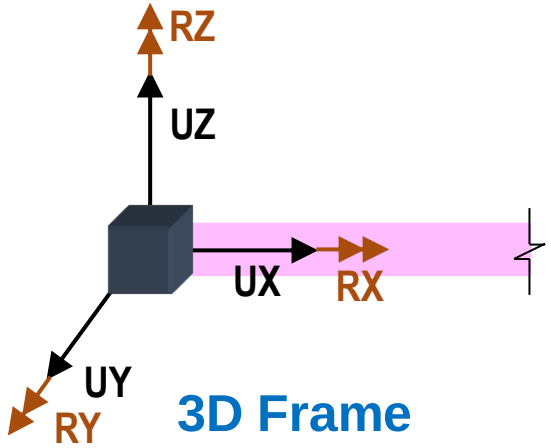

3D Truss




2D Frame

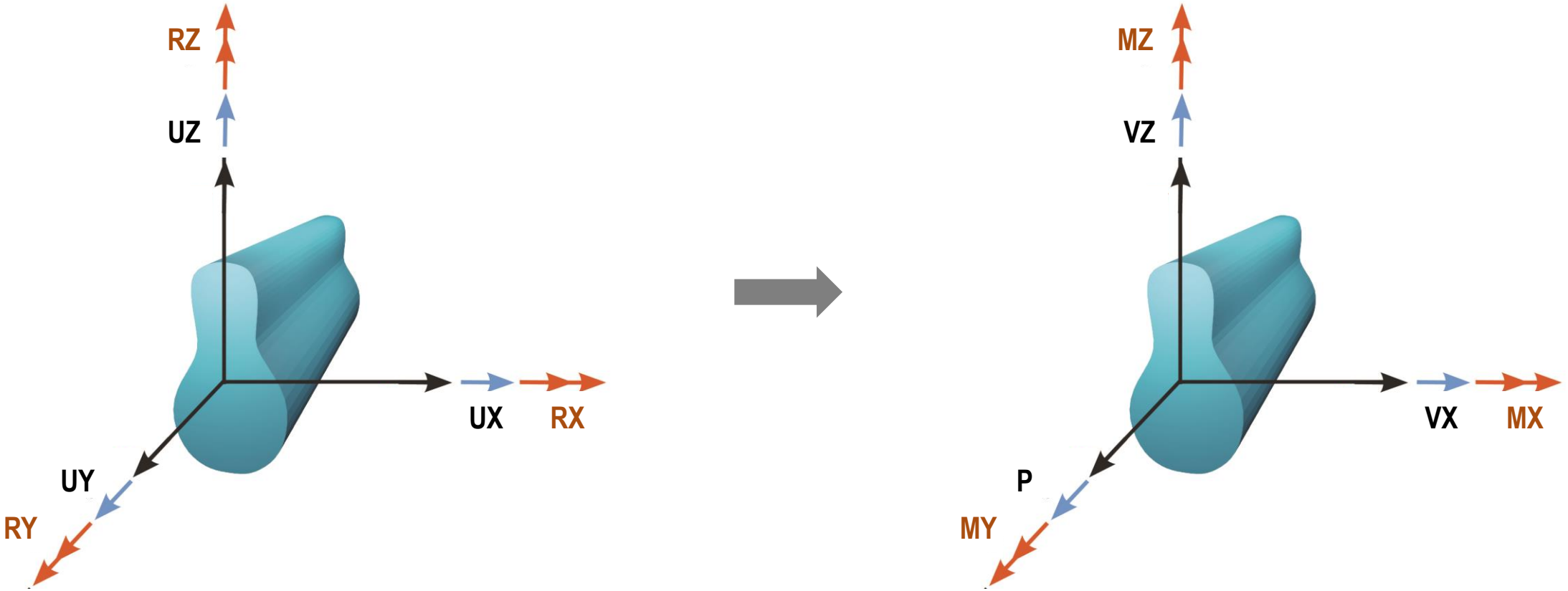



2D Grid

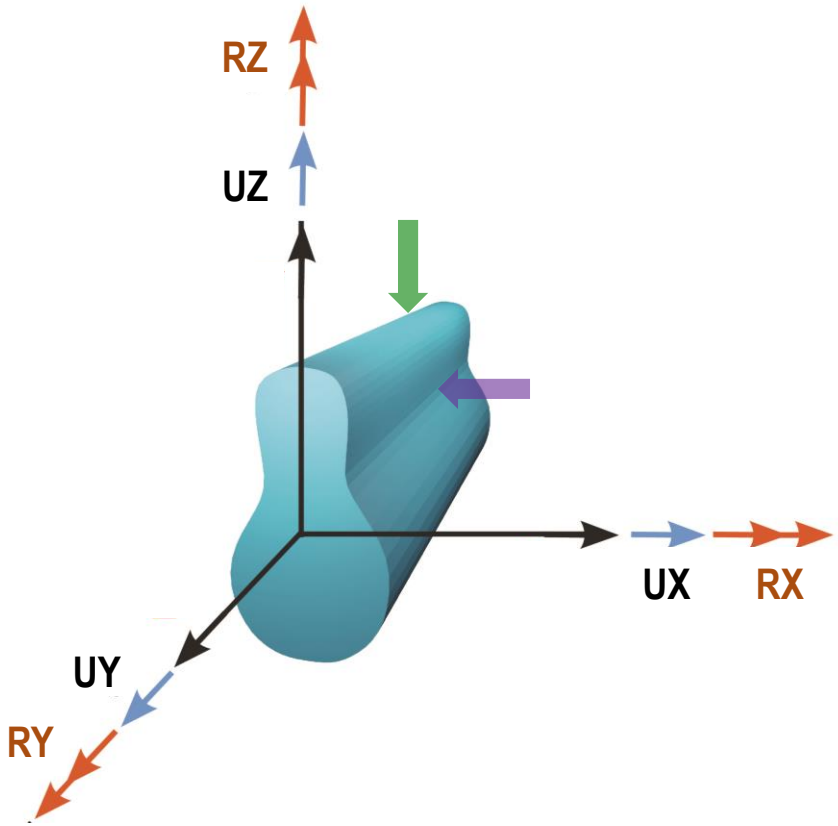



3D Frame

# Degrees of Freedom (DOFs) and Actions

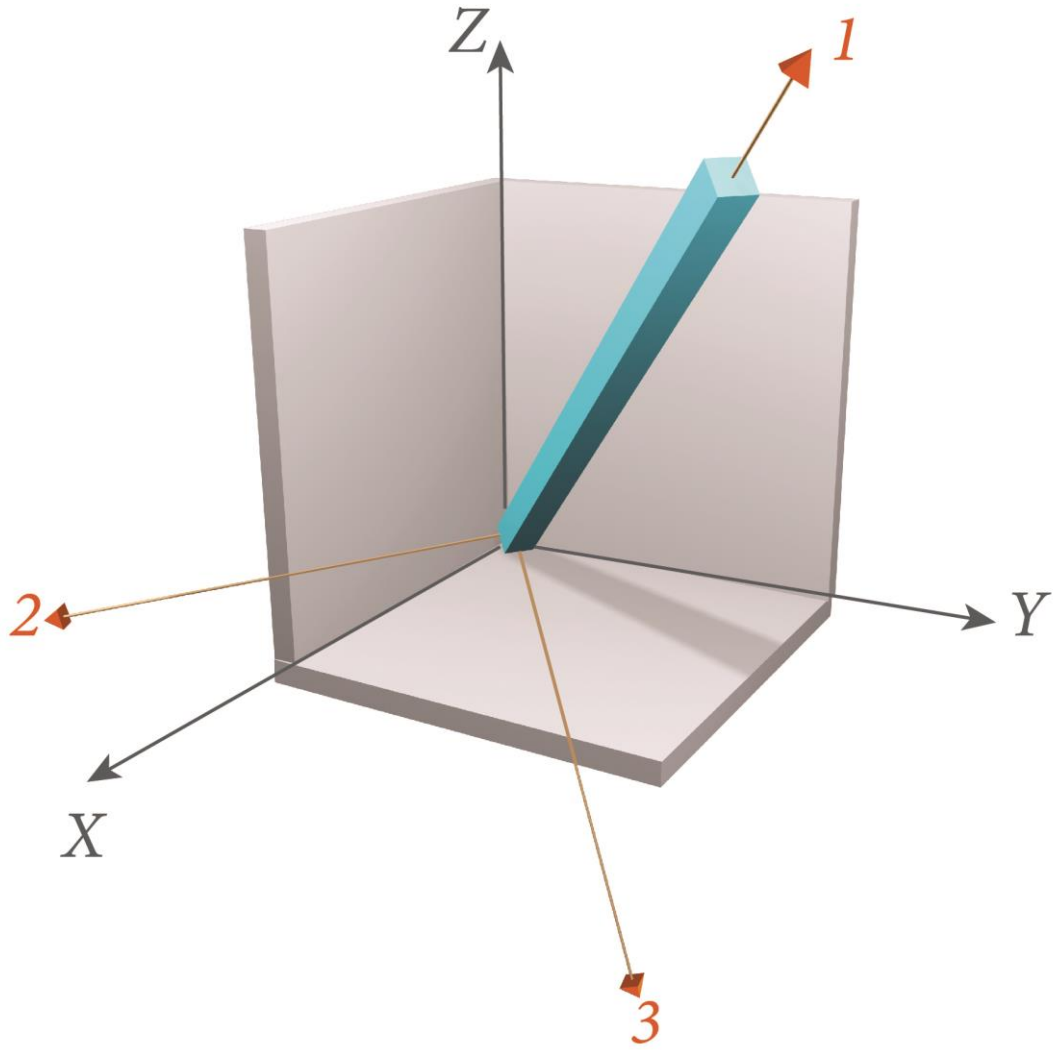


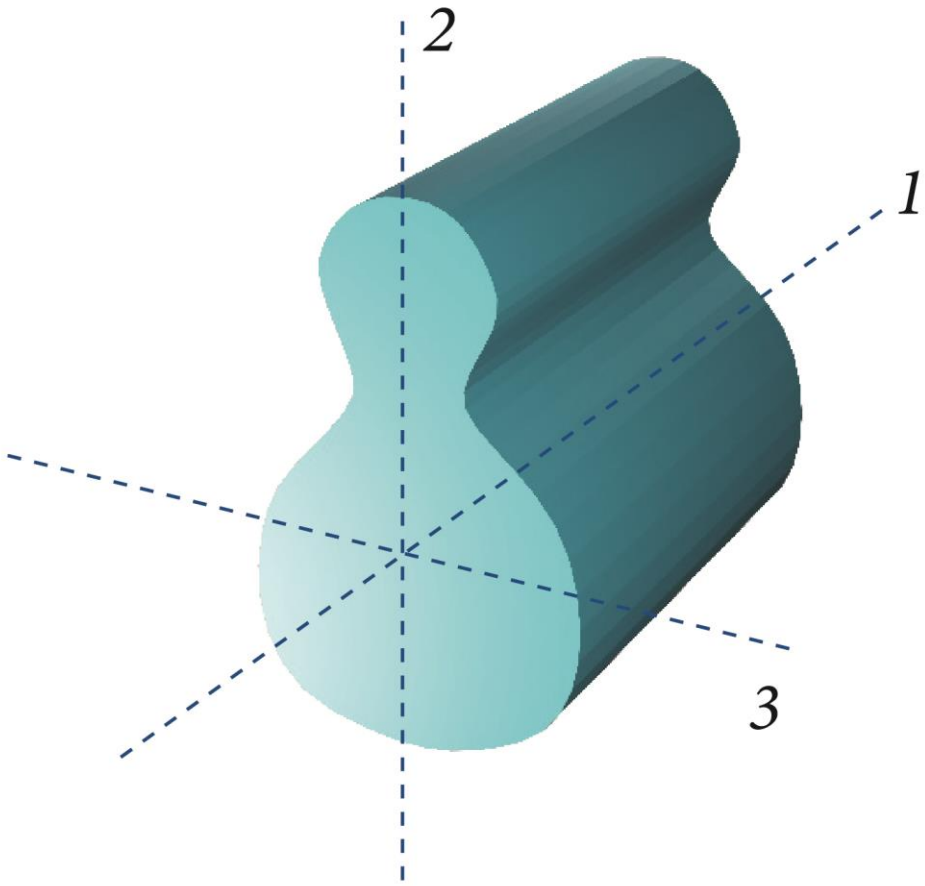
# DOFs and Corresponding Deformations (and Actions)




| | | | | |
|---|---|---|---|---|
| UY | Axial Deformation | Axial Strain | Axial Stress | Axial Force (P) |
| UZ | Shear Deformation | Shear Strain | Shear Stress | Shear Force (VZ) |
| UX | Shear Deformation | Shear Strain | Shear Stress | Shear Force (VX) |
| RY | Torsional Deformation (Twist) | Combined Shear and Axial Strains | Combined Shear and Axial Stress | Shear and Axial Forces caused by Torsion (T) |
| RX | Bending Deformation (Curvature) | Axial Strains (bending)<br>May also produce lateral shear stresses and strains | Axial Stresses (bending) | Bending Moment about X axis (MX)<br>Primary bending moment caused by vertical gravity loading |
| RZ | Bending Deformation (Curvature) | Axial Strains (bending)<br>May also produce lateral shear stresses and strains | Axial Stresses (bending) | Bending Moment about Z axis (MZ)<br>Primary bending moment caused by horizontal loading |

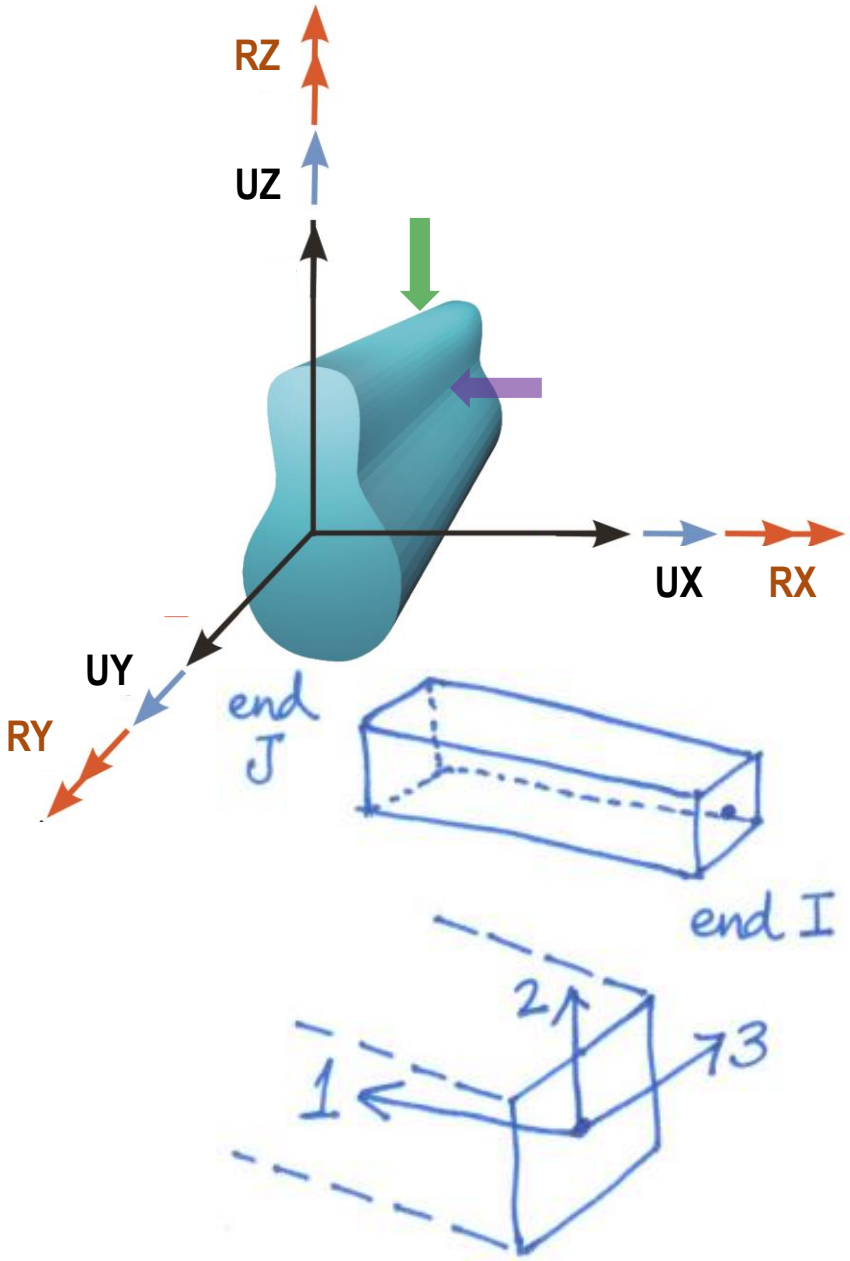
# Global and Local Coordinates

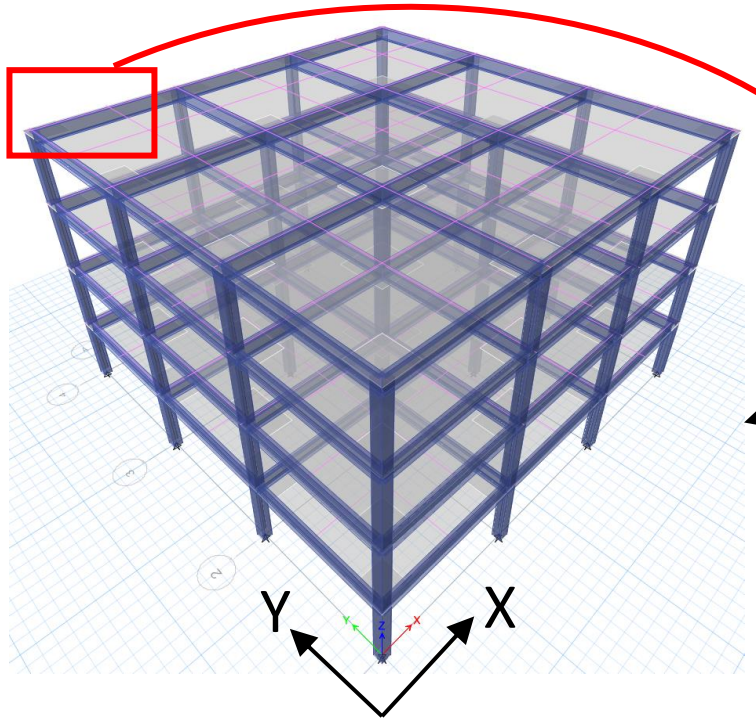


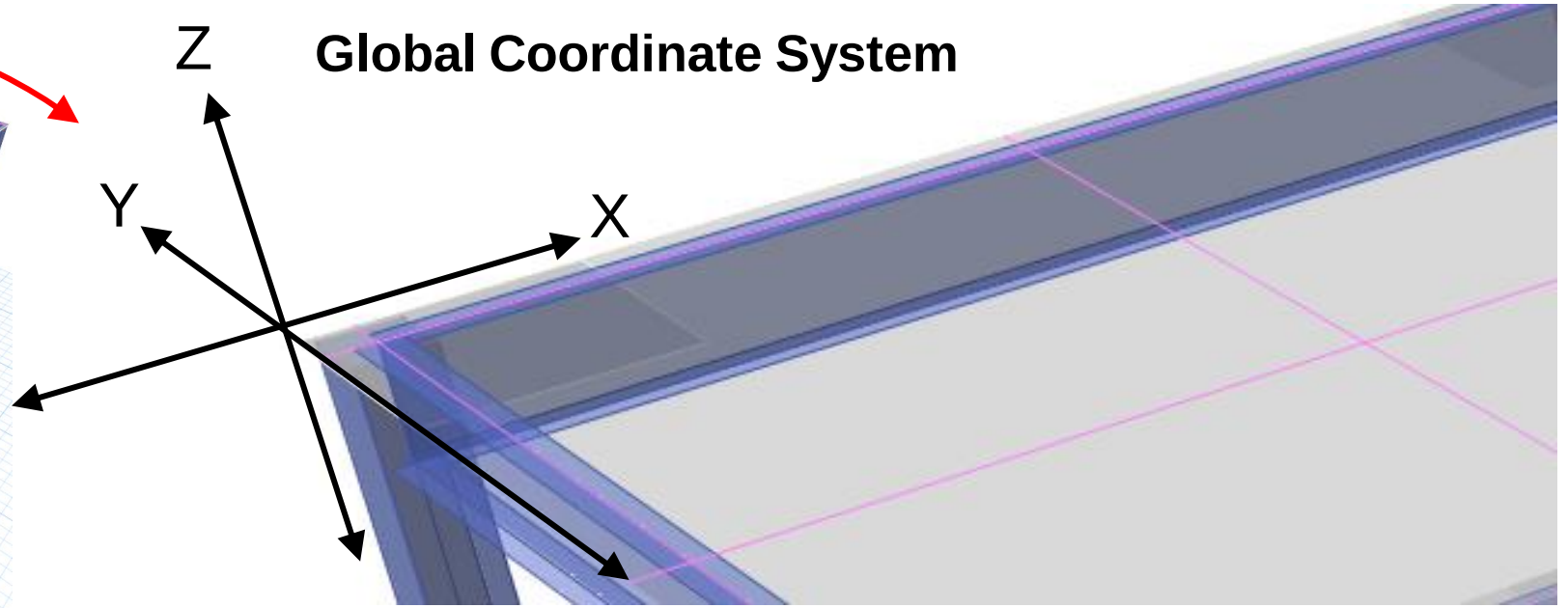



# Frame Local Coordinate System in SAP/ETABS

- The two ends of the element are denoted end I and end J, respectively.
- **The local axis 1 is always the longitudinal axis of the element**, the positive direction being directed from End I to End J. This axis is always located at the centroid of the cross section, and connects joint I to joint j.
- The default orientation of the local 2 and 3 axes is determined by the relationship between the local 1 axis and the global Z axis.
- The local 2 axis is taken to have an up ward (+Z) sense unless the element is vertical, in which case the local 2 axis is taken to be horizontal along the global +X direction.
- The local 3 axis is horizontal, i.e., it lies in the X-Y plane. This means that the local 2 axis points vertically up ward for horizontal elements.

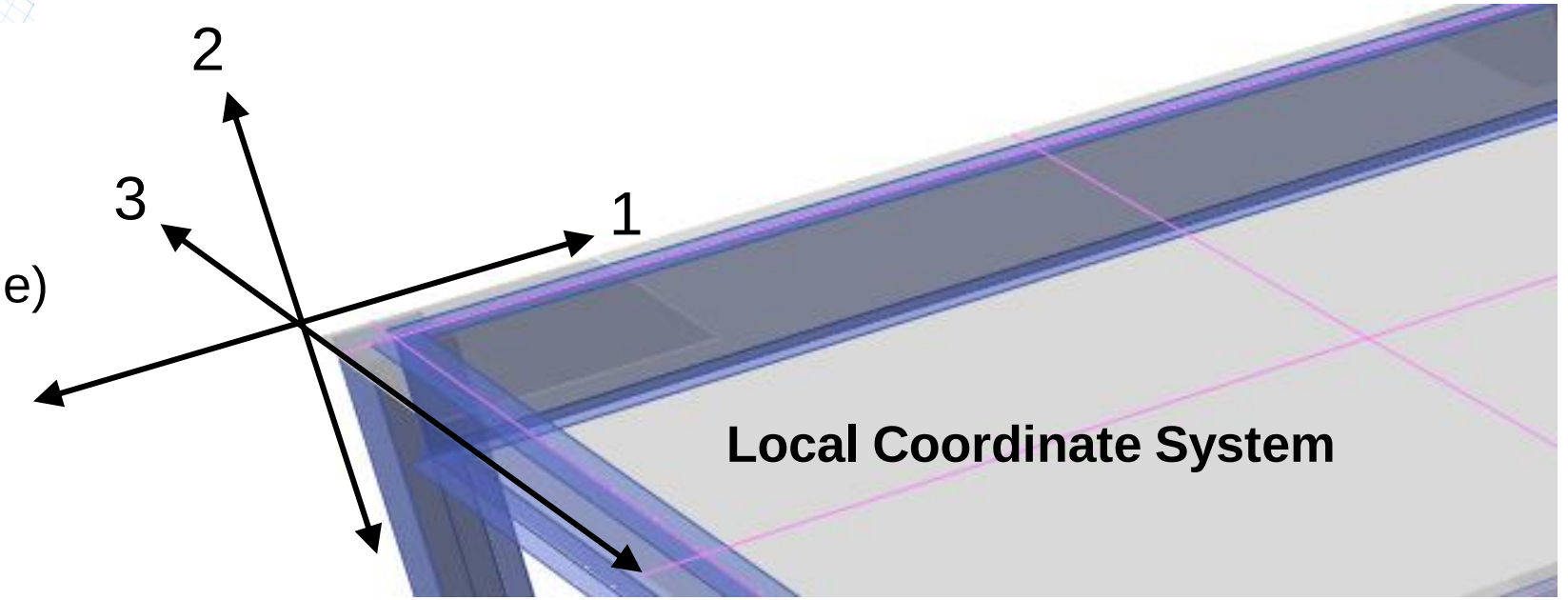


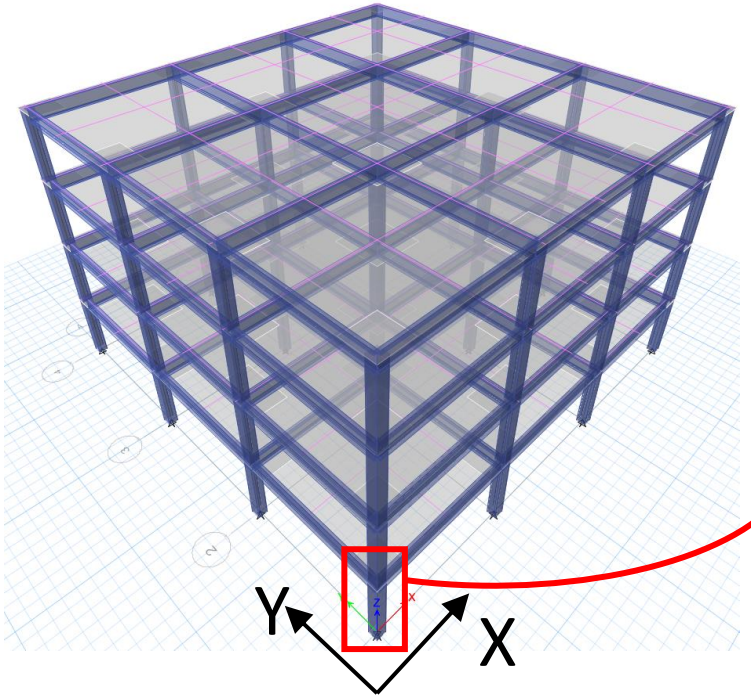


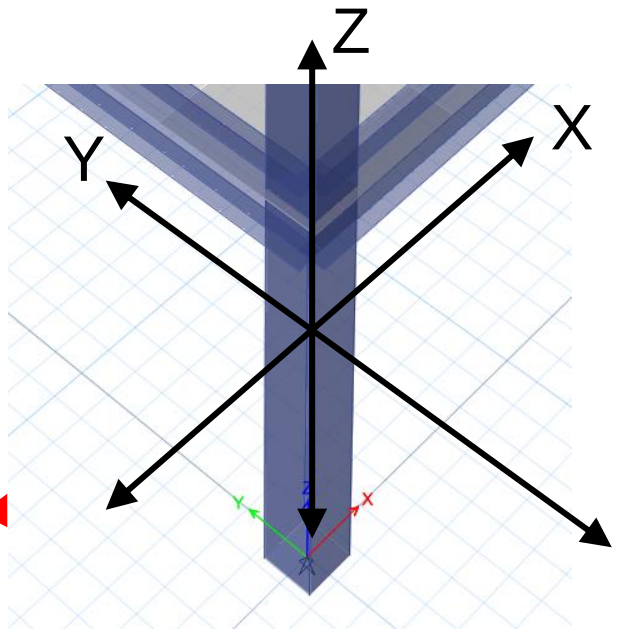




A Horizontal 3D Frame

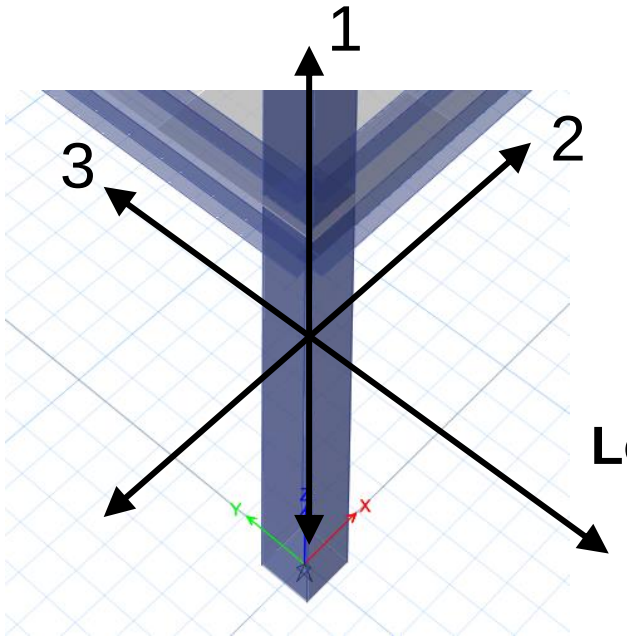
Element (in Global XY Plane)





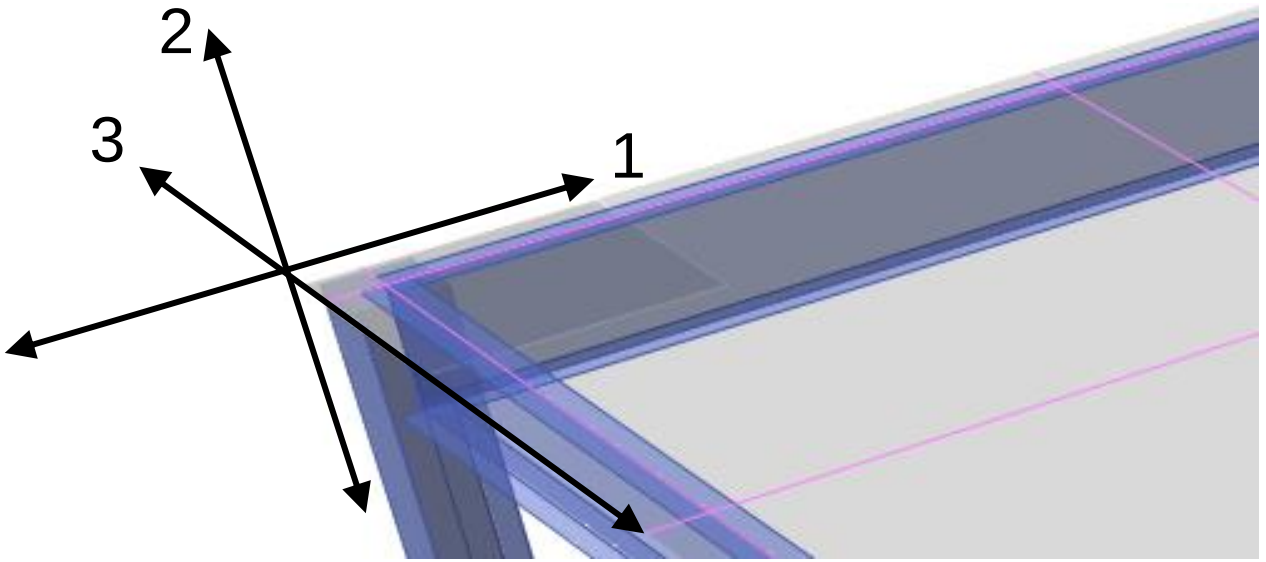

**Global Coordinate System**




**Local Coordinate System**

A Vertical 3D Frame Element

(in Global XZ Plane)




U1 → P

U2 → V2

U3 → V3

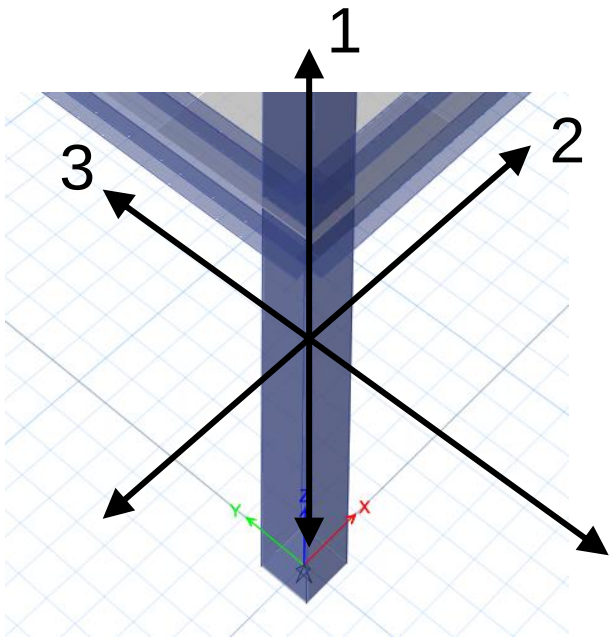
R1 → T

R2 → M2

R3 → M3

**Not all combinations are important in all the practical cases !!!**

| | |
|---|---|
| **M3** | → A beam subjected to pure bending moment caused by gravity load |
| **P, M3** | → A beam subjected to axial load and bending caused by gravity load |
| **V2** | → A beam subjected to pure shear force caused by gravity load |
| **P, V2, M3** | → A beam subjected to axial load, shear and bending caused by gravity load |
| **T** | → A beam subjected to pure torsion |
| **M3, V2, P, T** | → A beam subjected to bending moment and shear caused by gravity load as well as axial load and torsion |




**P** → A (perfect) concentrically loaded column

**P, M3** → A column under uniaxial bending. A column subjected to axial load and bending moment about axis 3. This bending moment can be transferred to this column from beam connecting in axis 2 direction.

**P, M2** → A column under uniaxial bending. A column subjected to axial load and bending moment about axis 2. This bending moment can be transferred to this column from beam connecting in axis 3 direction.

**P, M2, M3** → A column under biaxial bending. A column subjected to axial load and two bending moments (i.e. both about axis 2 and axis 3). These bending moment can be transferred to this column from beams connecting in axis 2 and axis 3 directions.

U1 → P

U2 → V2

U3 → V3

R1 → T

R2 → M2

R3 → M3

*The shear in beam along axis 2 direction (due to gravity load) will be transferred to this column as axial force P. Any axial force in this beam (along axis 2) will be transferred to this column as shear V2. Similarly, the shear in beam along axis 3 direction (due to gravity load) will be transferred to this column as axial load P. Any axial force in this beam (along axis 3) will be transferred to this column as shear V3.*

**P, M3, V2, M2, V3** → A column subjected to biaxial bending as well as biaxial shear.

**P, M3, V2, M2, V3, T** → The most unlucky column, subjected to everything, i.e. biaxial bending, biaxial shear as well as torsion.

# Cross-section Stiffness and Cross-section Properties

- The action along each degree of freedom is related to the corresponding deformation by the member stiffness, which in turn, depends on the cross-section stiffness.
- So there is a particular cross-section property corresponding to member stiffness for each degree of freedom. Therefore, for the six degrees of freedom defined earlier, the related cross-section properties are:

$UY$ ⇨ Cross-section area, $A$

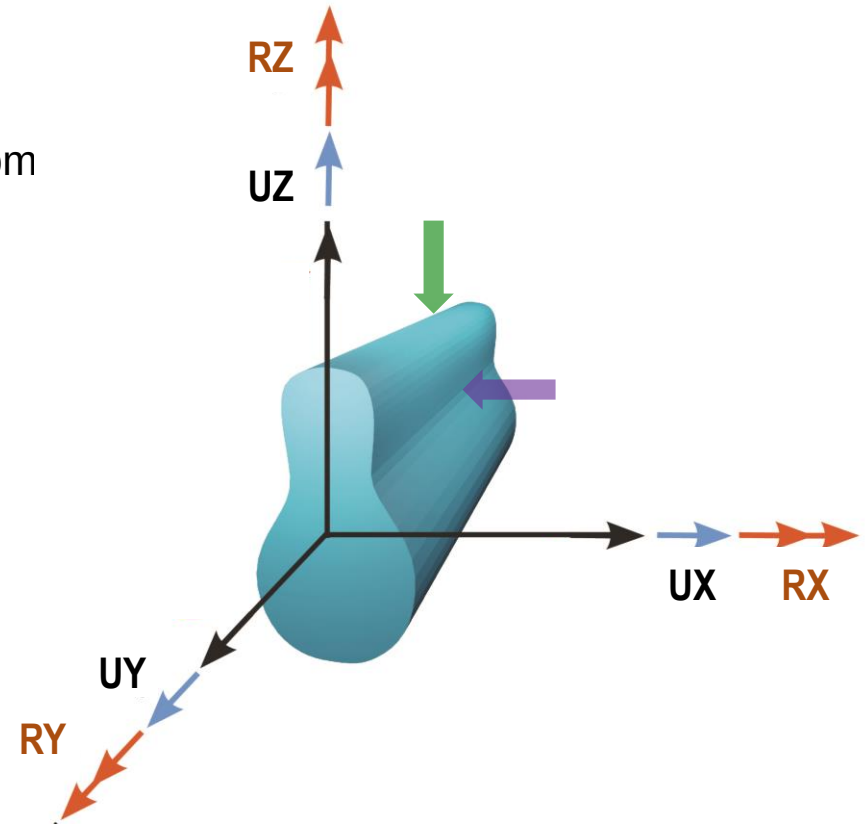
$UX$ ⇨ Shear Area along x, $SA_X$

$UZ$ ⇨ Shear Area along y, $SA_Z$

$RY$ ⇨ Torsional Constant, $J$

$RX$ ⇨ Moment of Inertia, $I_X$

$RZ$ ⇨ Moment of Inertia, $I_Z$



## Section Properties

- A **Frame Section** is a set of material and geometric properties that describe the cross-section of one or more Frame elements.
- Sections are defined independently of the Frame elements, and are **assigned** to the elements.
- Section properties are of two basic types:
  - Prismatic — all properties are constant along the full element length
  - Non-prismatic — the properties may vary along the element length

# Basic and Derived Properties

- Difference between Geometric and Section Properties
    - Geometric properties – No regard to material stiffness
    - Cross-section Properties: Due regard to material stiffness
- Cross-sectional properties can be categorized in many ways. From the computational point of view, we can look at the properties in terms of;
    - Basic or Intrinsic Properties
    - Derived Properties
    - Specific Properties for Reinforced Concrete Sections
    - Specific Properties for Pre-stressed Concrete Sections
    - Specific Properties for Steel Sections

## Cross-sectional Properties

- The area of the cross-section, $A$
- The first moment of area about a given axis, ($A.y$ or $A.x$ etc.)
- The second moment of area about a given axis, ($A.y^2$ or $A.x^2$ etc.)
- The moment of inertia about a given axis, $I$
- The shear area along a given axis, $SA$
- The torsional constant about an axis, $J$
- The warping constant about an axis, $W_z$ or $C_w$
- The plastic section modulus about a given axis, $Z_P$
- The shear center, $SC$

## Cross-sectional Properties

- The geometric center with reference to the given axis, $x_0$ , $y_0$
- The plastic center with reference to the given axis, $x_p$, $y_p$
- The elastic section modulus with reference to the given axis, $S_x$ , $S_y$
- The radius of gyration with reference to the given axis, $r_x$ , $r_y$
- Moment of inertia about the principle axis of bending, $I_{11}$ , $I_{22}$
- The orientation of the principal axis of bending, $\theta_p$
- The warping constant, $C_w$

# Basic Properties about x-y

The cross-sectional area $A = \int_A dA$

First Moment of Area about x and y:

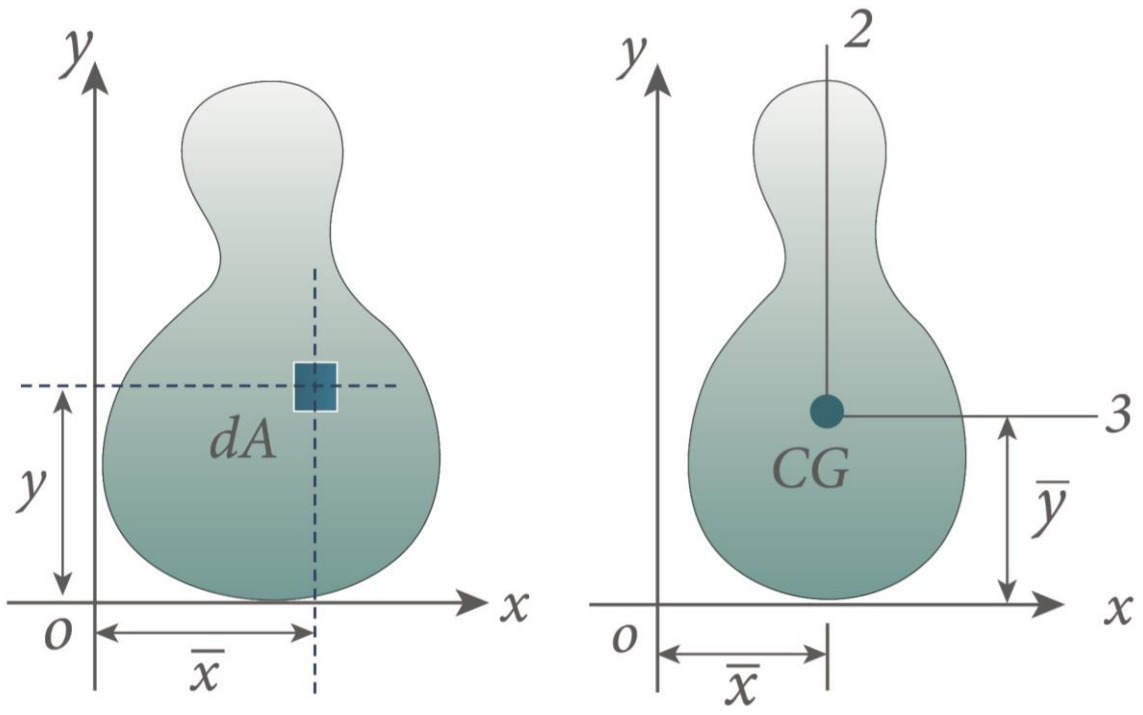
$A.y = y \int_A dA$

$A.x = x \int_A dA$

Second Moment of Area about x, y:

$A.y^2 = y^2 \int_A dA$

$A.x^2 = x^2 \int_A dA$



# Basic Properties about x-y

Centroid:

$$\bar{x} = \frac{A.x}{A}$$

$$\bar{y} = \frac{A.y}{A}$$

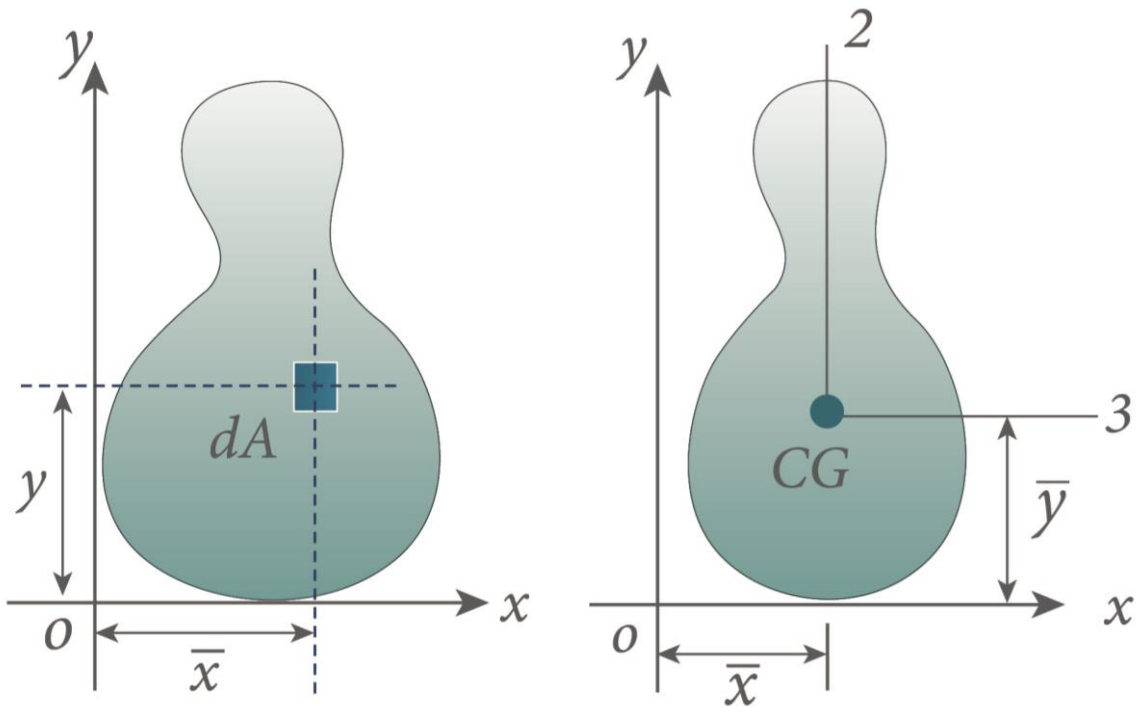
Moment of Inertia about x and y axis:

$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

Product of Inertia about xy-axes:

$$I_{xy} = \int_A x\, y\, dA$$



# Basic Properties about 2-3 Axis

Properties about the x-y coordinate system can be transformed to any other location using the Transformation Rule. Thus:

Area $A_o = A$

Moment of Inertia about the local 33 axis:

$$I_{33} = I_{xx} - A\,\bar{y}^2$$

Moment of Inertia about the local 22 axis:

$$I_{22} = I_{yy} - A\,\bar{x}^2$$

Product of Inertia:

$$I_{32} = I_{xy} - A\,\bar{x}\,\bar{y}$$

$$S_{3(top)} = \frac{I_{33}}{\bar{y}_{top}}$$

$$S_{3(bot)} = \frac{I_{33}}{\bar{y}_{bot}}$$

$$S_{2(left)} = \frac{I_{22}}{\bar{x}_{left}}$$

$$S_{2(right)} = \frac{I_{22}}{\bar{x}_{right}}$$

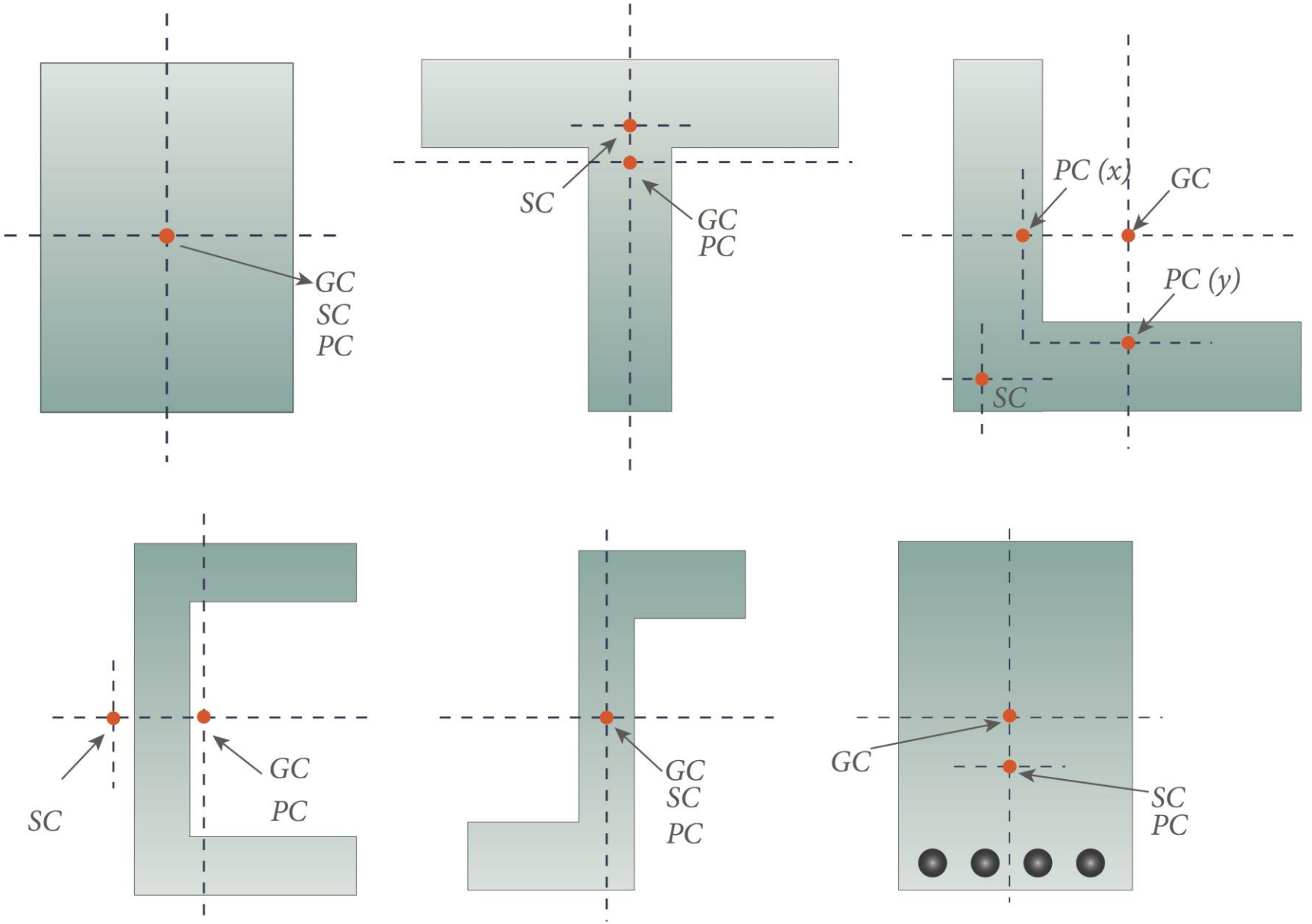
Radius of Gyration about 2-3 Axis:

$$r_3 = \sqrt{\frac{I_{33}}{A}}$$

$$r_2 = \sqrt{\frac{I_{22}}{A}}$$

# Centroids

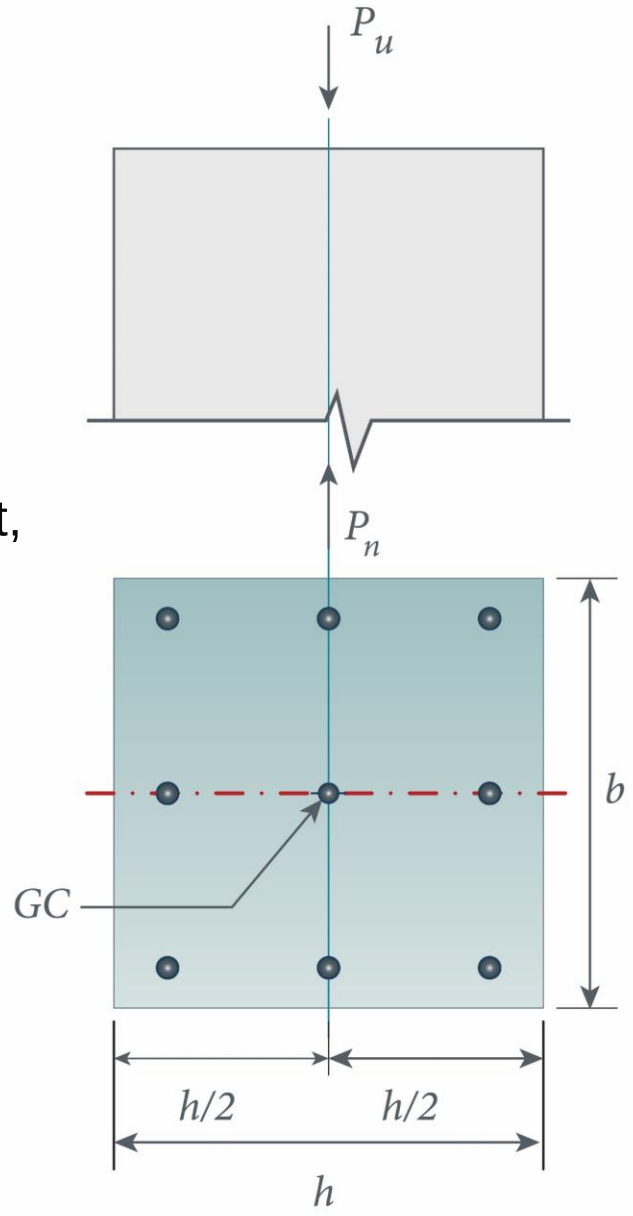
CG – Center of Gravity
SC – Shear Center
PC – Plastic Center



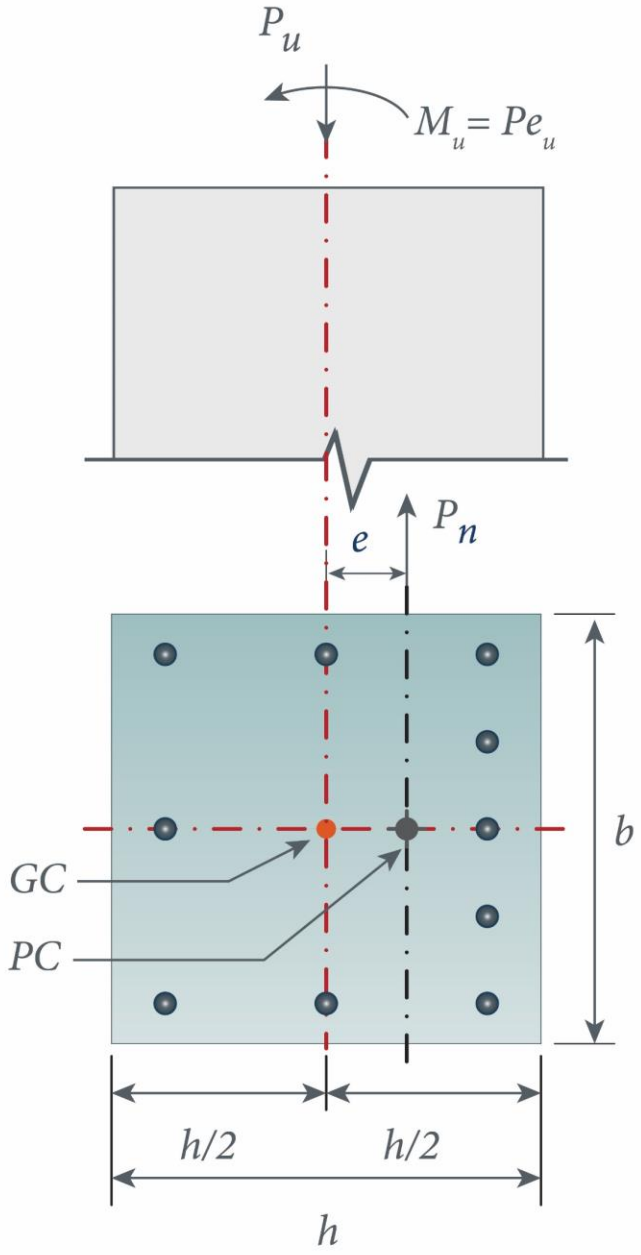
## The significance of geometric and plastic centroid in columns

(a) Symmetric rebar arrangement,

(b) Un-symmetric rebar arrangement

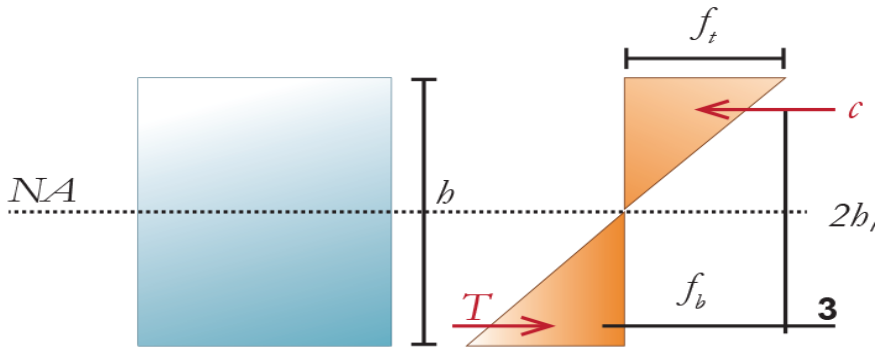



(a)



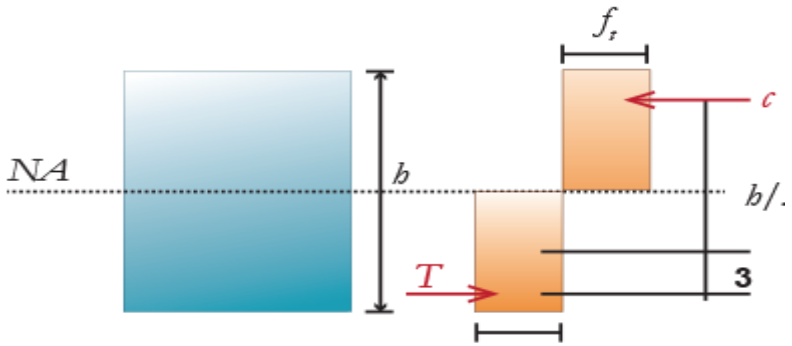

(b)

# Section Modulus




$$S_x = \frac{I_{xx}}{\bar{y}}$$

**Elastic**




$$Z_{Px} = \frac{bh^2}{4}$$

**Plastic**

# Shear Area

Shear Area about 2-3 axis:

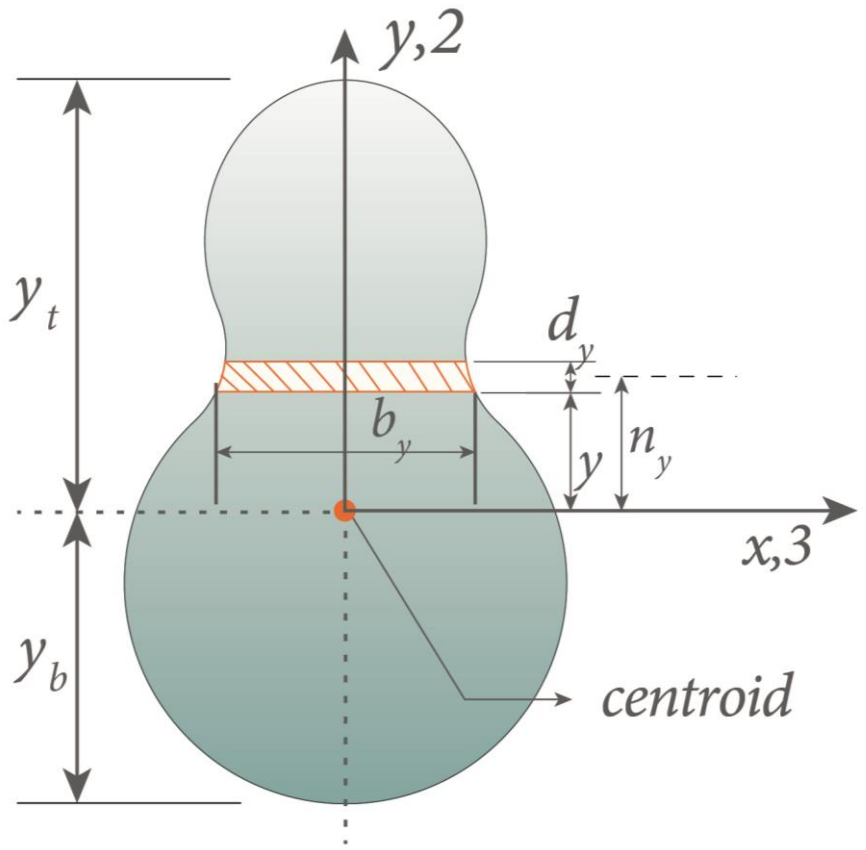
$$SA_3 = \frac{I_{33}^2}{\int_{y_b}^{y_t} \frac{Q^2(y)dy}{b(y)}}$$

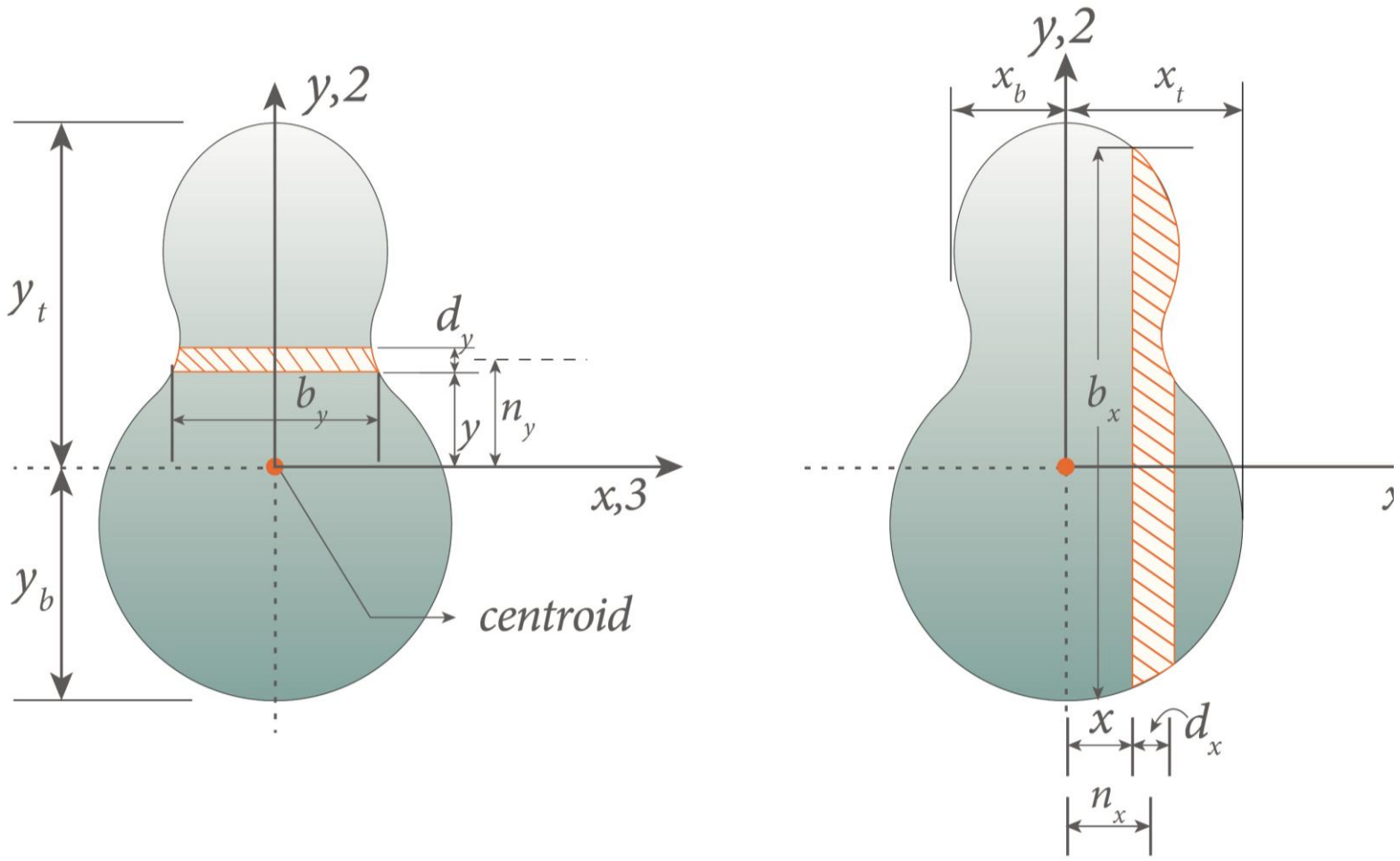
$$SA_2 = \frac{I_{22}^2}{\int_{x_b}^{x_t} \frac{Q^2(x)dx}{b(x)}}$$

where

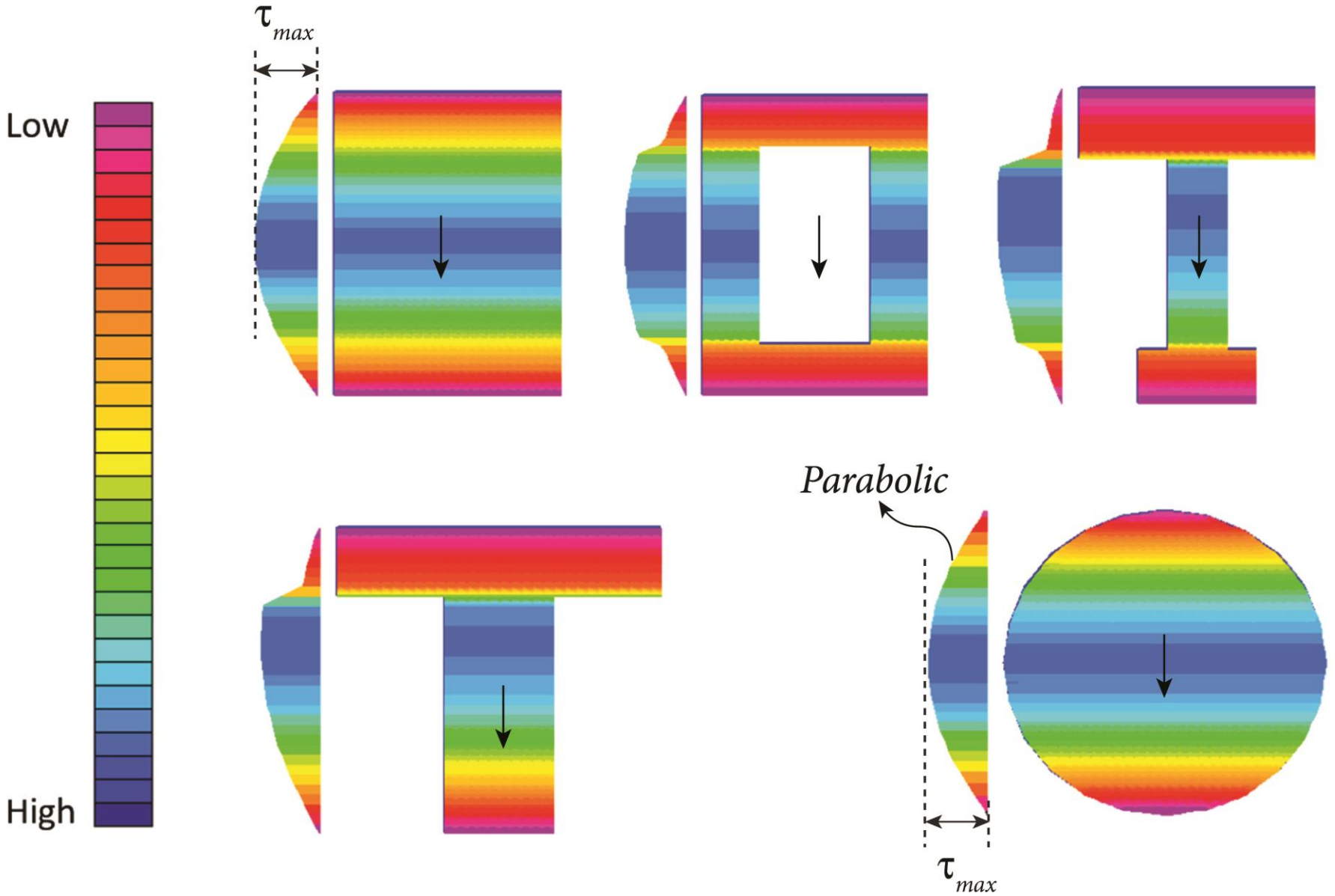
$$Q(y) = \int_{y}^{y_t} n\,b(n)dn$$

$$Q(x) = \int_{x}^{x_t} n\,b(n)dn$$






The shear area is obtained by integration of the shear stress distribution. The shear stress distribution is obtained by integration of static moment of area above the location on which shear stress is being determined.




Shear stress distribution due to shear along the y-axis in some common cross-section shapes.

# Torsional Constant, J

For a beam of uniform cross-section along its length:

$$\theta = \frac{TL}{JG}$$

where

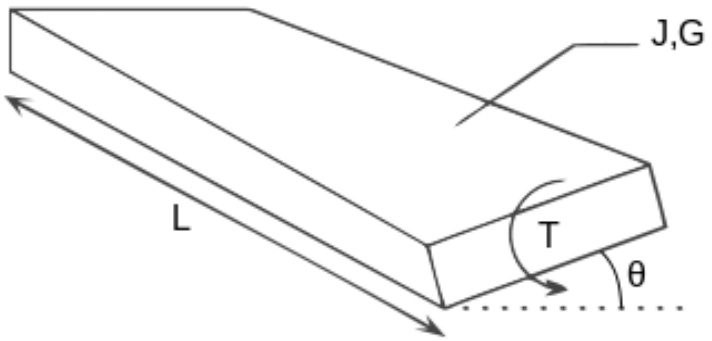
$\theta$ is the angle of twist in radians

$T$ is the applied torque

$L$ is the beam length

$J$ is the torsional constant

$G$ is the Modulus of rigidity (shear modulus) of the material

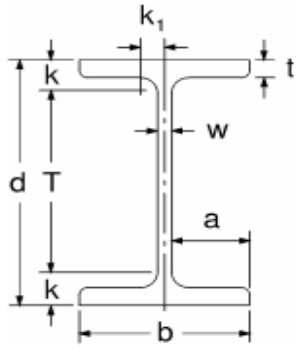



Circle $J = \frac{\pi D^4}{32}$

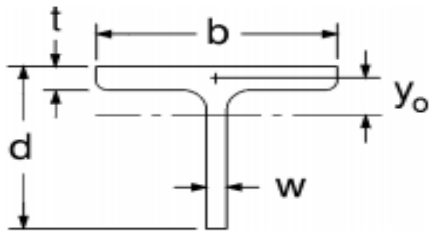
Square $J \approx 2.25a^4$

A finite element solution is need for general sections

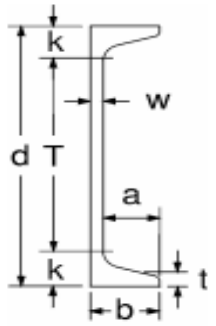
# Warping Constant, Cw



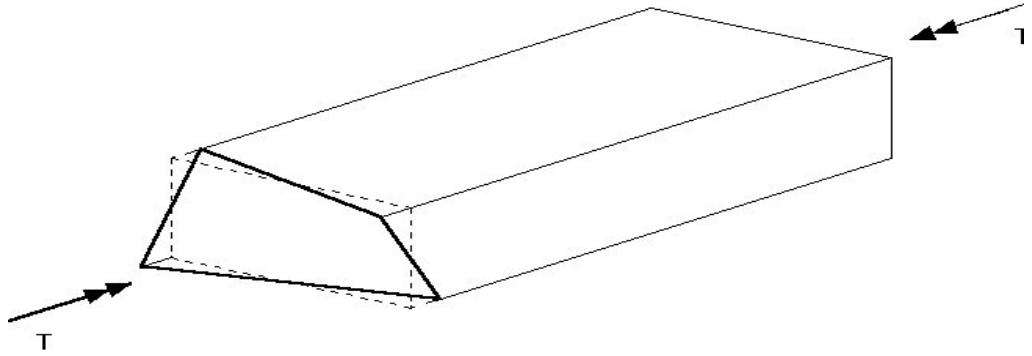

$$C_w = \frac{(d')^2 b^3 t}{24}$$



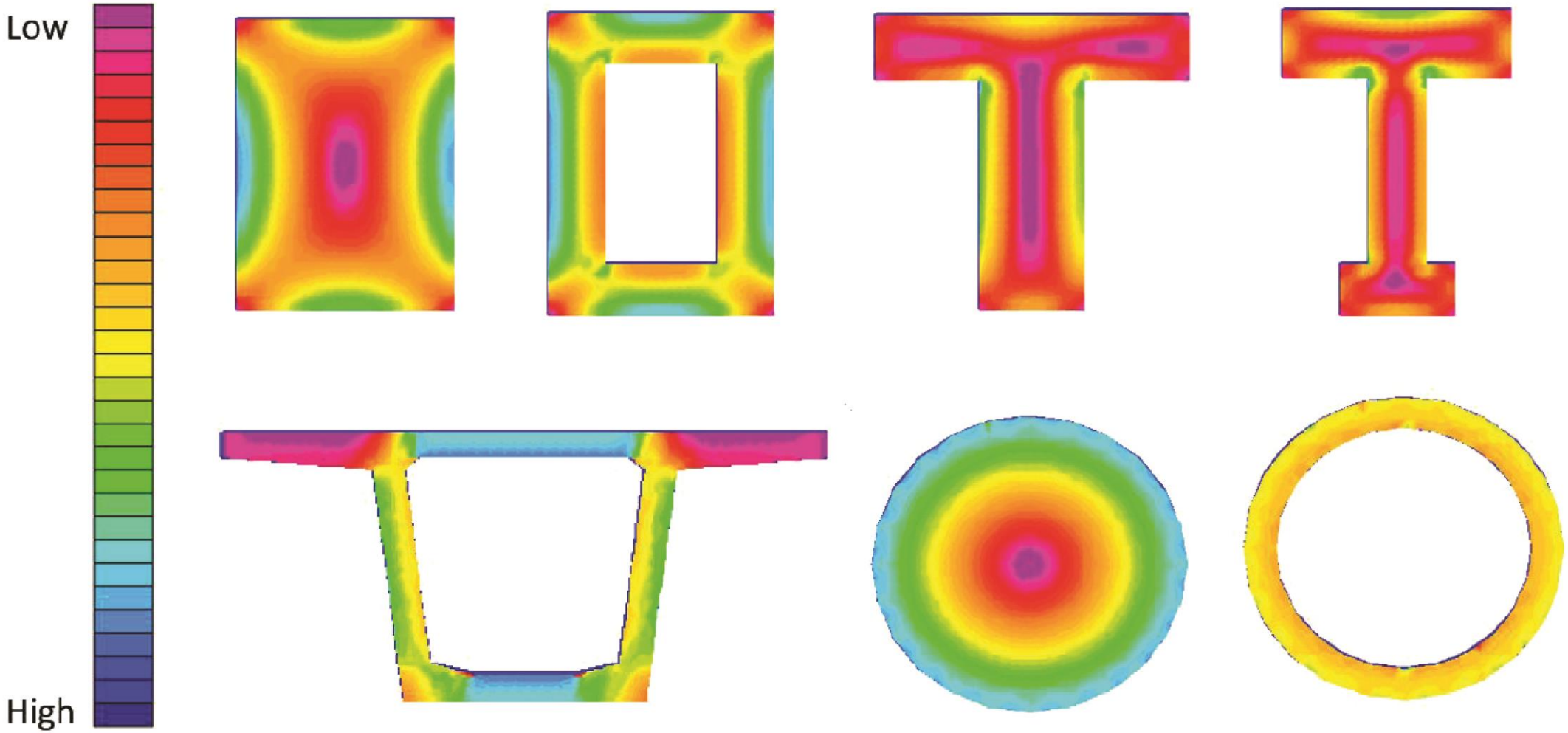

$$C_w = \frac{b^3 t^3}{144} + \frac{(d')^3 w^3}{36}$$




$$C_w = (d')^2 (b')^3 t \left[\frac{1-3\alpha}{6} + \frac{\alpha^2}{2}\left(1 + \frac{d'w}{6b't}\right)\right]$$




A finite element solution is need for general sections




Shear stress distribution due to torsion in some common cross-section shapes.

# Principal Properties

Moment of Inertia about Principal Axis *uu*

$$I_{uu} = \frac{I_{33} + I_{22}}{2} + \frac{I_{33} - I_{22}}{2}\cos 2\theta_p - I_{32}\sin 2\theta_p$$

Moment of Inertia about Principal Axis *vv*

$$I_{vv} = \frac{I_{33} + I_{22}}{2} - \frac{I_{33} - I_{22}}{2}\cos 2\theta_p + I_{32}\sin 2\theta_p$$

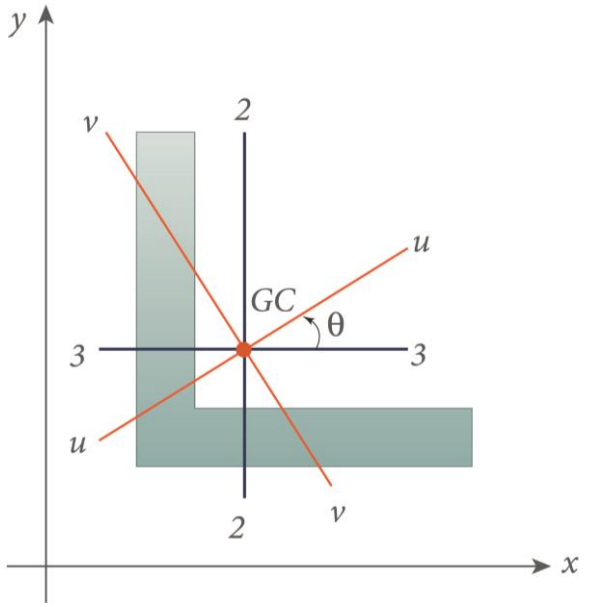
Principal Axis Angle:

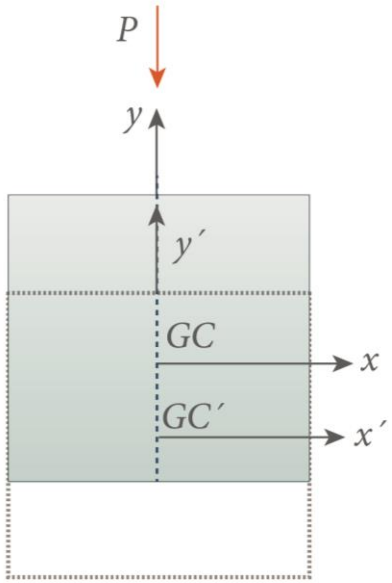
$$\tan 2\theta_p = \frac{2I_{32}}{I_{33} - I_{22}}$$

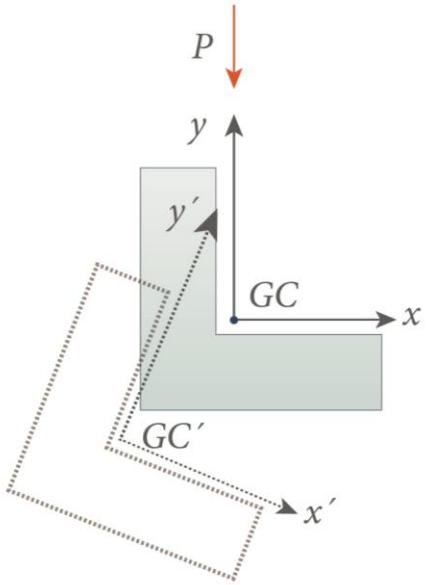
Radius of Gyration about Principal Axis:

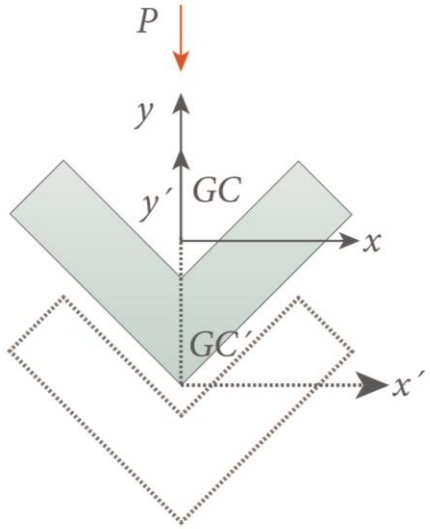
$$r_u = \sqrt{\frac{I_{uu}}{A}}$$

$$r_v = \sqrt{\frac{I_{vv}}{A}}$$



# Cracked Section Properties – RC Section

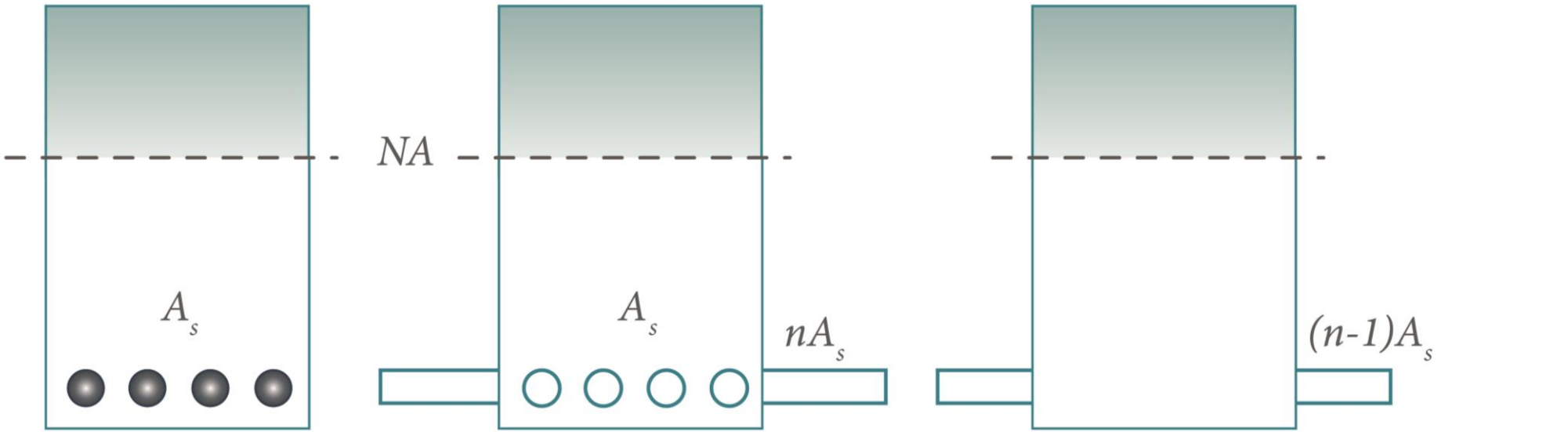
$$I_e = \left(\frac{M_{cr}}{M_a}\right)^3 I_g + \left[1 - \frac{M_{cr}}{M_a}\right]^3 I_{cr}$$

where, $M_{cr} = \dfrac{f_r I_g}{y_t}$

and $f_r = 0.62\lambda\sqrt{f_c'}$

| | | |
|---|---|---|
| $I_{cr}$ | = | Moment of inertia of cracked section transformed to concrete, $mm^4$ |
| $I_e$ | = | Effective moment of inertia for computation of deflection, $mm^4$ |
| $I_g$ | = | Moment of inertia of gross concrete section about centroidal axis, neglecting reinforcement, $mm^4$ |
| $M_{cr}$ | = | Cracking Moment, N-mm |
| $M_a$ | = | Applied Moment, N-mm |
| $f_c'$ | = | Compressive strength of concrete, Mpa |
| $f_r$ | = | Modulus of rupture of concrete, Mpa |
| $\lambda$ | = | Factor for lightweight aggregate concrete |
| $y_t$ | = | Distance from centroidal axis of gross section, neglecting reinforcement, to tension face, mm |

# Transformed Section Properties for RC



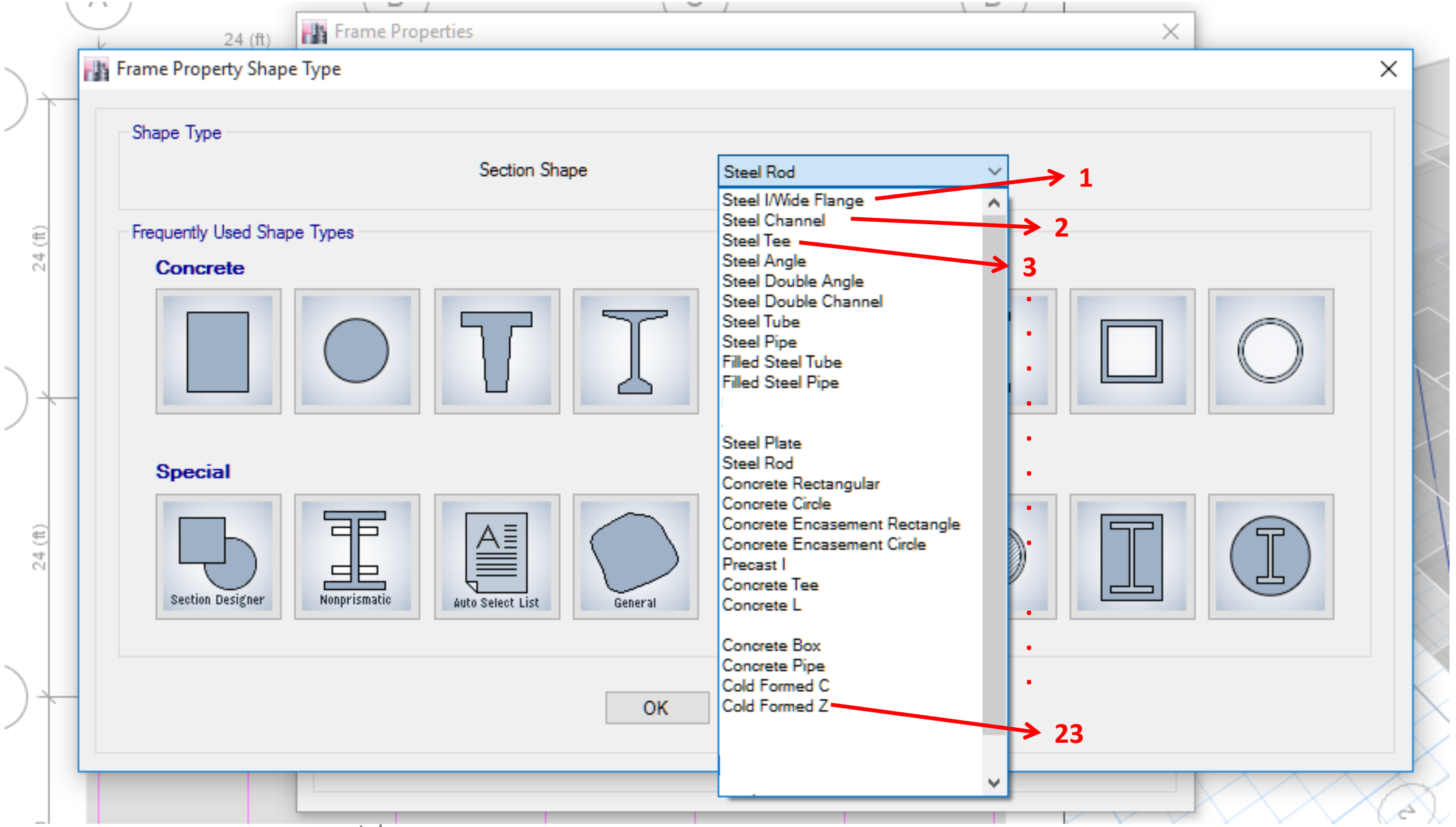
## Learning/Reading Assignment No. 2

# Learn to use CSI Section Designer in ETABS 2016/17/18

## Assignment No. 2

- Write a general computer program (in any programming language or MS Excel) to determine all cross-sectional properties of following cross-sections.

- Compare your results with SD ETABS 2016/17/18.

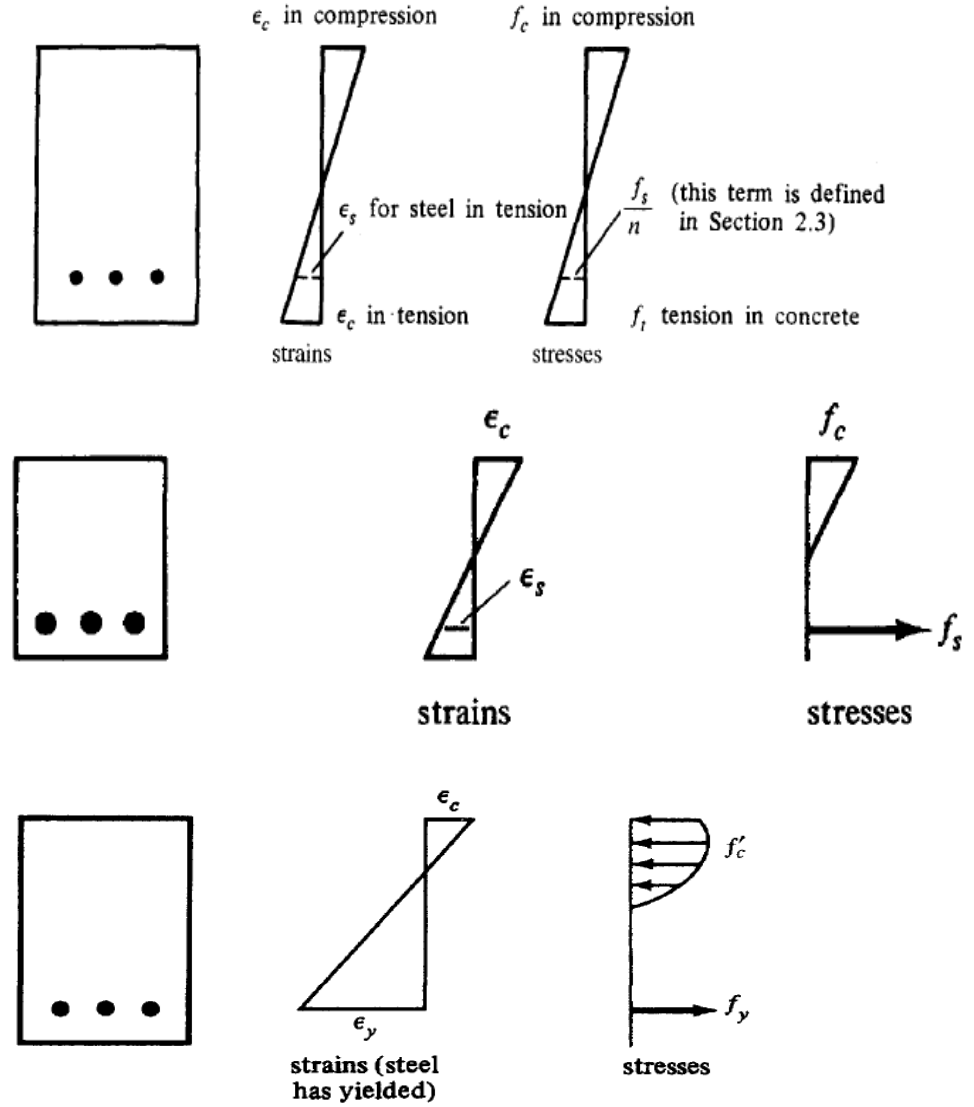

Frame Properties
Frame Property Shape Type
Shape Type
Section Shape
Steel Rod
Steel I/Wide Flange
Steel Channel
Steel Tee
Steel Angle
Steel Double Angle
Steel Double Channel
Steel Tube
Steel Pipe
Filled Steel Tube
Filled Steel Pipe
Steel Plate
Steel Rod
Concrete Rectangular
Concrete Circle
Concrete Encasement Rectangle
Concrete Encasement Circle
Precast I
Concrete Tee
Concrete L
Concrete Box
Concrete Pipe
Cold Formed C
Cold Formed Z
1
2
3
23
Frequently Used Shape Types
Concrete
Special
Section Designer
Nonprismatic
Auto Select List
General
OK
24 (ft)
24 (ft)
24 (ft)

# Flexural Analysis for Reinforced Concrete Design

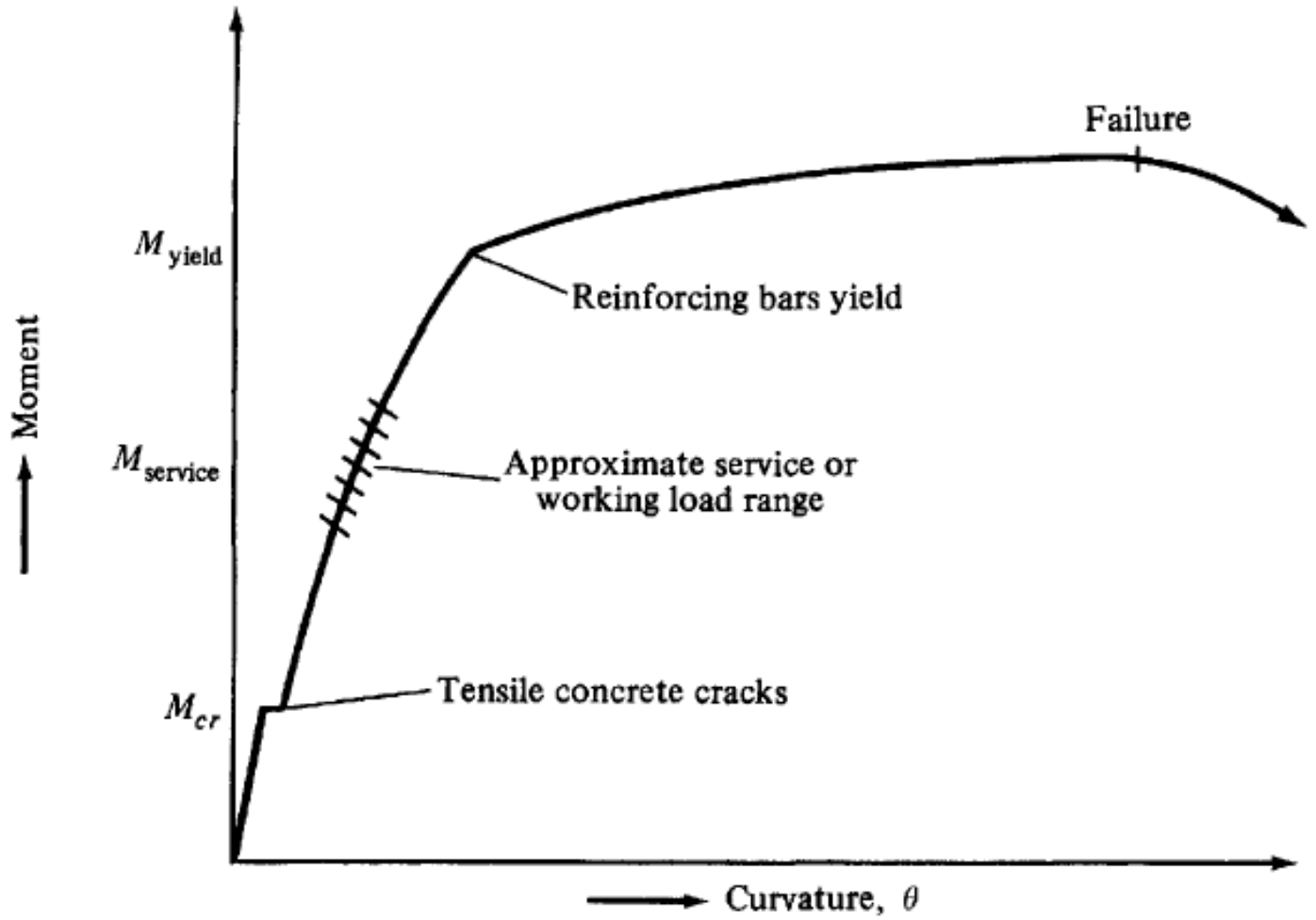
## – A Special Case of a more General Approach

# Stress Stages of a Beam - Stress and Strain Profiles

- The horizontal structural member, supporting the loads through flexure are known as beams
- Once load is applied to a beam, the stress stages may vary as
  - Uncracked Concrete Stage
  - Concrete Cracked – Elastic Stage
  - Ultimate Strength Stage



# Stress stages of RC beams



# Example 1: Flexural Analysis of a Beam (Uncracked Stage)

**(a)** Assuming the concrete is uncracked, compute the bending stresses in the extreme fibers of the beam of Figure 2.5 for a bending moment of 25 ft-k. The concrete has an $f'_c$ of 4000 psi and a modulus of rupture $f_r = 7.5\sqrt{4000} = 474$ psi.

**(b)** Determine the cracking moment of the section.

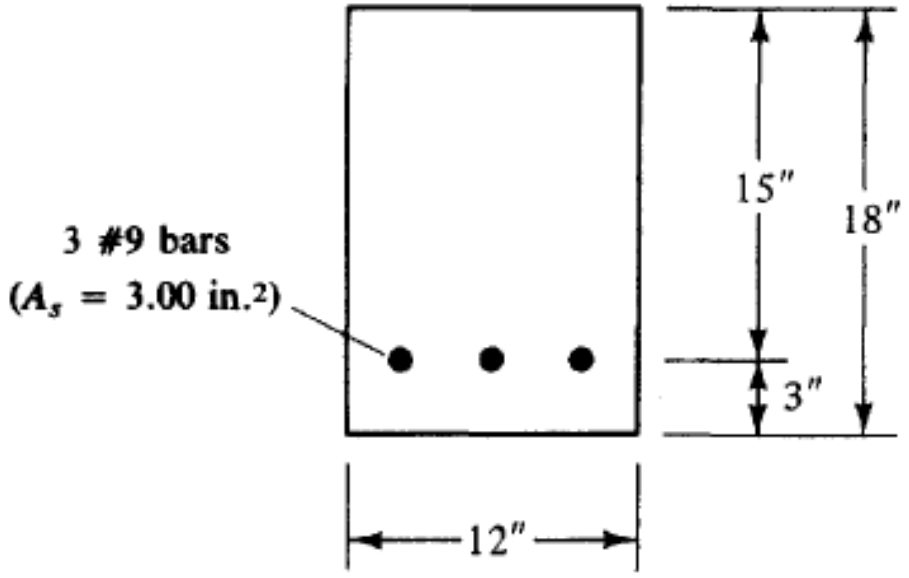



**Figure 2.5**

# Example 1: Flexural Analysis of a Beam (Uncracked Stage)

***SOLUTION*** **(a)** Bending stresses:

$$I_g = \frac{1}{12} bh^3 \text{ with } b = 12 \text{ in. and } h = 18 \text{ in.}$$

$$I_g = \left(\frac{1}{12}\right)(12)(18)^3 = 5832 \text{ in.}^4$$

$$f = \frac{My}{I_g} \text{ with } M = 25 \text{ ft-k} = 25{,}000 \text{ ft-lb}$$

Then multiply the 25,000 by 12 to obtain in.-lbs as shown below:

$$f = \frac{(12 \times 25{,}000)(9.00)}{5832} = 463 \text{ psi}$$

Since this stress is less than the tensile strength or modulus of rupture of the concrete of 474 psi, the section is assumed not to have cracked.

**(b)** Cracking moment:

$$M_{cr} = \frac{f_r I_g}{y_t} = \frac{(474)(5832)}{9.00} = 307{,}152 \text{ in.-lb} = \underline{25.6 \text{ ft-k}}$$

## Concrete Cracked-Elastic Stage

- If the bending stresses at bottom fibers of beam exceed the modulus of rupture then concrete will crack
- Stresses are transferred to steel reinforcing bars
- Steel bars are supposed to be in place due to bondage with concrete
- The strains are same not stresses
- Stresses are computed with modular ratio $E_s/E_c$
- The calculation is done through transformed area concept
- Can continue unless concrete stresses are less than $0.50f_c$' and steel stresses less than yield
- The stage is applicable for service loads

# Example 2: Flexural Analysis of a Beam (Cracked-Elastic Stage)

Calculate the bending stresses in the beam shown in Figure 2.7 by using the transformed area method: $n = 9$ and $M = 70$ ft-k.

Determine the allowable resisting moment of the beam of Example 2.2 if the allowable stresses are $f_c = 1350$ psi and $f_s = 20{,}000$ psi.

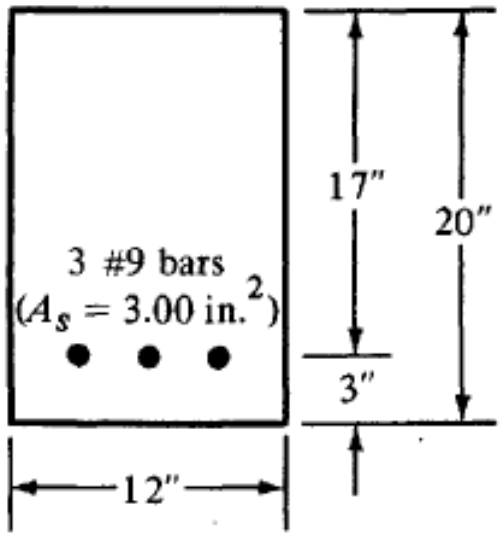



**Figure 2.7**

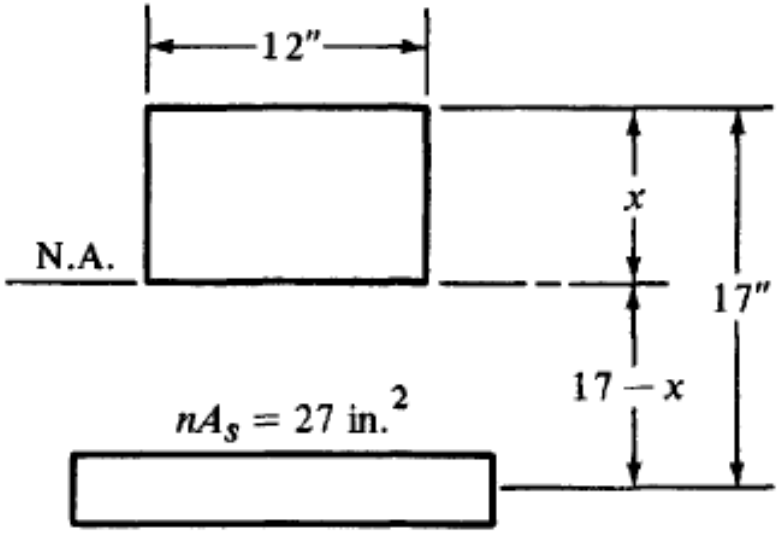



**Figure 2.8**

# Example 2: Flexural Analysis of a Beam (Cracked-Elastic Stage)

**Taking Moments about Neutral Axis (Referring to Figure 2.8)**

$$(12x)\left(\frac{x}{2}\right) = (9)(3.00)(17 - x)$$

$$6x^2 = 459 - 27.00x$$

**Solving by Completing the Square**

$$6x^2 + 27.00x = 459$$

$$x^2 + 4.50x = 76.5$$

$$(x + 2.25)(x + 2.25) = 76.5 + (2.25)^2$$

$$x + 2.25 = \sqrt{76.5 + (2.25)^2} = 9.03''$$

$$x = 6.78''$$

**Moment of Inertia**

$$I = (\tfrac{1}{3})(12)(6.78)^3 + (9)(3.00)(10.22)^2 = 4067 \text{ in.}^4$$

# Example 2: Flexural Analysis of a Beam (Cracked-Elastic Stage)

**Bending Stresses**

$$f_c = \frac{My}{I} = \frac{(12)(70{,}000)(6.78)}{4067} = \underline{1400 \text{ psi}}$$

$$f_s = n\frac{My}{I} = (9)\frac{(12)(70{,}000)(10.22)}{4067} = \underline{18{,}998 \text{ psi}}$$

$$M_c = \frac{f_c I}{y} = \frac{(1350)(4067)}{6.78} = 809{,}800 \text{ in.-lb} = \underline{67.5 \text{ ft-k}} \leftarrow$$

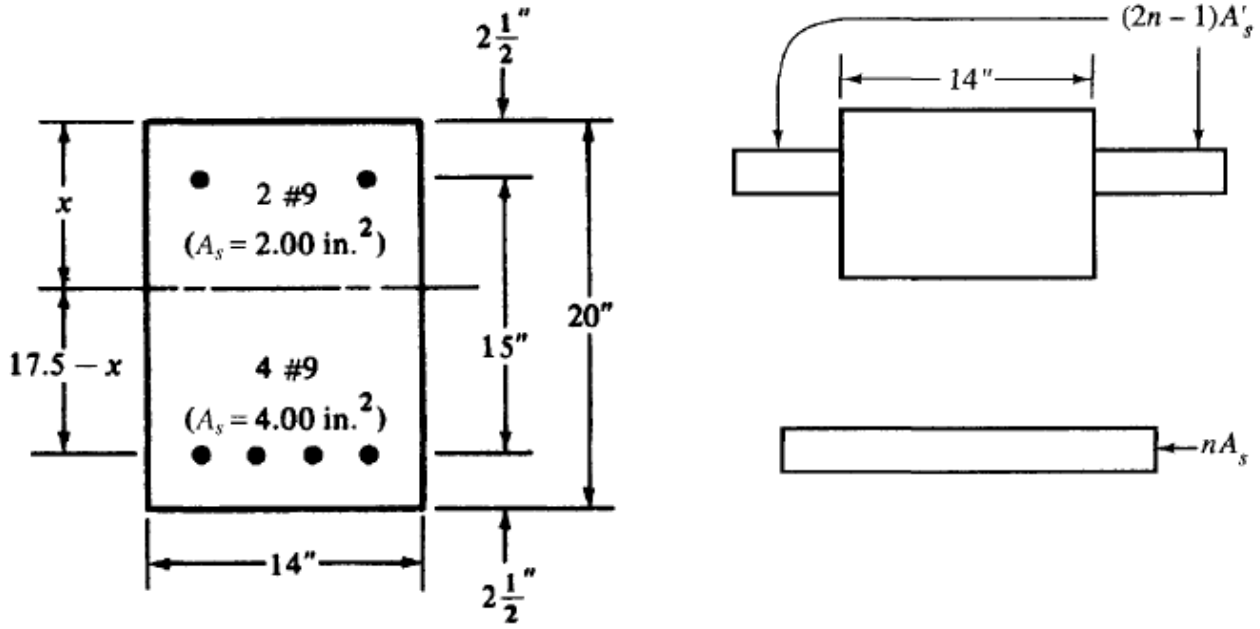
$$M_s = \frac{f_s I}{ny} = \frac{(20{,}000)(4067)}{(9)(10.22)} = 884{,}323 \text{ in.-lb} = 73.7 \text{ ft-k}$$

# Concrete Cracked-Elastic Stage – Beams with Compression Bars

- Reinforcement added to the top (compression) side of the beam is called compression reinforcement or negative steel
- Used for fixed and continuous supports
- Is very useful even for long term deflections
- Can reduce the cross-sectional size of the members
- Long term sustained loads can cause creep & negative bars may get overstressed
- Transformed area is computed by $(2n-1)A'_s$
  - $n$ = Modular ratio
  - $A'_s$ = Compression steel area

# Example 3: Flexural Analysis of a Doubly-reinforced Beam (Cracked-Elastic Stage)

Compute the bending stresses in the beam shown in Figure 2.10; $n = 10$ and $M = 118$ ft-k.




(a) Actual section (b) Transformed section

**Figure 2.10**

# Example 3: Flexural Analysis of a Doubly-reinforced Beam (Cracked-Elastic Stage)

***SOLUTION*** **Locating Neutral Axis**

$$(14x)\left(\frac{x}{2}\right) + (20 - 1)(2.00)(x - 2.5) = (10)(4.00)(17.5 - x)$$

$$7x^2 + 38x - 95 = 700 - 40x$$

$$7x^2 + 78x = 795$$

$$x^2 + 11.14x = 113.57$$

$$x + 5.57 = \sqrt{113.57 + (5.57)^2} = 12.02$$

$$x = 6.45 \text{ in.}$$

**Moment of Inertia**

$$I = (\tfrac{1}{3})(14)(6.45)^3 + (20 - 1)(2.00)(3.95)^2 + (10)(4.00)(11.05)^2 = 6729 \text{ in.}^4$$

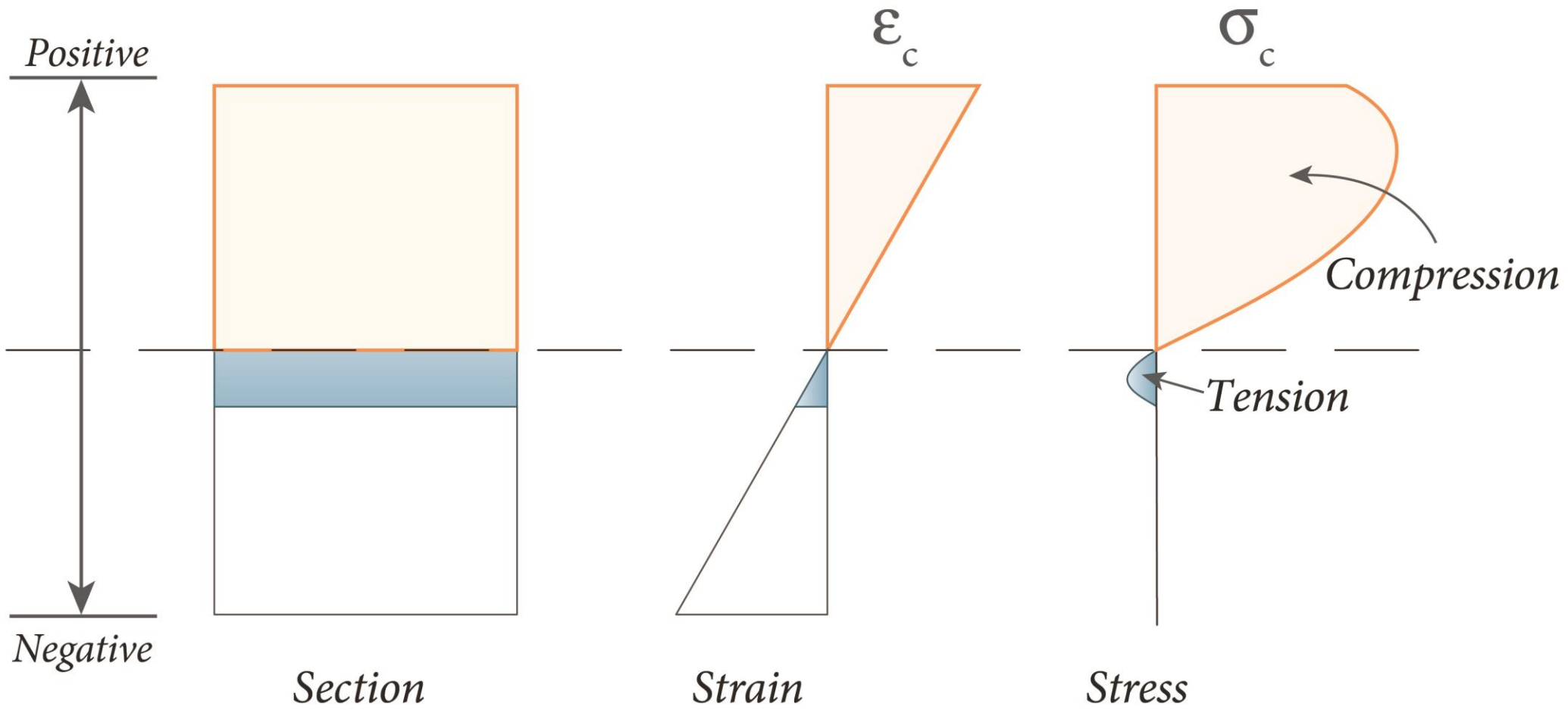
**Bending Stresses**

$$f_c = \frac{(12)(118{,}000)(6.45)}{6729} = \underline{1357 \text{ psi}}$$

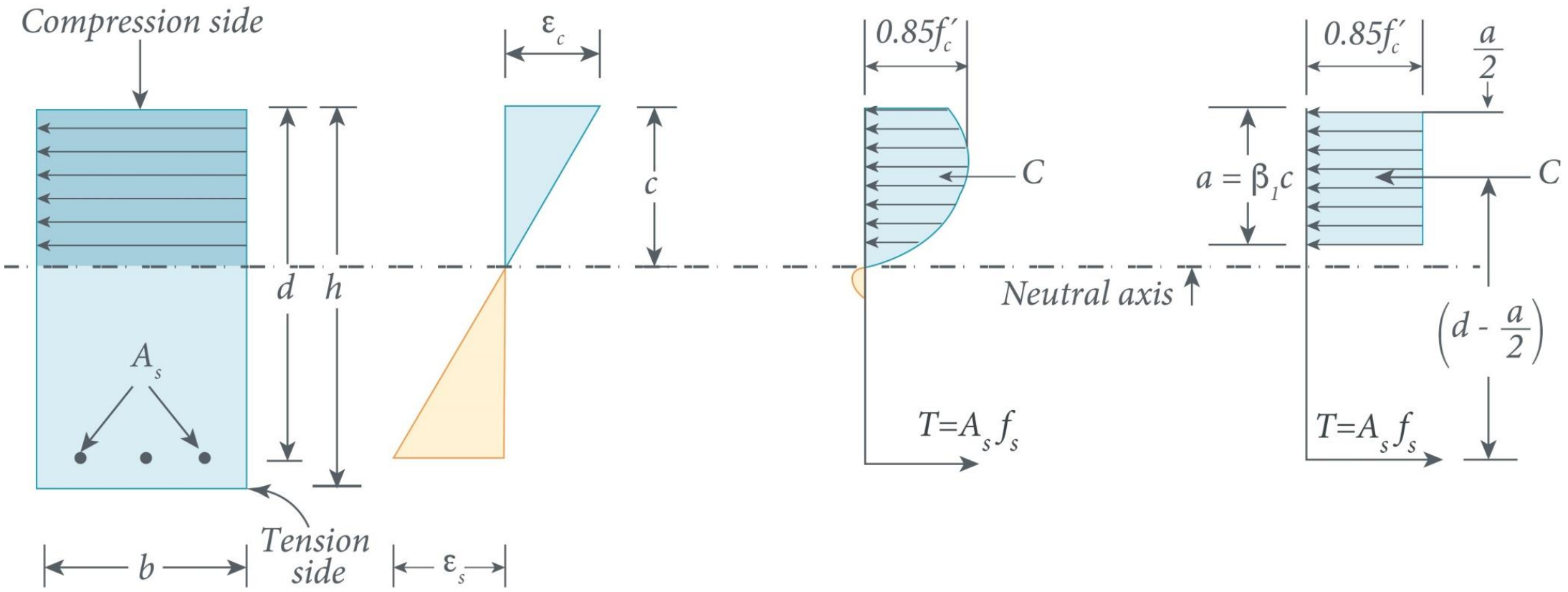
$$f'_s = 2n\frac{My}{I} = (2)(10)\frac{(12)(118{,}000)(3.95)}{6729} = \underline{16{,}624 \text{ psi}}$$

$$f_s = (10)\frac{(12)(118{,}000)(11.05)}{6729} = \underline{23{,}253 \text{ psi}}$$

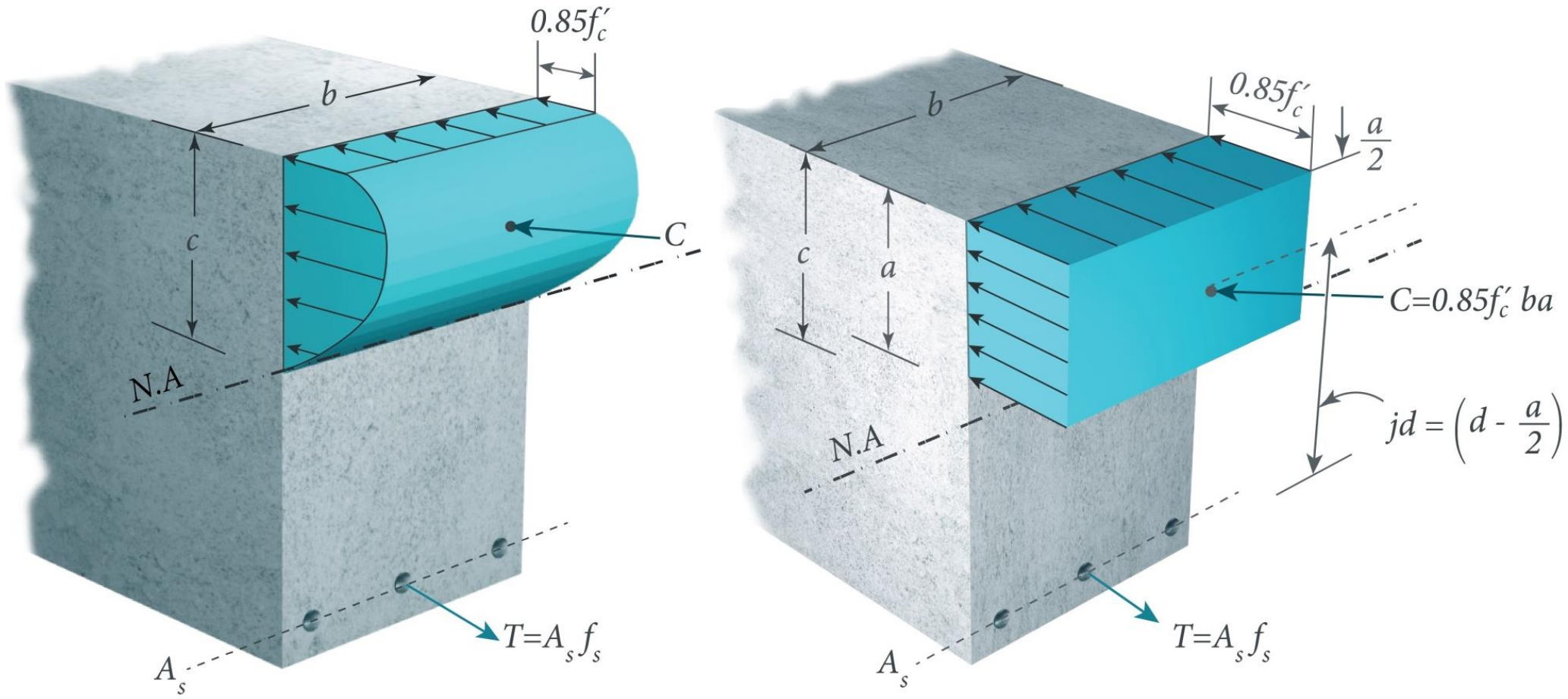
# Ultimate Strength Stage - Stress and Strain Profiles



# Equivalent Rectangular Stress Block

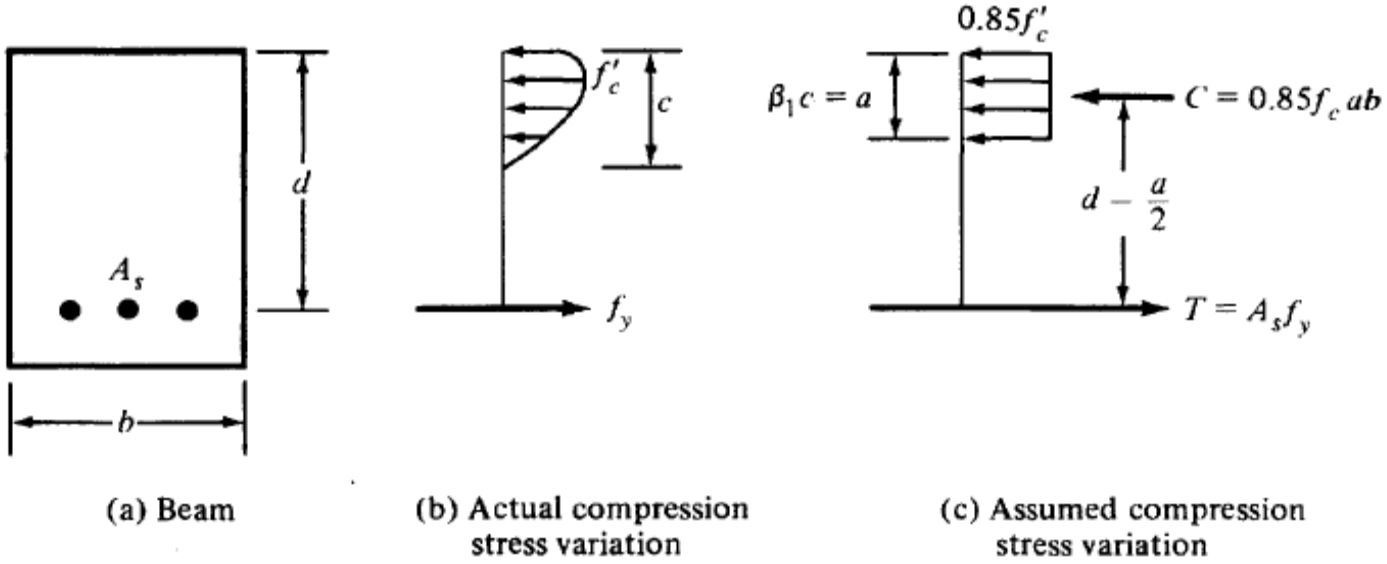


# Equivalent Rectangular Stress Block



# Ultimate Strength Stage - Equivalent Rectangular Stress Block

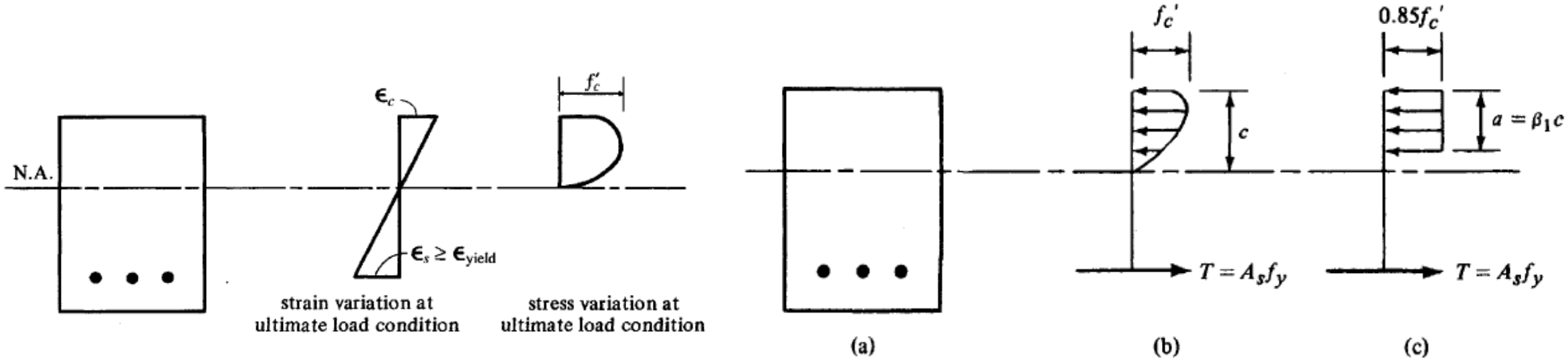
- Once the stresses in beam exceeds 0.50fc' or yield strength of steel
  - Concrete stress distribution is nonlinear from N.A
  - Parabolic distribution
  - May be approximated by simplified rectangular distribution
    - Same C.G
    - Same magnitude
  - Whitney stress block is used
- The stage is most common for design of all ultimate strength designs as per ACI




(a) Beam (b) Actual compression stress variation (c) Assumed compression stress variation

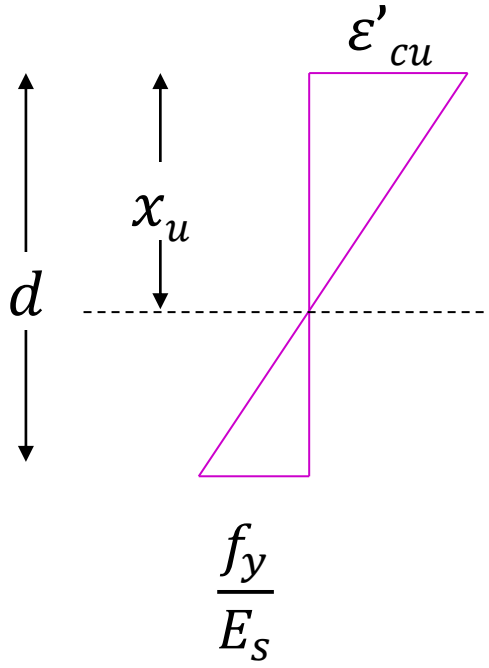
# Flexural Analysis of a Singly Reinforced Beam (Ultimate Strength Stage)

- Compression stresses vary linearly from neutral axis unless the stress increases beyond 0.50fc'
- The actual stress distribution is parabolic
- The approximation be required such that equivalent shape of stress block should give same centroid and resultant force
- Whitney gave the best results, proved from various tests



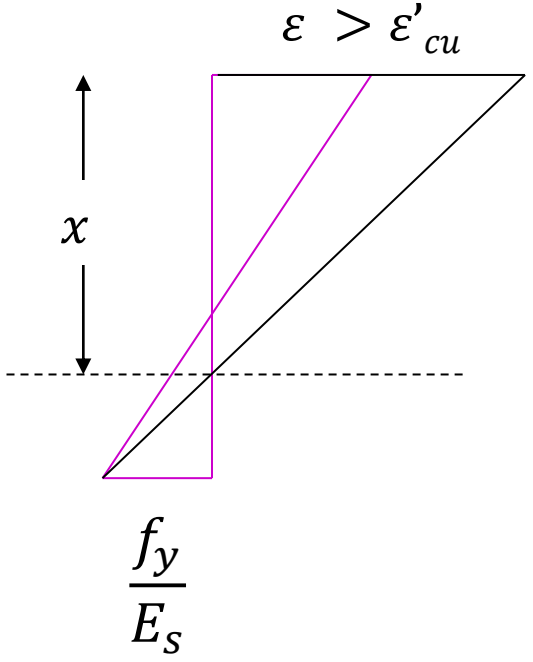
## Balanced State

Concrete and steel reach their failure strain simultaneously



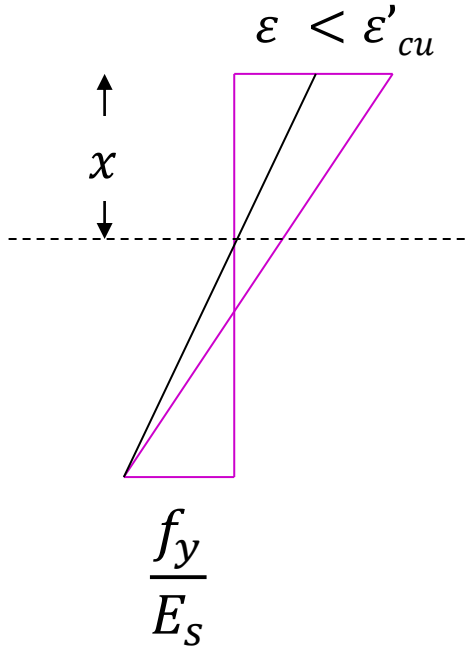

## Over-reinforced State

Concrete reaches failure strain prior to Steel ($x > x_u$)




## Under-reinforced State

Steel reaches failure strain prior to concrete ($x < x_u$)



# Determining $A_{st}$ for Singly Reinforced Beam

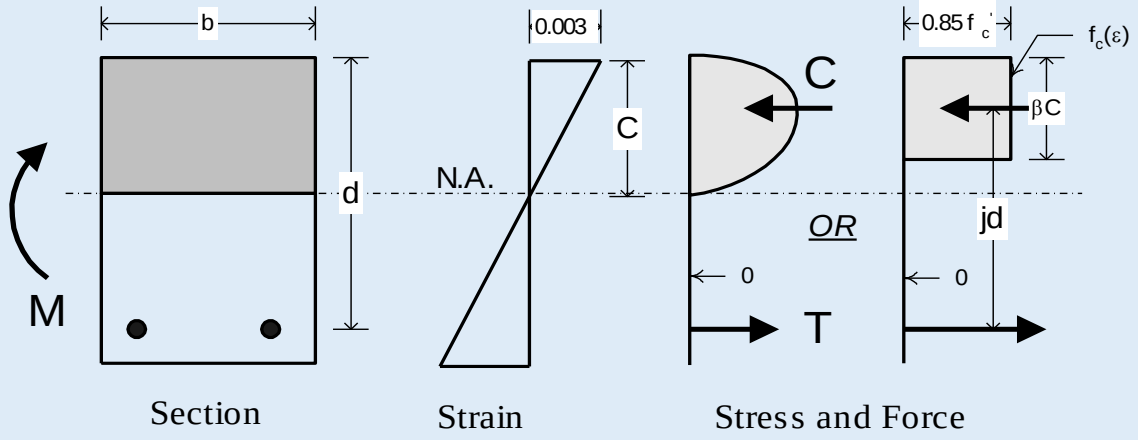
$$A_{st} = \frac{M_u}{\phi f_y \left( d - \frac{a}{2} \right)}$$

$$where,\ a = c.\beta(f_c')$$

$$a = \frac{A_{st} f_y}{.85 f_c b}$$

$$M_n = \phi\,(.85 f_c')ab\left( d - \frac{a}{2} \right)$$

This procedure for $A_{st}$ is iterative




$$A_{st} = \frac{\beta f_c b}{f_y}\left( d + \sqrt{d^2 - \frac{2M_u}{\phi \beta f_c b}} \right)$$

$\beta$ = 0.85 to 0.65 , $\phi$ = 0.65 - 0.9

## ACI 318 Design Approach - Determine $M_n$

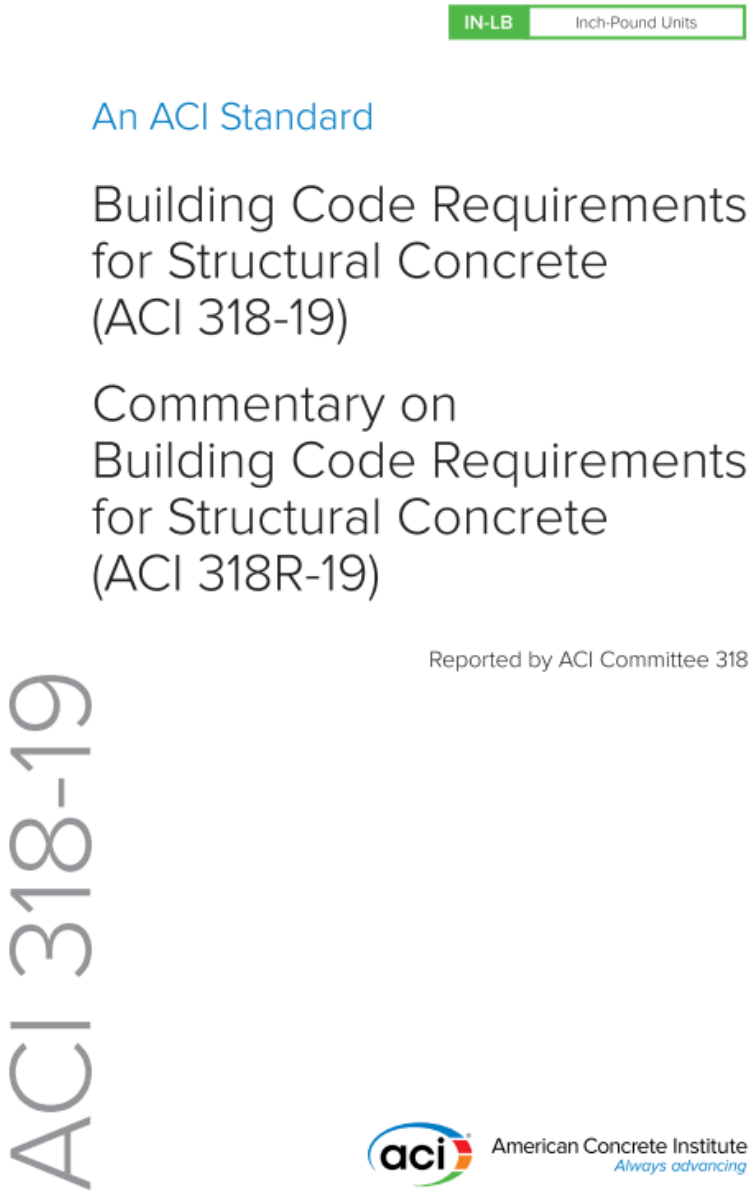
$$M_n(f_c', f_y, b, d)$$

---

$$\varepsilon_c = 0.003, \quad \varepsilon_s = \frac{f_y}{E_s}$$

$$c \longrightarrow c_b(\varepsilon_c, \varepsilon_s), \quad a_b = c_b \beta(f_c)$$

- $M_n$ is a measure of the capacity of concrete in compression to resist moment .
- It also ensures some ductility by forcing failure in tension
- It primarily depends on $f_c, b, d$



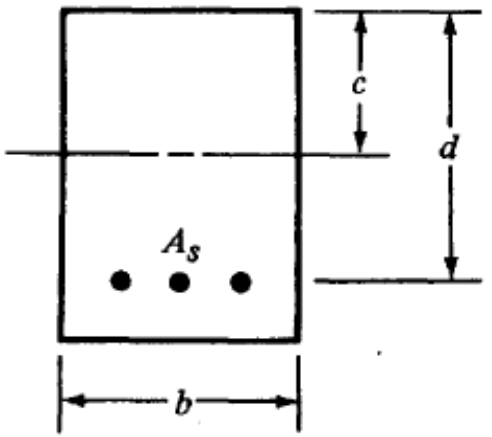
# ACI 318 Design Approach

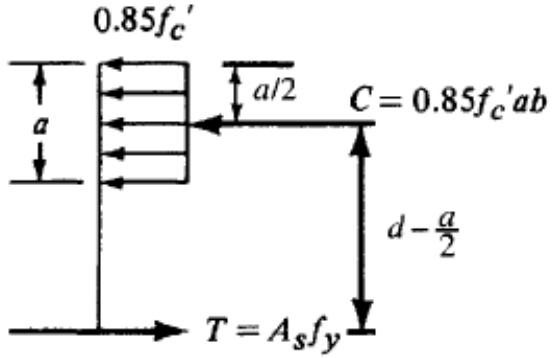
- The various parameters are defined as follows
  - For fc' 4000 psi, $\beta_1 = 0.85$
  - For concrete with fc' >4000 psi, it is calculated by

$$\beta_1 = 0.85 - \left(\frac{f_c' - 4000}{1000}\right)(0.05) \geq 0.65$$

  - Nominal moment is calculated for a section

$$M_n = T\left(d - \frac{a}{2}\right) = A_s f_y\left(d - \frac{a}{2}\right)$$



# ACI 318-19 Design Approach

## Table 9.3.1.1—Minimum depth of nonprestressed beams

| Support condition | Minimum $h$[1] |
|---|---|
| Simply supported | $\ell/16$ |
| One end continuous | $\ell/18.5$ |
| Both ends continuous | $\ell/21$ |
| Cantilever | $\ell/8$ |

[1]Expressions applicable for normalweight concrete and $f_y$ = 60,000 psi. For other cases, minimum $h$ shall be modified in accordance with 9.3.1.1.1 through 9.3.1.1.3, as appropriate.

**9.3.1.1.1** For $f_y$ other than 60,000 psi, the expressions in Table 9.3.1.1 shall be multiplied by $\mathbf{(0.4 + f_y/100{,}000)}$.

## 9.5—Design strength

### 9.5.1 *General*

**9.5.1.1** For each applicable factored load combination, design strength at all sections shall satisfy $\phi S_n \geq U$ including (a) through (d). Interaction between load effects shall be considered.

(a) $\phi M_n \geq M_u$
(b) $\phi V_n \geq V_u$
(c) $\phi T_n \geq T_u$
(d) $\phi P_n \geq P_u$

**9.5.1.2** $\phi$ shall be determined in accordance with 21.2.

# ACI 318-19 Design Approach

## Table 21.2.1—Strength reduction factors ϕ

| Action or structural element | | ϕ | Exceptions |
|---|---|---|---|
| (a) | Moment, axial force, or combined moment and axial force | 0.65 to 0.90 in accordance with 21.2.2 | Near ends of pretensioned members where strands are not fully developed, ϕ shall be in accordance with 21.2.3. |
| (b) | Shear | 0.75 | Additional requirements are given in 21.2.4 for structures designed to resist earthquake effects. |
| (c) | Torsion | 0.75 | — |
| (d) | Bearing | 0.65 | — |
| (e) | Post-tensioned anchorage zones | 0.85 | — |
| (f) | Brackets and corbels | 0.75 | — |
| (g) | Struts, ties, nodal zones, and bearing areas designed in accordance with strut-and-tie method in Chapter 23 | 0.75 | — |
| (h) | Components of connections of precast members controlled by yielding of steel elements in tension | 0.90 | — |
| (i) | Plain concrete elements | 0.60 | — |
| (j) | Anchors in concrete elements | 0.45 to 0.75 in accordance with Chapter 17 | — |

## Table 22.2.2.4.3—Values of $\beta_1$ for equivalent rectangular concrete stress distribution

| $f_c'$, psi | $\beta_1$ | |
|---|---|---|
| $2500 \le f_c' \le 4000$ | 0.85 | (a) |
| $4000 < f_c' < 8000$ | $0.85 - \frac{0.05(f_c' - 4000)}{1000}$ | (b) |
| $f_c' \ge 8000$ | 0.65 | (c) |

## Table 21.2.2—Strength reduction factor $\phi$ for moment, axial force, or combined moment and axial force

| Net tensile stain $\varepsilon_t$ | Classification | $\phi$ | | | |
|---|---|---|---|---|---|
| | | Type of transverse reinforcement | | | |
| | | Spirals conforming to 25.7.3 | | Other | |
| $\varepsilon_t \le \varepsilon_{ty}$ | Compression-controlled | 0.75 | (a) | 0.65 | (b) |
| $\varepsilon_{ty} < \varepsilon_t < \varepsilon_{ty} + 0.003$ | Transition[1] | $0.75 + 0.15\frac{(\varepsilon_t - \varepsilon_{ty})}{(0.003)}$ | (c) | $0.65 + 0.25\frac{(\varepsilon_t - \varepsilon_{ty})}{(0.003)}$ | (d) |
| $\varepsilon_t \ge \varepsilon_{ty} + 0.003$ | Tension-controlled | 0.90 | (e) | 0.90 | (f) |

[1]For sections classified as transition, it shall be permitted to use $\phi$ corresponding to compression-controlled sections.

## ACI 318-19 Design Approach

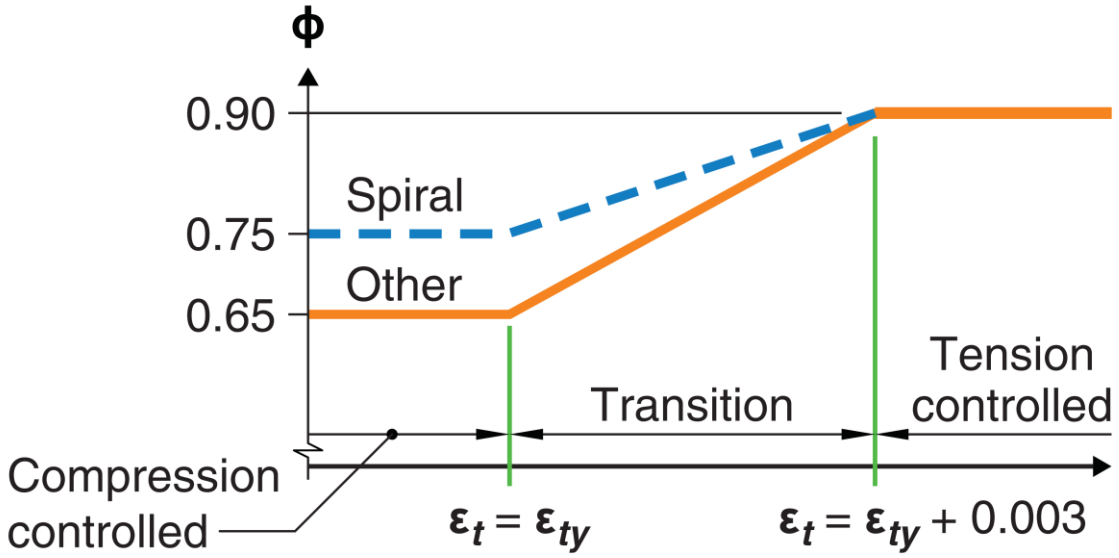



***Fig. R21.2.2b***—*Variation of $\phi$ with net tensile strain in extreme tension reinforcement, $\varepsilon_t$.*

# Reinforcement Limits for Flexure (9.6.1.2 ACI 318-19)

- Minimum Steel
  - For Rectangular Beams and Tee beams with flange in compression

$$A_{s,\min} = \frac{3\sqrt{f_c'}}{f_y} b_w d \qquad \text{not less than} \frac{200 b_w d}{f_y}$$

  - For Tee beams with flange in tension (All values in psi and inches)

$$A_{s,\min} = \frac{6\sqrt{f_c'}}{f_y} b_w d$$

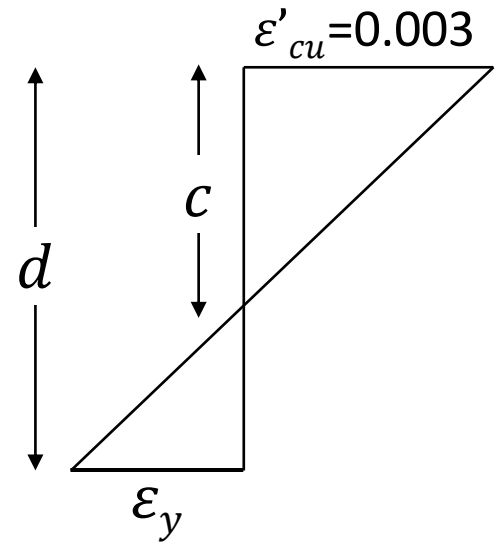
- Maximum Steel

$$\rho_{\max} = \rho' + 0.75\rho_b$$

$$\rho' = \frac{A_{sc}}{bd}$$

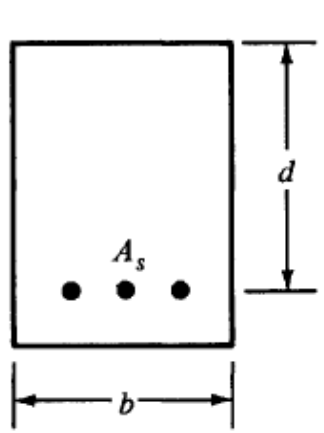
# Balanced Condition and Reinforcement

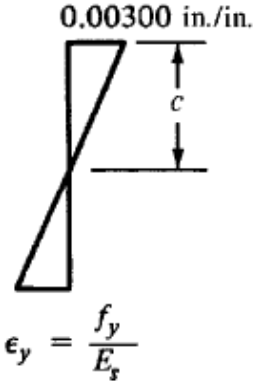
- For balanced condition, the concrete Crushing and yielding of reinforcing bars take place simultaneously.




$$\frac{0.003}{\left(\frac{f_y}{E_s}\right)} = \frac{c}{d-c}$$

$$c = \frac{0.003}{0.003 + \left(\frac{f_y}{E_s}\right)} d$$




$$\frac{c}{d} = \frac{0.00300}{0.00300 + (f_y/E_s)} = \frac{0.00300}{0.003 + (f_y/29 \times 10^6)}$$

$$c = \frac{87{,}000}{87{,}000 + f_y} d$$

$$\rho_b = \left(\frac{0.85\beta_1 f'_c}{f_y}\right)\left(\frac{87{,}000}{87{,}000 + f_y}\right)$$

$x_u$ = Neutral axis for balanced condition

# Maximum & Minimum Steel

- Both limits need to be exercised
- Minimum steel is required for
  - Deep architectural beams: May not require steel reinforcement
  - Slabs and footings for shrinkage and creep
- Maximum steel need to be limited
  - To ensure ductile failure
  - To maintain the economy
  - Warning to the occupants

# Example 4: Analysis of a Singly Reinforced Beam (Ultimate Strength Stage)

Determine $M_n$ the nominal or theoretical ultimate moment strength of the beam section shown in Figure 2.12 if $f_y = 60{,}000$ psi and $f'_c = 3000$ psi.

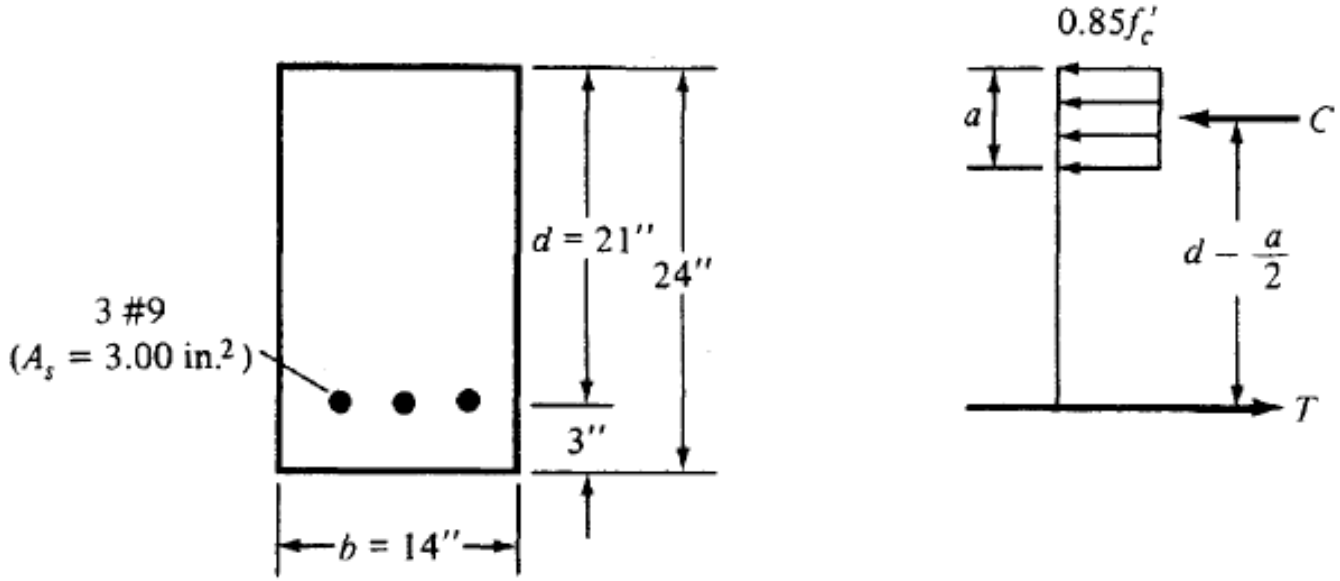



**Figure 2.12**

# Example 4: Analysis of a Singly Reinforced Beam (Ultimate Strength Stage)

***SOLUTION*** Computing Tensile and Compressive Forces $T$ and $C$

$$T = A_s f_y = (3.00)(60) = 180 \text{ k}$$
$$C = 0.85 f'_c ab = (0.85)(3)(a)(14) = 35.7a$$

Equating $T$ and $C$ and Solving for $a$

$$T = C \text{ for equilibrium}$$
$$180 = 35.7a$$
$$a = 5.04 \text{ in.}$$

Computing $d - \frac{a}{2}$ and $M_n$

$$d - \frac{a}{2} = 21 - \frac{5.04}{2} = 18.48 \text{ in.}$$
$$M_n = (180)(18.48) = 3326.4 \text{ in.-k} = \underline{277.2 \text{ ft-k}}$$

# Example 5: Design of a Singly Reinforced Beam

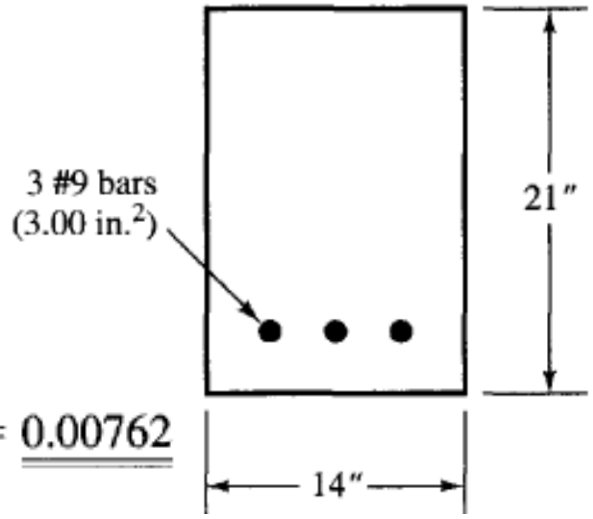
Determine the values of $a$, $c$, and $\epsilon_t$ for the beam shown in Figure 3.4. $f_y = 60{,}000$ psi and $f'_c = 3000$ psi.

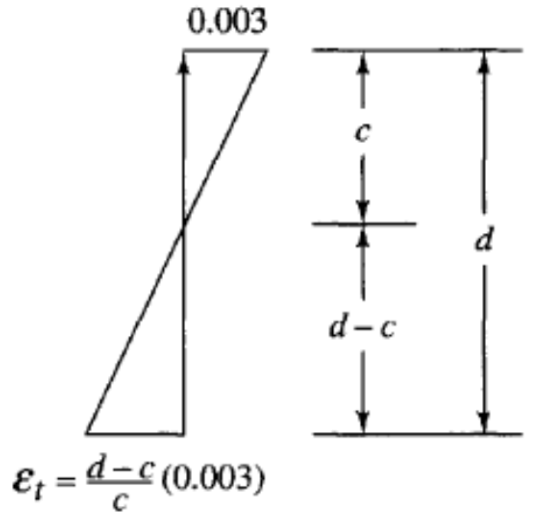
$$a = \frac{A_s f_y}{0.85 f'_c b} = \frac{(3.00)(60)}{(0.85)(3)(14)} = \underline{5.04''}$$

$\beta_1 = 0.85$ for 3000 psi concrete

$$c = \frac{a}{\beta_1} = \frac{5.04}{0.85} = \underline{5.93''}$$

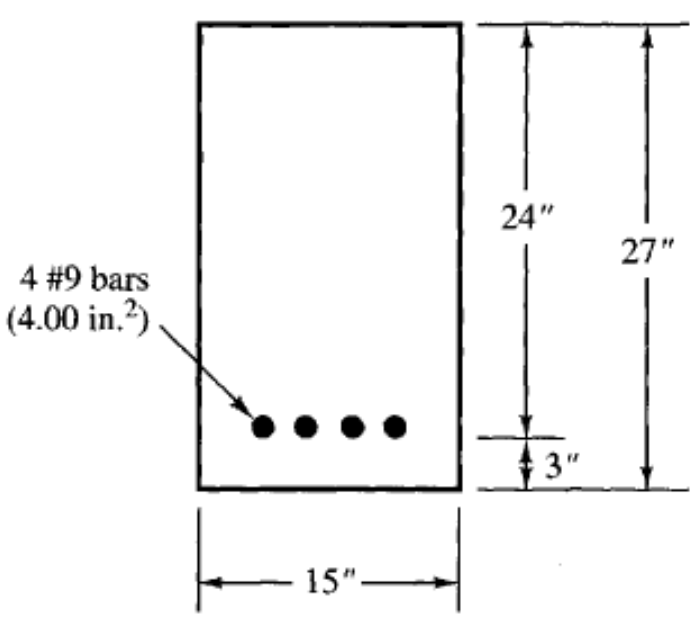
$$\epsilon_t = \frac{d-c}{c}(0.003) = \left(\frac{21 - 5.93}{5.93}\right)(0.003) = \underline{0.00762}$$



# Example 6: Design of a Singly Reinforced Beam

Determine the ACI design moment capacity $\phi M_n$ of the beam shown in Figure 3.7 if $f'_c = 4000$ psi and $f_y = 60{,}000$ psi.



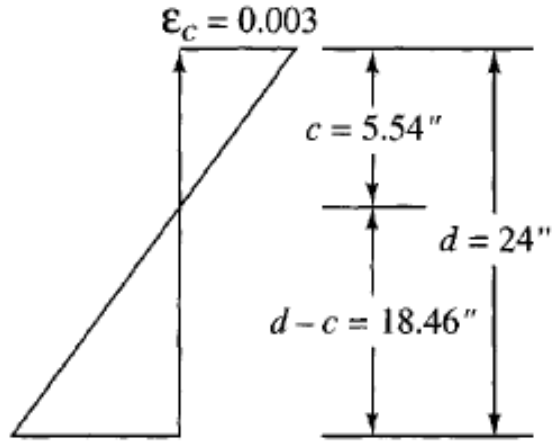




**Figure 3.8**

# Example 5: Design of a Singly Reinforced Beam

**Checking Steel Percentage**

$$\rho = \frac{A_s}{bd} = \frac{4.00}{(15)(24)} = 0.0111$$

$$\left.\begin{array}{l} > \rho_{\min} = 0.0033 \\ < \rho_{\max} = 0.0181 \end{array}\right\} \text{both from Appendix Table A.7}$$

$$a = \frac{A_s f_y}{0.85 f'_c b} = \frac{(4.00)(60{,}000)}{(0.85)(4000)(15)} = 4.71 \text{ in.}$$

$$\beta_1 = 0.85 \text{ for 4000 psi concrete}$$

$$c = \frac{a}{\beta_1} = \frac{4.71}{0.85} = 5.54 \text{ in.}$$

**Drawing Strain Diagram** (Figure 3.8)

$$\epsilon_t = \frac{d - c}{c}(0.003) = \frac{18.46}{5.54}(0.003) = 0.0100$$

$$> 0.005 \therefore \text{tension controlled}$$

$$M_n = A_s f_y \left(d - \frac{a}{2}\right) = (4.00)(60)\left(24 - \frac{4.71}{2}\right)$$

$$= 5194.8 \text{ in. k} = 432.9 \text{ ft-k}$$

$$\phi M_n = (0.9)(432.9) = \underline{389.6 \text{ ft-k}}$$

# Beam Design: Considerations

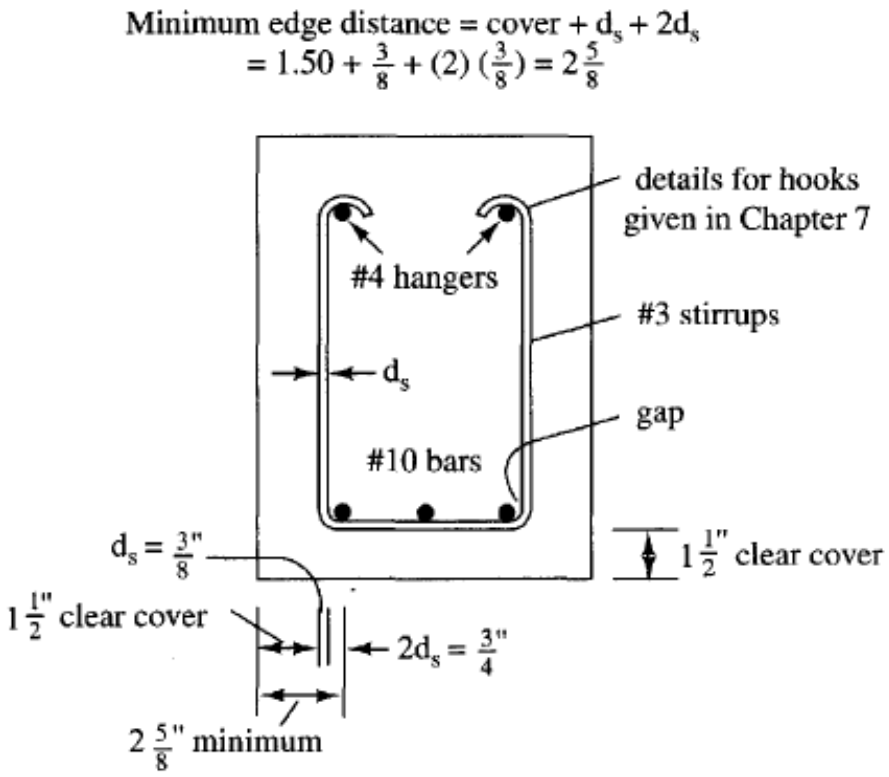
- Beam proportioning
  - For short spans (20 to 25ft), depth is kept to 1.5 to 2 times of width b
  - For larger spans, usually deeper sections be preferably used
  - Size of beams be selected based on available formworks and easy workmanship
- Deflections
  - Care be exercised for safer structures
  - To be proportioned for minimum depths as shown in table 4.1

**Table 4.1** Minimum Thickness of Nonprestressed Beams or One-Way Slabs Unless Deflections Are Computed[1,2]

| | Minimum thickness, $h$ | | | |
|---|---|---|---|---|
| | Simply supported | One end continuous | Both ends continuous | Cantilever |
| Member | Members not supporting or attached to partitions or other construction likely to be damaged by large deflections | | | |
| Solid one-way slabs | $\ell/20$ | $\ell/24$ | $\ell/28$ | $\ell/10$ |
| Beams or ribbed one-way slabs | $\ell/16$ | $\ell/18.5$ | $\ell/21$ | $\ell/8$ |

# Beam Design: Considerations

- Estimated beam weights
- Selection of reinforcing bars
  - Ideally, same bar sizes be used in each layer
  - Care be exercised to select the locally available sizes
- Cover
  - To protect the steel from direct weathering effects
  - To provide the minimum insulation
  - 1.5” is minimum for internal beams from centroid of bars
- Minimum spacing of bars
  - To ensure the passing of concrete and accommodate the aggregates
  - Must be > 1.00”



# Example 7: Design of a Singly Reinforced Rectangular Beam

Design a rectangular beam for a 22-ft simple span if a dead load of 1 k/ft (not including the beam weight) and a live load of 2 k/ft are to be supported. Use $f'_c = 4000$ psi and $f_y = 60{,}000$ psi.

**Estimating Beam Dimensions and Weight**

$$\text{Assume } h = (0.10)(22) = 2.2 \text{ ft } \underline{\text{Say 27 in.}} \ (d = 24.5 \text{ in.})$$

$$\text{Assume } b = \frac{1}{2}h = \frac{24.5}{2} = \underline{\text{Say 12 in.}}$$

$$\text{Beam } wt = \frac{(12)(27)}{144}(150) = 337.5 \text{ lb/ft}$$

**Computing $w_u$ and $M_u$**

$$w_u = (1.2)(1 + 0.3375) + (1.6)(2) = 4.805 \text{ k/ft}$$

$$M_u = \frac{w_u L^2}{8} = \frac{(4.805)(22)^2}{8} = 290.7 \text{ ft-k}$$

## Example 7: Design of a Singly Reinforced Rectangular Beam

Assuming $\phi = 0.90$ and computing $\rho$ with the following expression which was derived in Section 3.4.

$$\rho = \frac{0.85f'_c}{f_y}\left(1 - \sqrt{1 - \frac{2R_n}{0.85f'_c}}\right)$$

$$R_m = \frac{M_u}{\phi bd^2} = \frac{(12)(290{,}700)}{(0.90)(12)(24.5)^2} = 538.11$$

$$\rho = \frac{(0.85)(4000)}{60{,}000}\left(1 - \sqrt{1 - \frac{(2)(538.11)}{(0.85)(4000)}}\right) = 0.00982$$

**Selecting Reinforcing**

$$A_s = \rho bd = (0.00982)(12)(24.5) = 2.89 \text{ in.}^2$$

Use three #9 bars ($A_s = 3.00$ in.$^2$)

# Example 7: Design of a Singly Reinforced Rectangular Beam

**Checking Solution**

$$\rho = \frac{A_s}{bd} = \frac{3.00}{(12)(24.5)} = 0.01020 > \rho_{\min} = 0.0033$$

$< \rho_{\max} = 0.0181$ ($\rho$ values from Appendix Table A.7). $\therefore$ section is ductile and $\phi = 0.90$.

$$a = \frac{A_s f_y}{0.85 f'_c b} = \frac{(3.00)(60)}{(0.85)(4)(12)} = 4.41 \text{ in.}$$

$$\phi M_n = \phi A_s f_y \left(d - \frac{a}{2}\right) = (0.90)(3.00)(60)\left(24.5 - \frac{4.41}{2}\right)$$

$= 3612$ in.-k $= 301$ ft-k $> 290.7$ ft-k OK

**Final Section (Figure 4.2)**

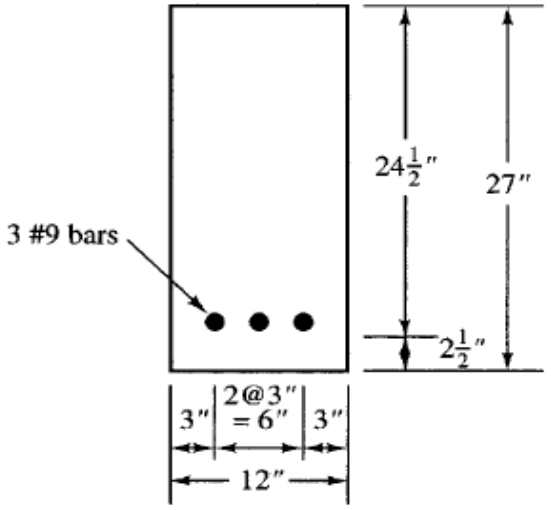
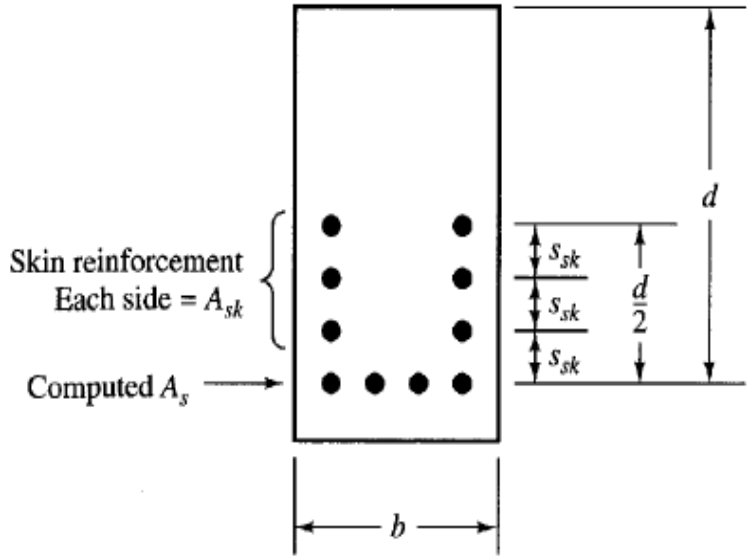



**Figure 4.2**

# Other Important Considerations for Beams

- Lateral Support: Required unless the spacing is more than 50 x b (compression flange or face).
- Skin Reinforcement:
  - For deep beams having web more than 36”
  - It is provided to prevent the cracks due to tension in bottom fibers
  - Must extend at a distance of d/2 from bottom steel
  - Spacing must be less than d/6, 12” or 1000Ab/(d-30)
- Bundled Bars:
  - Done once required steel area doesn’t fit in cross section
  - Not recommended for bar sizes > #11
  - The area should be considered as fictitious area of one bar
  - Cutoff be done with stagger @ 40 x bar diameter (ACI 7.6.6.4)






# One-way Slabs

- The slabs having larger to shorter span > 2 are one way slabs
- Are designed as rectangular slab with strip of 12” width
- Poisson ratio is almost zero?
- The flexural steel be provided as per design but not less than 18” c/c (ACI 7.6.5)
- Thickness be selected as ACI code table for deflection control. They are rounded to nearest ¼” for up to 6” thick and ½” for more than 6” thick slabs
- Minimum shrinkage or temperature reinforcement be also provided
- ACI 7.12 recommends
    - For grade 40 or 50 steel, minimum area be 0.002bh but < 5h or 18”
    - For grade 60, minimum area is 0.0018bh
    - For higher grades, ((0.0018 x 60000)/fy) > 0.0014

# Example 8: Design of One-way Slabs

Design a one-way slab for the inside of a building using the span, loads, and other data given in Figure 4.8.

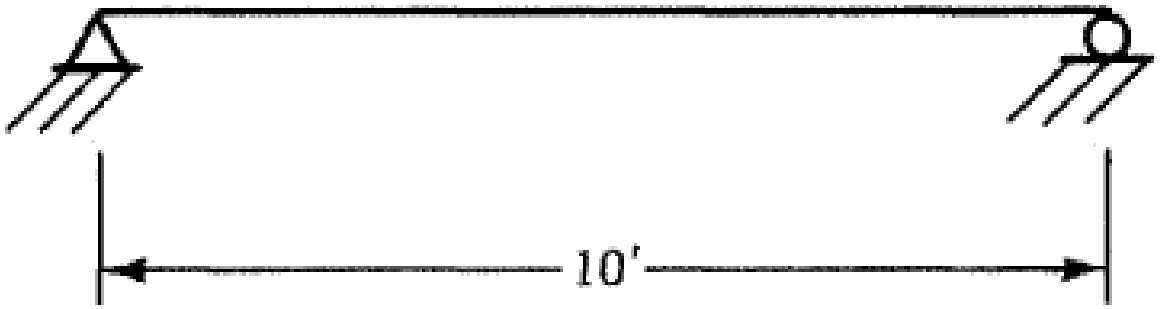



**Figure 4.8**

LL = 200 psf
$f'_c$ = 4000 psi
$f_y$ = 60,000 psi

## Example 8: Design of One-way Slabs

**Minimum Total Slab Thickness $h$ if Deflections Are Not Computed (See Table 4.1)**

$$h = \frac{\ell}{20} = \frac{(12)(10)}{20} = 6''$$

**Assume 6-in. slab (with $d$ = approximately $6'' - \frac{3}{4}''$ Cover $- \frac{1}{4}''$ for estimated half-diameter of bar size = 5.0″).** The moment is calculated, and then the amount of steel required is determined. If this value seems unreasonable, a different thickness is tried.

Design a 12-in.-wide strip of the slab.

$$DL = \text{slab wt} = \left(\frac{6}{12}\right)(150) = 75 \text{ psf}$$

$$LL = 200 \text{ psf}$$

$$w_u = (1.2)(75) + (1.6)(200) = 410 \text{ psf}$$

$$M_u = \frac{(0.410)(10)^2}{8} = 5.125 \text{ ft-k}$$

$$\frac{M_u}{\phi b d^2} = \frac{(12)(5125)}{(0.9)(12)(5.00)^2} = 227.8 \text{ psi}$$

$$\rho = 0.00393 \text{ (from Appendix Table A.13)}$$

$$> \rho_{\min} = 0.0033$$

$$A_s = (0.00393)(12)(5.0) = 0.236 \text{ in.}^2/\text{ft}$$

Use #4 @ 10″ from Appendix Table A.6 ($A_s = 0.24$ in$^2$/ft)

spacing < maximum of 18 in. as per ACI 7.6.5

**Transverse Direction—Shrinkage and Temperature Steel**

$$A_s = 0.0018bh = (0.0018)(12)(6) = 0.1296 \text{ in.}^2/\text{ft}$$

Use #3 @ 10″(0.13 in.$^2$/ft) as selected from Appendix Table A.6

Spacing < maximum of 18 in. as per ACI 7.12.2.2 OK

# T Beams

- For RC construction, beams are usually T beams since they are casted as monolithically with slab
- Some portion of the slab act as flange to the beam
- Effective flange width is an approximation to make the distribution of compression stresses uniform. ACI 8.10.4 recommends effective flange width as smaller of
    - 1/4th of beam span
    - Overhang width < 8 times slab thickness
    - ½ clear distance to the web of consecutive beams
- Neutral axis may fall inside the flange

# Example 9: Design of T Beams

Design a T beam for the floor system shown in Figure 5.9 for which $b_w$ and $d$ are given. $M_D = 80$ ft-k, $M_L = 100$ ft-k, $f'_c = 4000$ psi, $f_y = 60{,}000$ psi, and simple span = 20 ft.

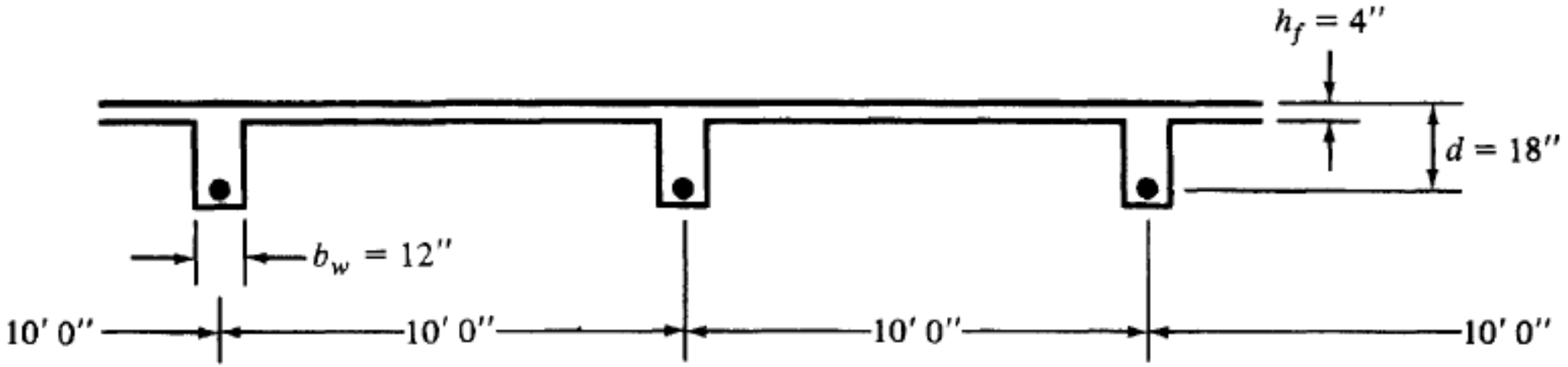



**Figure 5.9**

# Example 9: Design of T Beams

**Effective Flange Width**

**(a)** $\frac{1}{4} \times 20 = 5'0'' = \underline{60''}$

**(b)** $12 + (2)(8)(4) = 76''$

**(c)** $10'0'' = 120''$

**Computing Moments Assuming $\phi = 0.90$**

$$M_u = (1.2)(80) + (1.6)(100) = 256 \text{ ft-k}$$

$$M_n = \frac{M_u}{\phi} = \frac{256}{0.90} = 284.4 \text{ ft-k}$$

**Assuming a Lever Arm $z$ Equal to the Larger of $0.9d$ or $d - (h_f/2)$**

$$z = (0.9)(18) = \underline{\underline{16.20 \text{ in.}}}$$

$$z = 18 - \frac{4}{2} = 16.00 \text{ in.}$$

**Trial Steel Area**

$$A_s f_y z = M_n$$

$$A_s = \frac{(12)(284.4)}{(60)(16.20)} = 3.51 \text{ in.}^2$$

# Example 9: Design of T Beams

**Computing Values of $a$ and $z$**

$$0.85f'_cA_c = A_sf_y$$
$$(0.85)(4)(A_c) = (3.51)(60)$$
$$A_c = 61.9 \text{ in.}^2 < (4)(60) = 240 \text{ in.}^2 \therefore \text{N.A. in flange}$$
$$a = \frac{61.9}{60} = 1.03 \text{ in.}$$

Therefore, the neutral axis is in the flange.

$$z = 18 - \frac{1.03}{2} = 17.48 \text{ in.}$$

**Calculating $A_s$ with this Revised $z$**

$$A_s = \frac{(12)(284.4)}{(60)(17.48)} = 3.25 \text{ in.}^2$$

**Computing Values of $a$ and $z$**

$$A_c = \frac{(3.25)(60)}{(0.85)(4)} = 57.4 \text{ in.}^2$$
$$a = \frac{57.4}{60} = 0.96 \text{ in.}$$
$$z = 18 - \frac{0.96}{2} = 17.52 \text{ in.}$$

**Calculating $A_s$ with this Revised $z$**

$$A_s = \frac{(12)(284.4)}{(60)(17.52)} = 3.25 \text{ in.}^2 \quad \text{OK, close enough to previous value}$$

# Example 9: Design of T Beams

**Checking Minimum Reinforcing**

$$A_{s\,\min} = \frac{3\sqrt{f'_c}}{f_y} b_w d = \frac{3\sqrt{4000}}{60{,}000}(12)(18) = 0.68 \text{ in.}^2$$

but not less than

$$A_{s\,\min} = \frac{200 b_w d}{f_y} = \frac{(200)(12)(18)}{60{,}000} = 0.72 \text{ in.}^2 < 3.25 \text{ in.}^2 \quad \underline{\text{OK}}$$

or $\rho_{\min}$ from Appendix Table A.7 = 0.0033

$$A_{s\,\min} = (0.0033)(12)(18) = 0.71 \text{ in.}^2 < 3.25 \text{ in.}^2 \quad \underline{\text{OK}}$$

**Computing $c$, $\epsilon_t$, and $\phi$**

$$c = \frac{a}{\beta_1} = \frac{0.96}{0.85} = 1.13 \text{ in.}$$

$$\epsilon_t = \left(\frac{d-c}{c}\right)(0.003) = \left(\frac{18 - 1.13}{1.13}\right)(0.003)$$

$$= 0.045 > 0.005 \therefore \phi = 0.90 \text{ as assumed}$$

$$\underline{A_s \text{ reqd} = 3.25 \text{ in.}^2}$$

# Example 10: Design of T Beams

Design a T beam for the floor system shown in Figure 5.10, for which $b_w$ and $d$ are given. $M_D = 200$ ft-k, $M_L = 425$ ft-k, $f'_c = 3000$ psi, $f_y = 60{,}000$ psi, and simple span = 18 ft.

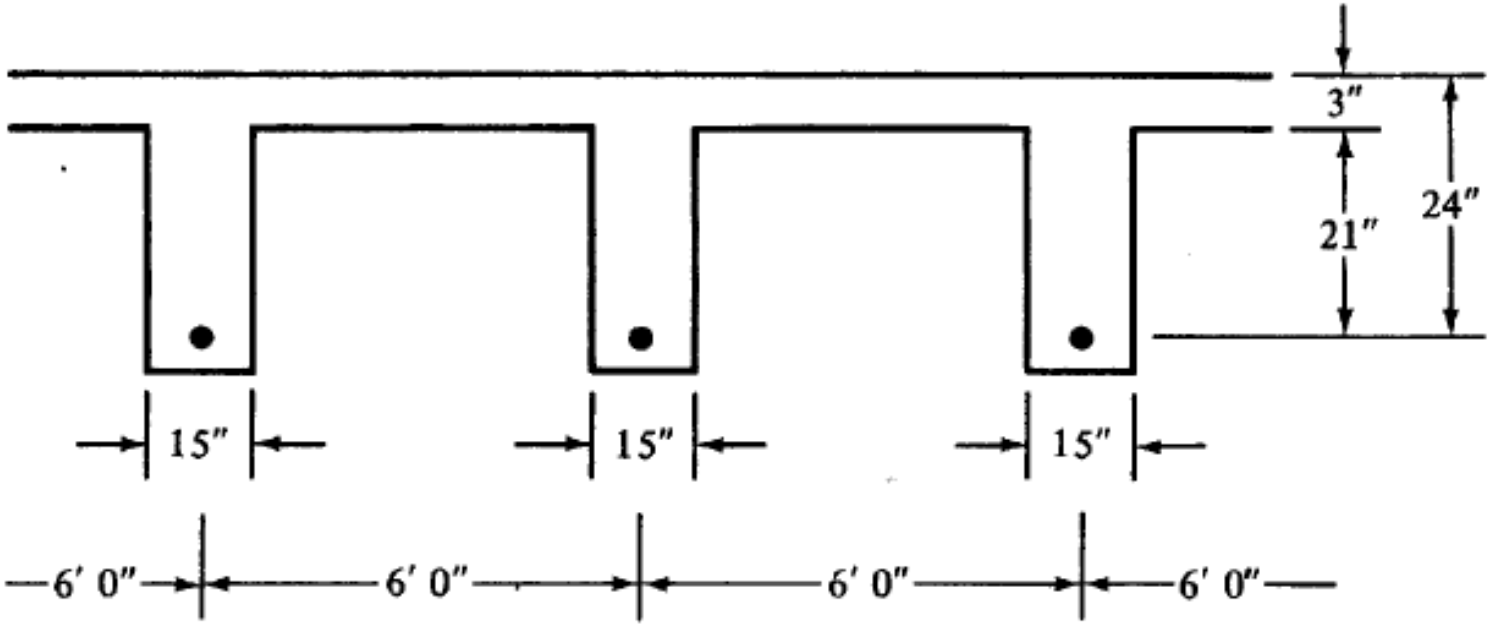



**Figure 5.10**

# Example 10: Design of T Beams

**Effective Flange Width**

**(a)** $\frac{1}{4} \times 18' = 4'6'' = \underline{54''}$

**(b)** $15'' + (2)(8)(3) = 63''$

**(c)** $6'0'' = 72''$

**Moments assuming $\phi = 0.90$**

$$M_u = (1.2)(200) + (1.6)(425) = 920 \text{ ft-k}$$

$$M_n = \frac{M_u}{0.90} = \frac{920}{0.90} = 1022 \text{ ft-k}$$

**Assuming a Lever Arm $z$** (Note that the compression area in the slab is very wide, and thus its required depth is very small.)

$$z = (0.90)(24) = 21.6''$$

$$z = 24 - \tfrac{3}{2} = \underline{22.5''}$$

**Trial Steel Area**

$$A_s = \frac{(12)(1022)}{(60)(22.5)} = 9.08 \text{ in.}^2$$

# Example 10: Design of T Beams

**Checking Values of $a$ and $z$**

$$A_c = \frac{(60)(9.08)}{(0.85)(3)} = 213.6 \text{ in.}^2$$

The stress block extends down into the web, as shown in Figure 5.11.

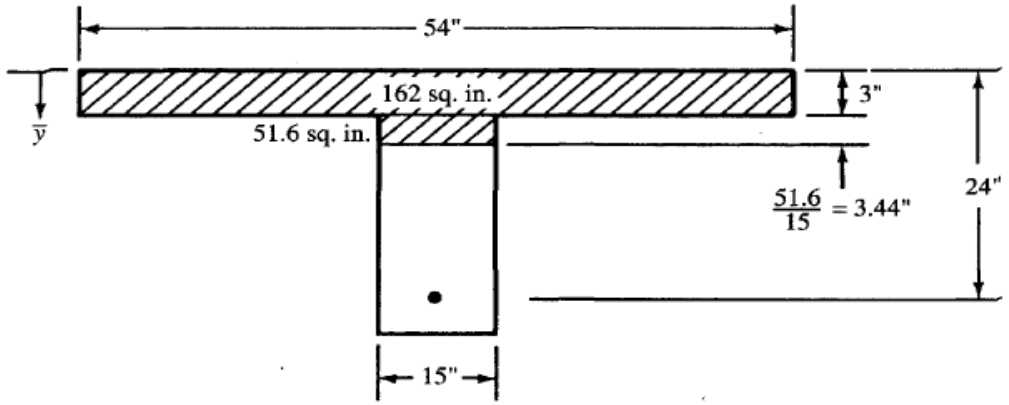
**Computing the Distance $\bar{y}$ from the Top of the Flange to the Center of Gravity of $A_c$**

$$\bar{y} = \frac{(162)(1.5) + (51.6)\left(3 + \frac{3.44}{2}\right)}{213.6} = 2.28 \text{ in.}$$

$$z = 24 - 2.28 = 21.72 \text{ in.}$$

$$A_s = \frac{(12)(1022)}{(60)(21.72)} = 9.41 \text{ in.}^2$$

The steel area required (9.41 in.$^2$) could be refined a little by repeating the design, but space is not used to do this. (If this is done $A_s = 9.51$ in.$^2$)



# Example 10: Design of T Beams

**Checking Minimum Reinforcing**

$$\rho_{min} \text{ from Appendix Table A.7} = 0.00333$$

or $$A_{s\,min} = (0.00333)(15)(24) = 1.20 \text{ in.}^2 < 9.58 \text{ in.}^2 \quad \underline{\text{OK}}$$

**Checking values of $\epsilon_t$ and $\phi$**

$$a = 3 + 3.44 = 6.44 \text{ in.}$$

$$c = \frac{a}{\beta_1} = \frac{6.44}{0.85} = 7.58 \text{ in.}$$

$$\epsilon_t = \left(\frac{d-c}{c}\right)(0.003) = \left(\frac{24 - 7.58}{7.58}\right)(0.003)$$

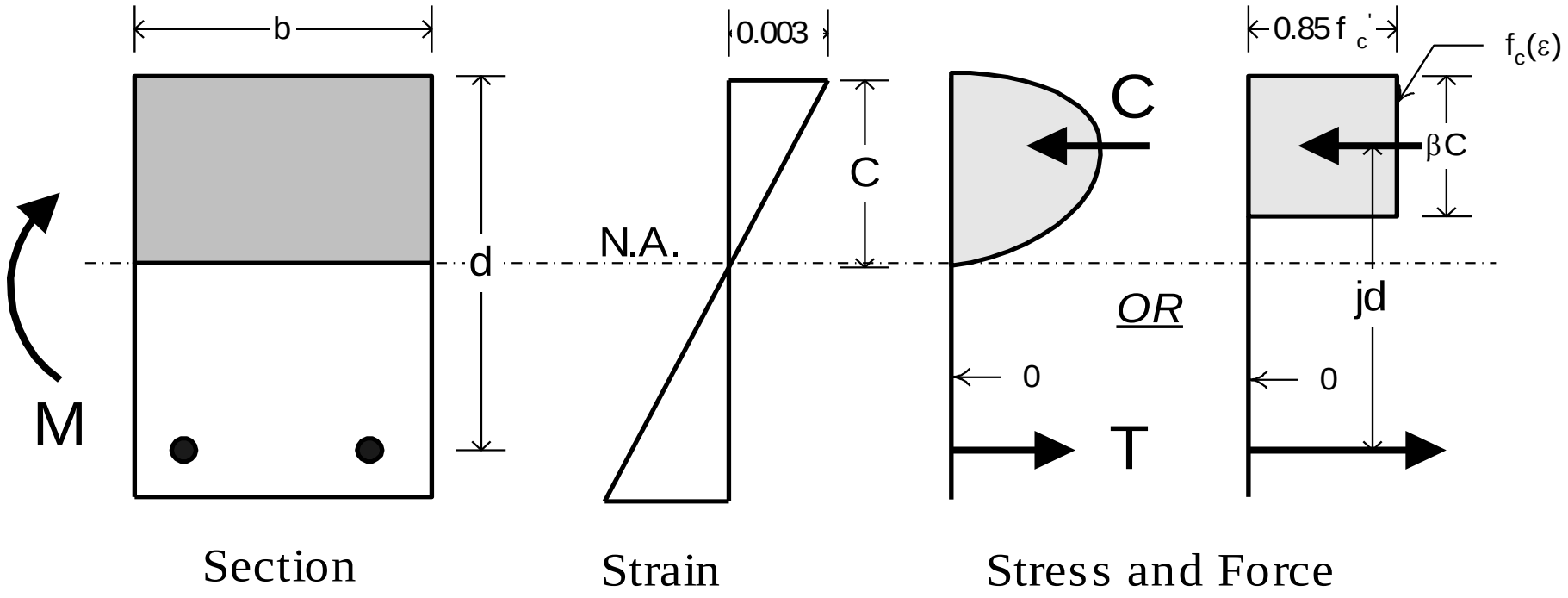
$$= 0.00650 > 0.005 \therefore \phi = 0.90 \text{ as assumed}$$

# Compression Steel in Beams – Doubly Reinforced

- Steel reinforcement in compression zone of RC beams is referred as compression steel
- Since steel is present at bottom and top fibers hence beam is called doubly reinforced
- Though for USD, addition of compression steel is not required but it is since:
  - Stress reversals
  - Effects of creep and shrinkage
  - Positioning the stirrups
  - Unless confined, they give better bending resistance

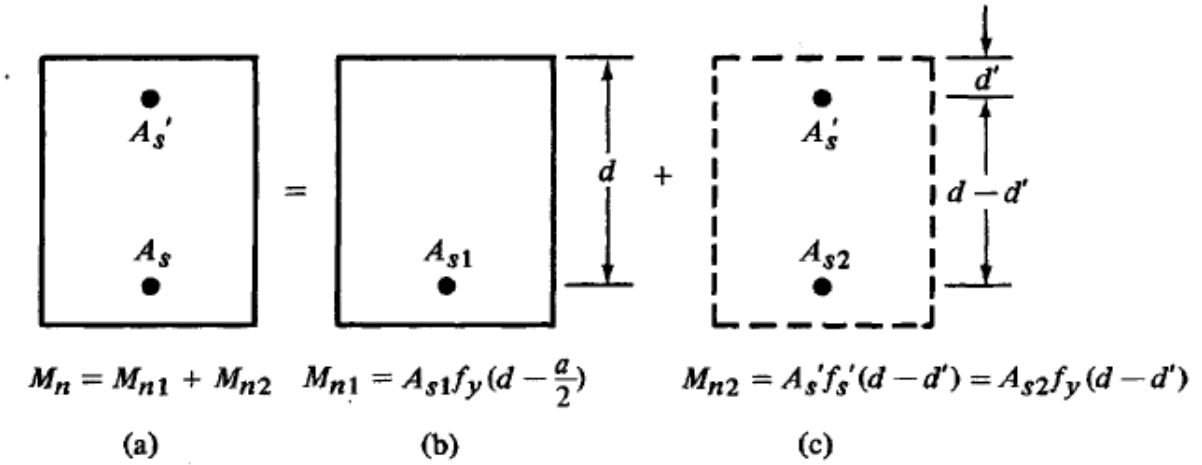


# $A_{st}$ and $A_{sc}$ for Doubly Reinforced Beams



# Compression Steel in Beams – Doubly Reinforced

- Doubly reinforced beams have two sets of resisting moments
  - Concrete compression zone vs. balancing tensile steel
  - Compression steel vs. additional tensile steel
- It has no good function unless the balancing additional tensile steel is present to counter the compression steel
- It is assumed that compression steel has also yielded




$$M_{n1} = A_{s1} f_y \left( d - \frac{a}{2} \right) \qquad M_{n2} = A_s' f_y (d - d')$$

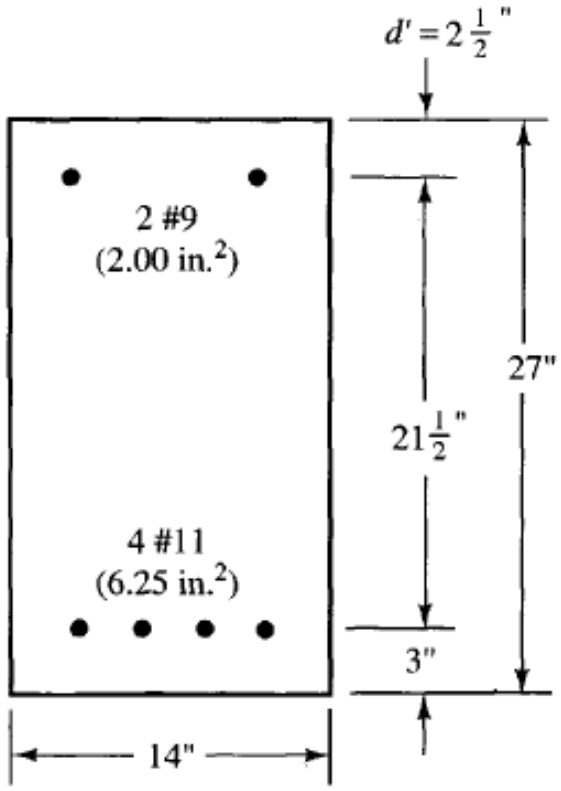
$$M_n = A_{s1} f_y \left( d - \frac{a}{2} \right) + A_{s2} f_y (d - d')$$

$$\phi M_n = \phi \left[ A_{s1} f_y \left( d - \frac{a}{2} \right) + A_{s2} f_y (d - d') \right]$$

$$A_s f_y = 0.85 f_c' \beta_1 c b + A_s' \left( \frac{c - d'}{c} \right) (0.003)(29{,}000)$$

# Example 11: Analysis of Doubly Reinforced Beams

Determine the design moment capacity of the beam shown in Figure 5.14 for which $f_y = 60{,}000$ psi and $f'_c = 3000$ psi.



# Example 11: Analysis of Doubly Reinforced Beams

**Writing the Quadratic Equation for Equilibrium**

$$(6.25)(60) = (0.85)(3)(14)(0.85c) + (2.00)\left(\frac{c - 2.5}{c}\right)(0.003)(29{,}000)$$

**Multiplying Through by $c$ and Simplifying**

$$c^2 - 6.624c = 14.335$$

**Solving Equation**

$$c = 8.34 \text{ in.}$$

$$a = (0.85)(8.34) = 7.09 \text{ in.}$$

**Computing Strains in Compression and Tensile Steel**

$$\epsilon_s' = \left(\frac{8.34 - 2.50}{8.34}\right)(0.003) = 0.00210$$

$$> \frac{60}{29{,}000} = 0.00207 \therefore f_s' = 60 \text{ ksi}$$

# Example 11: Analysis of Doubly Reinforced Beams

$$A_{s2} f_y = A_s' f_s'$$

$$A_{s2} = \frac{A_s' f_y}{f_s'} = \frac{(2.00)(60)}{60} = 2.00 \text{ in.}^2$$

$$A_{s1} = A_s - A_{s2} = 6.25 - 2.00 = 4.25 \text{ in.}^2$$

$$\epsilon_t = \left(\frac{24 - 8.34}{8.34}\right)(0.003) = 0.00563 > 0.005$$

$$\therefore \phi = 0.90$$

**Then the Design Moment Strength Is**

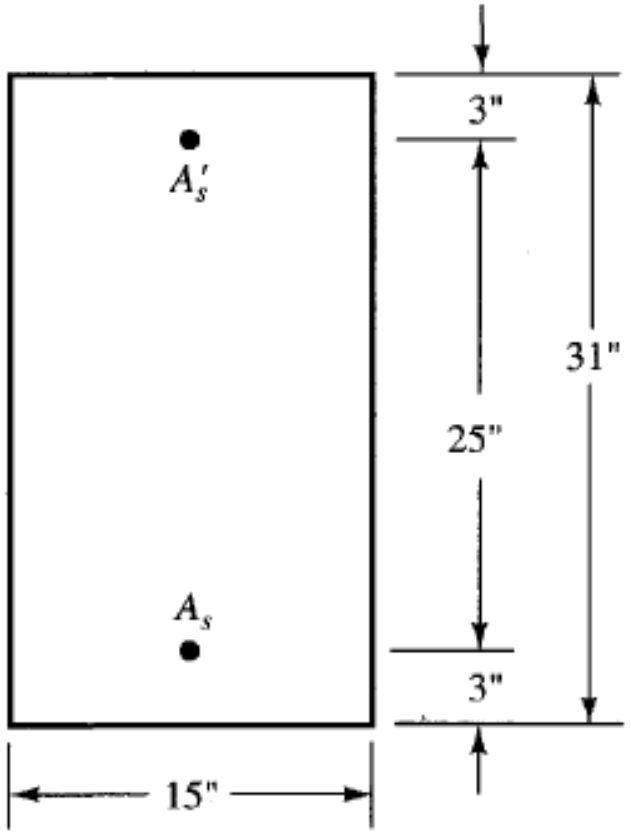
$$\phi M_n = \phi\left[A_{s1} f_y\left(d - \frac{a}{2}\right) + A_s' f_s'(d - d')\right]$$

$$= 0.90\left[(4.25)(60)\left(24 - \frac{7.09}{2}\right) + (2.00)(60)(21.5)\right]$$

$$= 7016 \text{ in.-k} = 584.7 \text{ ft-k}$$

# Example 12: Analysis of Doubly Reinforced Beams

Design a rectangular beam for $M_D = 325$ ft-k and $M_L = 400$ ft-k if $f'_c = 4000$ psi and $f_y = 60{,}000$ psi. The maximum permissible beam dimensions are shown in Figure 5.16.



# Example 12: Analysis of Doubly Reinforced Beams

$$M_u = (1.2)(325) + (1.6)(400) = 1030 \text{ ft-k}$$

**Assuming $\phi = 0.90$**

$$M_n = \frac{M_u}{\phi} = \frac{1030}{0.90} = 1144.4 \text{ ft-k}$$

Assuming maximum possible tensile steel with no compression steel and computing beams nominal moment strength

$$\rho_{max} \text{ from Appendix Table A.7} = 0.0181$$

$$A_{s1} = (0.0181)(15)(28) = 7.60 \text{ in.}^2$$

$$\text{For } \rho = 0.0181 \frac{M_u}{\phi b d^2} \text{ from Appendix Table A.13} = 912.0$$

$$M_{u1} = (912.0)(0.9)(15)(28)^2 = 9{,}652{,}608 \text{ in.-lb}$$

$$= 804.4 \text{ ft-k}$$

$$M_{n1} = \frac{804.4}{0.90} = 893.8 \text{ ft-k}$$

$$M_{n2} = M_n - M_{n1} = 1144.4 - 893.8 \text{ ft-k} = 250.6 \text{ ft-k}$$

# Example 12: Analysis of Doubly Reinforced Beams

**Checking to See Whether Compression Steel Has Yielded**

$$a = \frac{(7.60)(60)}{(0.85)(4)(15)} = 8.94 \text{ in.}$$

$$c = \frac{8.94}{0.85} = 10.52 \text{ in.}$$

$$\epsilon_s' = \left(\frac{10.52 - 3}{10.52}\right)(0.00300) = 0.00214 > 0.00207$$

Therefore, compression steel has yielded.

$$\text{Theoretical } A_s' \text{ required} = \frac{M_{n2}}{(f_y)(d - d')} = \frac{(12)(250.6)}{(60)(28 - 3)} = 2.00 \text{ in.}^2$$

$$A_s' f_s' = A_{s2} f_y$$

$$A_{s2} = \frac{A_s' f_s'}{f_y} = \frac{(2.00)(60)}{60} = \underline{\underline{2.00 \text{ in.}^2}} \quad \underline{\text{Try}} \quad 2 \#9 \ (2.00 \text{ in.}^2)$$

$$A_s = A_{s1} + A_{s2}$$

$$A_s = 7.60 + 2.00 = 9.60 \text{ in.}^2 \quad \underline{\text{Try}} \quad 8 \#10 \ (10.12 \text{ in.}^2)$$

# Example 12: Analysis of Doubly Reinforced Beams

If we had been able to select bars with exactly the same areas as calculated here, $\epsilon_t$ would have remained $= 0.005$ and $\phi = 0.90$, but such was not the case.

Writing the quadratic equation $c$ is found to equal 11.08 in. and $a = 9.43$ in.

$$\epsilon_s' = \left(\frac{11.80 - 3}{11.80}\right)(0.003) = 0.00224 > 0.00207 \qquad \underline{\text{OK}}$$

$$\epsilon_t = \left(\frac{28 - 11.80}{11.80}\right)(0.003) = 0.00412 < 0.005$$

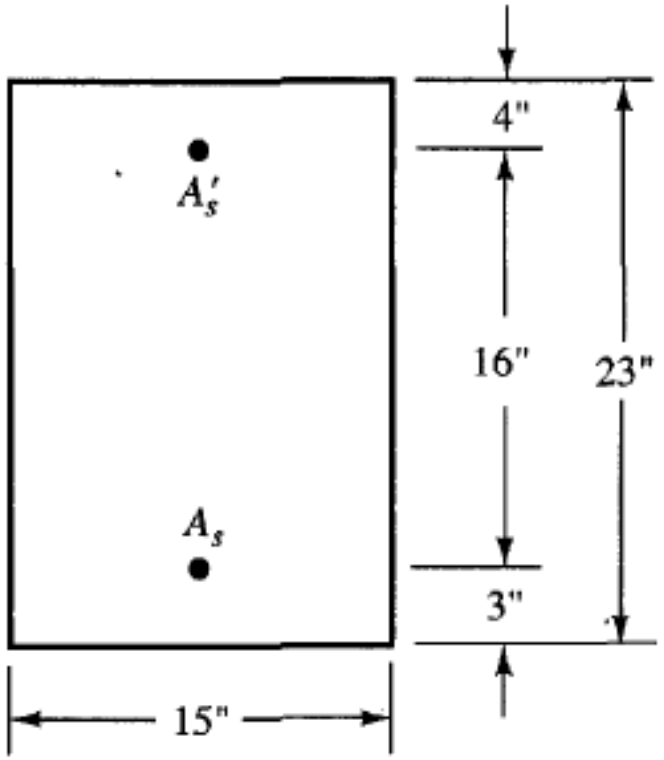
$$\phi = 0.65 + (0.00412 - 0.002)\left(\frac{250}{3}\right) = 0.0083$$

$$\phi M_n = 0.83\left[(10.12 - 2.00)(60)\left(28 - \frac{9.43}{2}\right) + (2.00)(60)(25)\right]$$

$$= 11{,}906 \text{ in. lbs} = 992 \text{ ft-k} < 1030 \text{ ft-k} \qquad \underline{\text{N.G.}}$$

# Example 13: Analysis of Doubly Reinforced Beams

A beam is limited to the dimensions shown in Figure 5.17. If $M_D = 170$ ft-k, $M_L = 225$ ft-k, $f'_c = 4000$ psi, and $f_y = 60{,}000$ psi, select the reinforcing required.



# Example 13: Analysis of Doubly Reinforced Beams

$$M_u = (1.2)(170) + (1.6)(225) = 564 \text{ ft-k}$$

**Assuming $\phi = 0.90$**

$$M_n = \frac{564}{0.90} = 626.7 \text{ ft-k}$$

$$\text{Max } A_{s1} = (0.0181)(15)(20) = 5.43 \text{ in.}^2$$

$$\text{For } \rho = 0.0181 \frac{M_u}{\phi b d^2} = 912.0 \text{ from Appendix Table A.13}$$

$$M_{u1} = (912)(0.90)(15)(20)^2 = 4{,}924{,}800 \text{ in.-lb} = 410.4 \text{ ft-k}$$

$$M_{n1} = \frac{410.4}{0.90} = 456.0 \text{ ft-k}$$

$$M_{n2} = 626.7 - 456.0 = 170.7 \text{ ft-k}$$

# Example 13: Analysis of Doubly Reinforced Beams

**Checking to See if Compression Steel Has Yielded**

$$a = \frac{A_{s1} f_y}{0.85 f'_c b} = \frac{(5.43)(60)}{(0.85)(4)(15)} = 6.39 \text{ in.}$$

$$c = \frac{6.39}{0.85} = 7.52 \text{ in.}$$

$$\epsilon'_s = \left(\frac{7.52 - 4.00}{7.52}\right)(0.003) = 0.00140 < \frac{60}{29{,}000} = 0.00207$$

$$\therefore f'_s = (0.00140)(29{,}000) = 40.6 \text{ ksi}$$

$$\text{Theoretical } A'_s \text{ reqd} = \frac{M_{n2}}{f'_s (d - d')}$$

$$= \frac{(12)(170.7)}{(40.6)(20 - 4)} = \underline{3.15 \text{ in.}^2} \quad \underline{\text{Try 4 \#8 (3.14 in.}^2\text{)}}$$

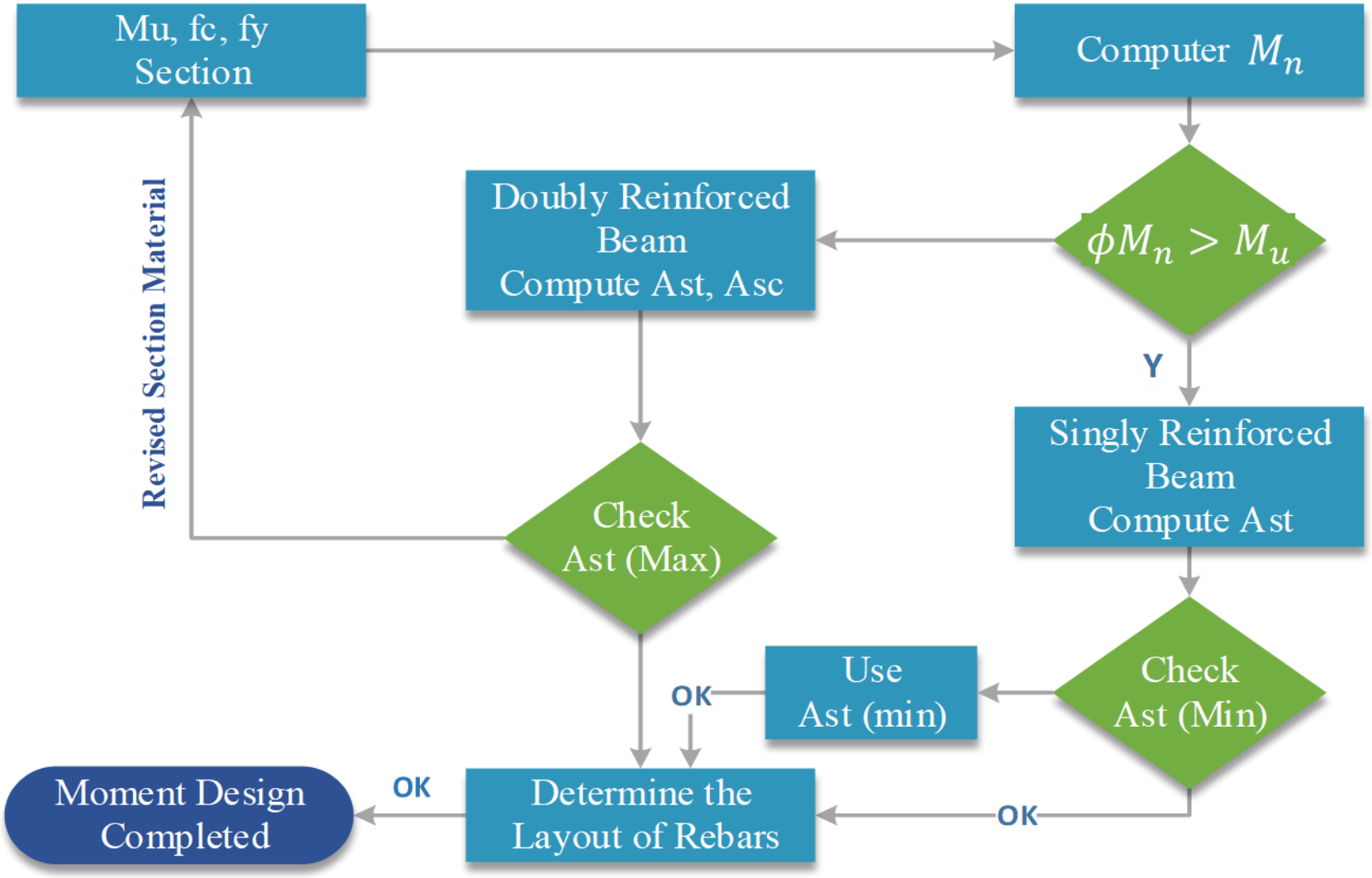
$$A'_s f'_s = A_{s2} f_y$$

$$A_{s2} = \frac{(3.15)(40.6)}{60} = 2.13 \text{ in.}^2$$

$$A_s = A_{s1} + A_{s2} = 5.43 + 2.13 = \underline{7.56 \text{ in.}^2} \quad \underline{\text{Try 6 \#10 (7.59 in.}^2\text{)}}$$

Subsequent checks for $c$, $\epsilon_t$, and so on, prove this design to be satisfactory.

# Summary – Design for Bending Moment



# A Unified Theory for Reinforced Concrete Design

# The Diversity of Concrete Cross-section Design Problem

| Types of Materials and Their Combinations | Types of Actions | Location of Reinforcement | Stress–Strain Curve | Cross-Section Shape | Design Approach and Method |
|---|---|---|---|---|---|
| Un-reinforced concrete | Uniaxial bending, $M_x$ or $M_y$ only | Singly reinforced | Simplified rectangular block | Rectangular | Allowable stress design (*ASD*) |
| Reinforced concrete | Uniaxial bending and axial force, $M_x$ or $M_y$ and $P$ | Doubly reinforced | Semiparabolic and full parabolic curves | Circular | Ultimate strength design (*USD*) |
| Partially prestressed concrete | Biaxial bending, $M_x$ and $M_y$ | Arbitrarily reinforced | Curves for confined and unconfined concrete | Flanged | Load resistance factor design (*LRFD*) |
| Fully prestressed concrete | Biaxial bending and axial force, $M_x$ and $M_y$ and $P$ | | Linear elastic | General polygonal | Performance-based design (*PBD*) |
| Fiber reinforced concrete | | | Bilinear elasto-plastic | Multicell sections | |
| Steel–concrete composite | | | Elastic, post elastic, strain hardening | | |

# Unified Theory for Concrete Design

- It is possible to develop a single theory for determining the axial flexural stress resultants of most types of concrete members for all design methods and for most design codes

  - Unifying Beams and Columns
  - Unifying Reinforced and Pre-stressed Concrete
  - Unifying WSD and USD Methods
  - Unifying different Cross-section Types
  - Incorporating various stress-strain models

# Unifying Beams and Columns

| | Actions | Sections |
|---|---|---|
| Beam | $M_x$ or $M_y$ | Rectangular, T, L, Box |
| Column | P, $M_x$ and/or $M_y$ | Circular, Polygonal, General Shape |

# Unifying Reinforced and Pre-stressed

| | Reinforced Steel | Pre-stressing Steel |
|---|---|---|
| Un-reinforced | No | No |
| Reinforced | Yes | No |
| Partially Pre-stressed | Yes | Yes |
| Fully Pre-stressed | No | Yes |

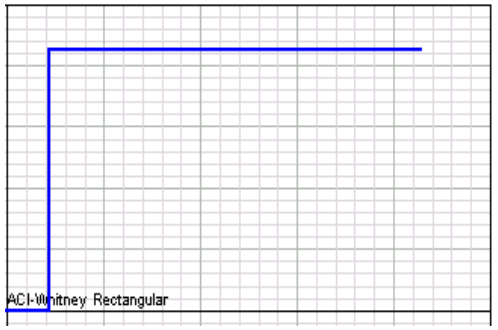
# Unifying Reinforced and Composite

| | Reinforced Steel | Pre-stressing Steel | Steel Section |
|---|---|---|---|
| Reinforced | Yes | No | No |
| Reinforced-Composite | Yes | No | Yes |
| Partially Pre-stressed - Composite | Yes | Yes | Yes |
| Fully Pre-stressed - Composite | No | Yes | Yes |

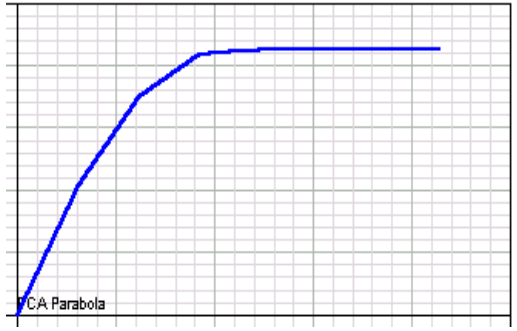
# Unifying Material Models



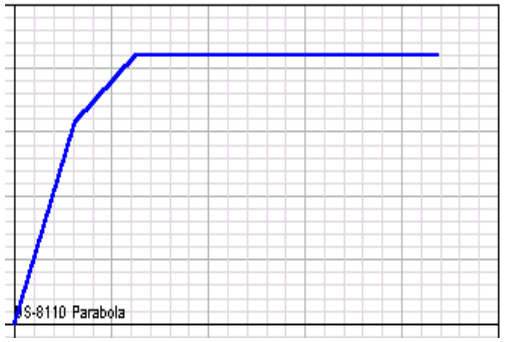

***Linear***



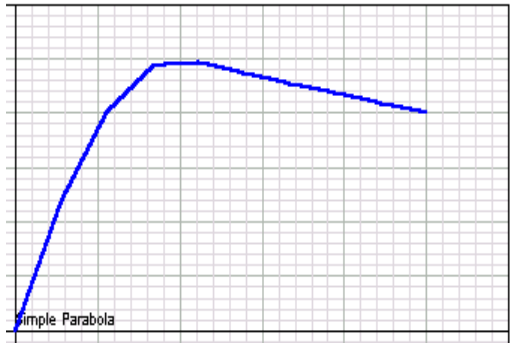

***Whitney***



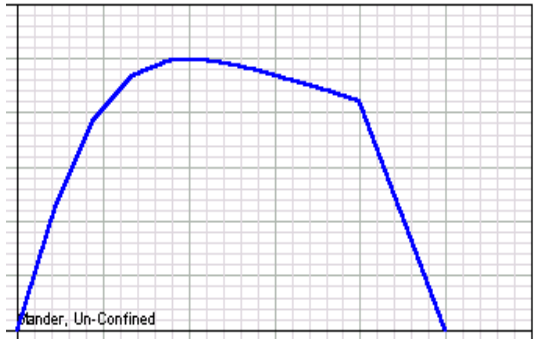

***PCA Parabola***



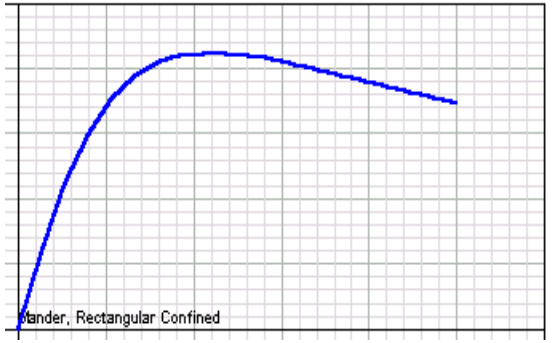

***BS-8110 Parabola***



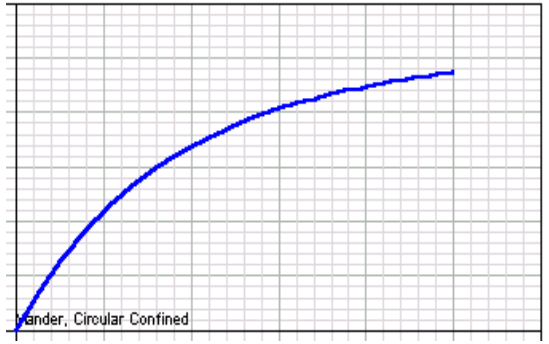

***Simple Parabola***




***Mander (Unconfined)***

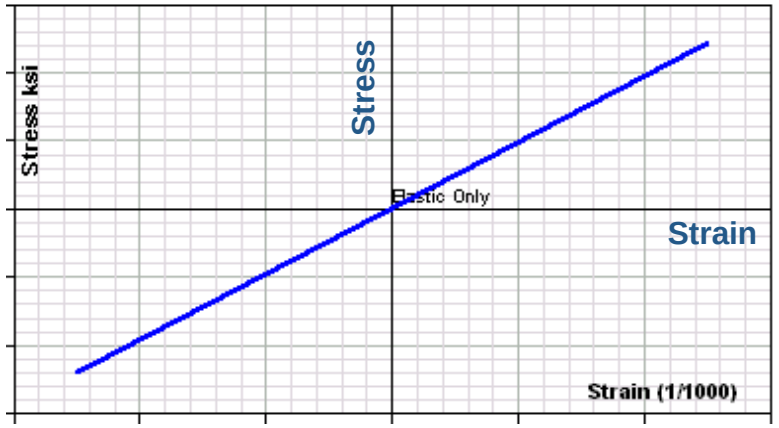



***Mander (Rectangular Confined)***

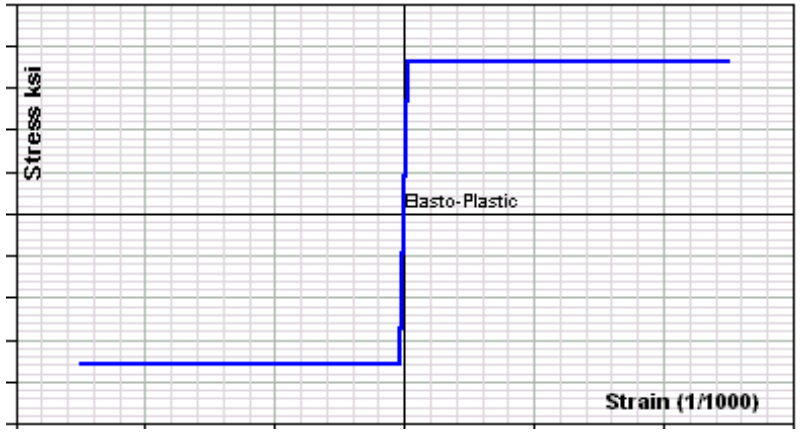



***Mander (Circular Confined)***

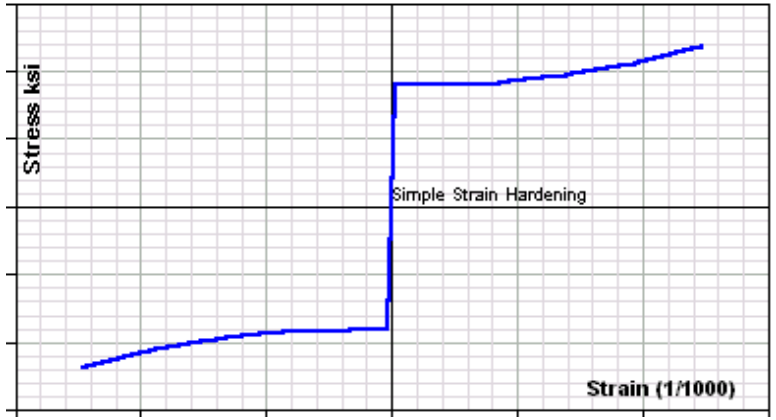
# Unifying Material Models



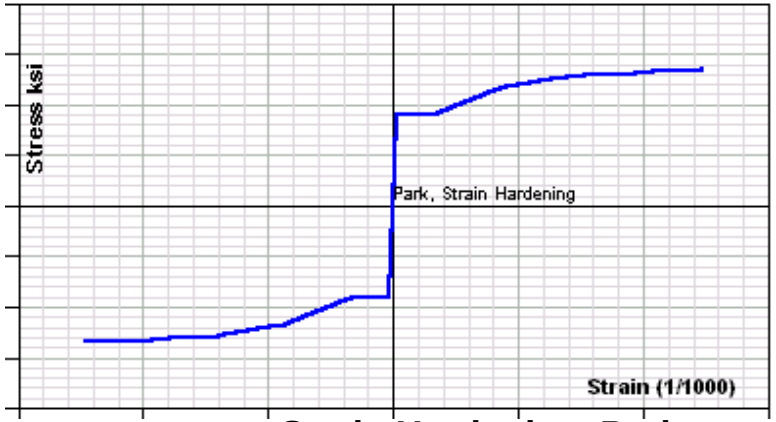

Linear - Elastic




Elasto-Plastic




Strain Hardening - Simple




Strain Hardening Park

## Unification of Cross-sections

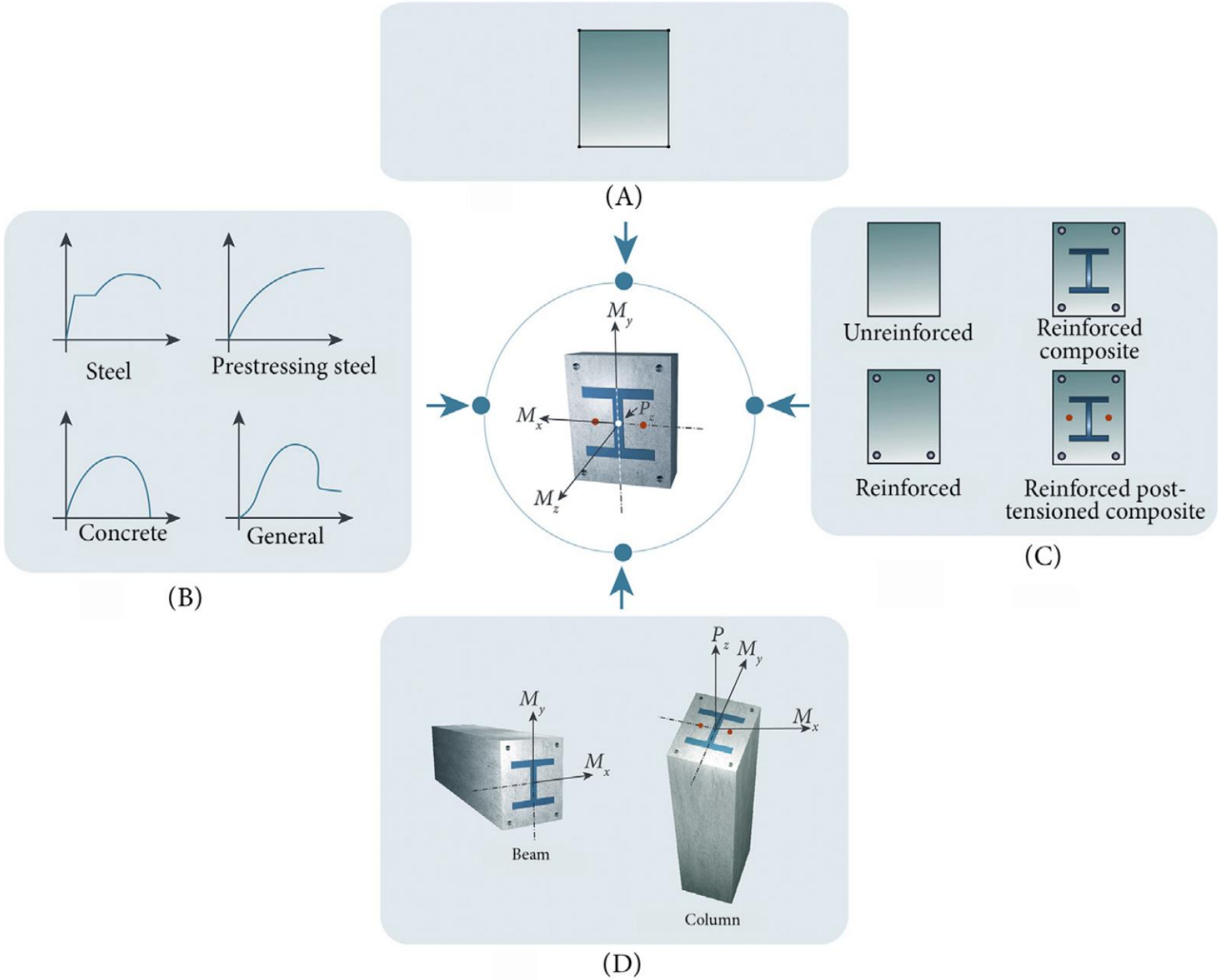



**Figure 3.15** A general partially prestressed, composite beam-column cross-section, subjected to bending about arbitrary axis $m-n$. (A) General polygon shape (B) Unified materials (C) Unified composition (D) Unified design type.

# A Generalized Cross-section

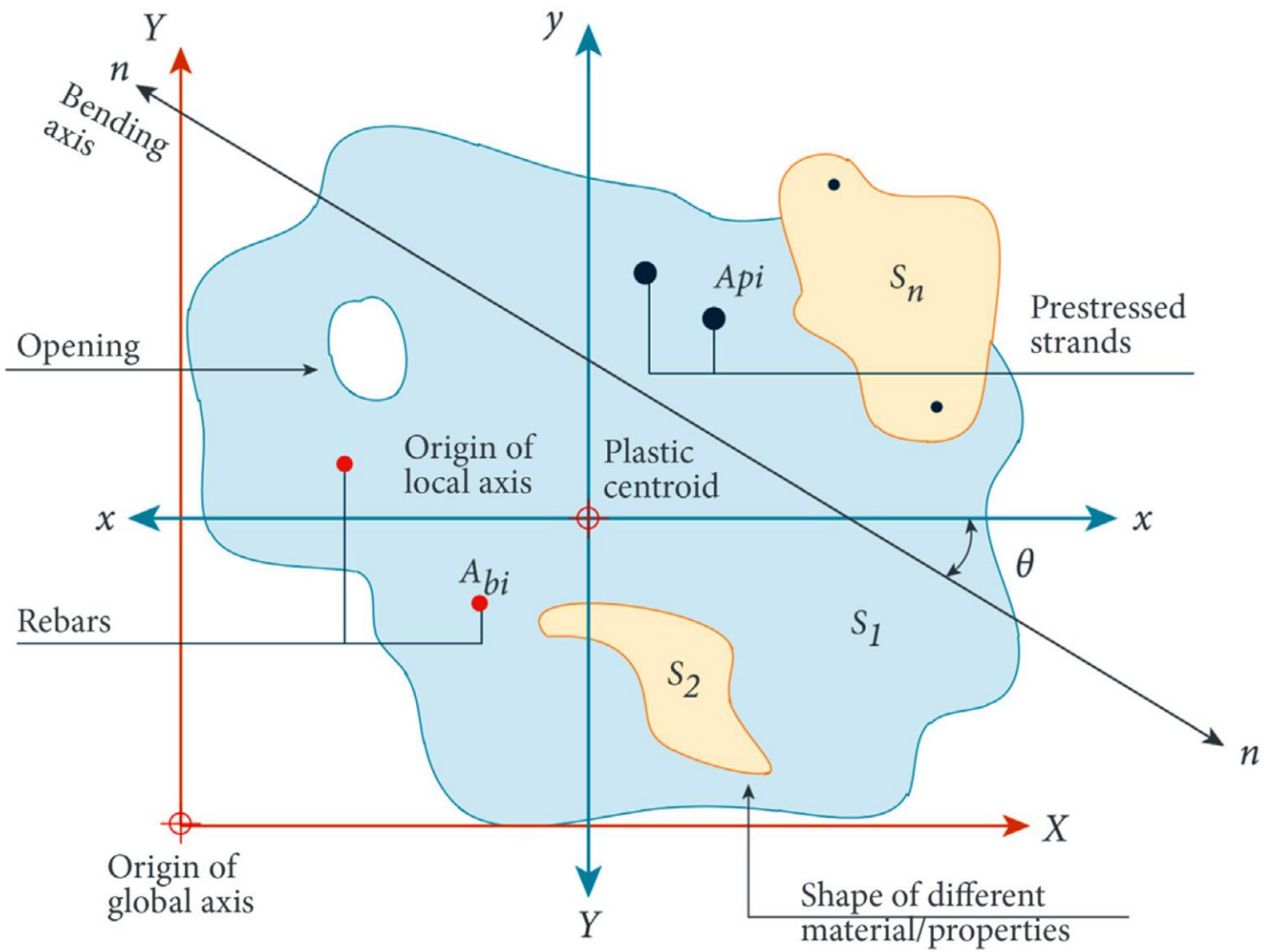



**Figure 3.22** A generalized representation of cross-section having several shapes and materials.

# Unifying Service and Ultimate State

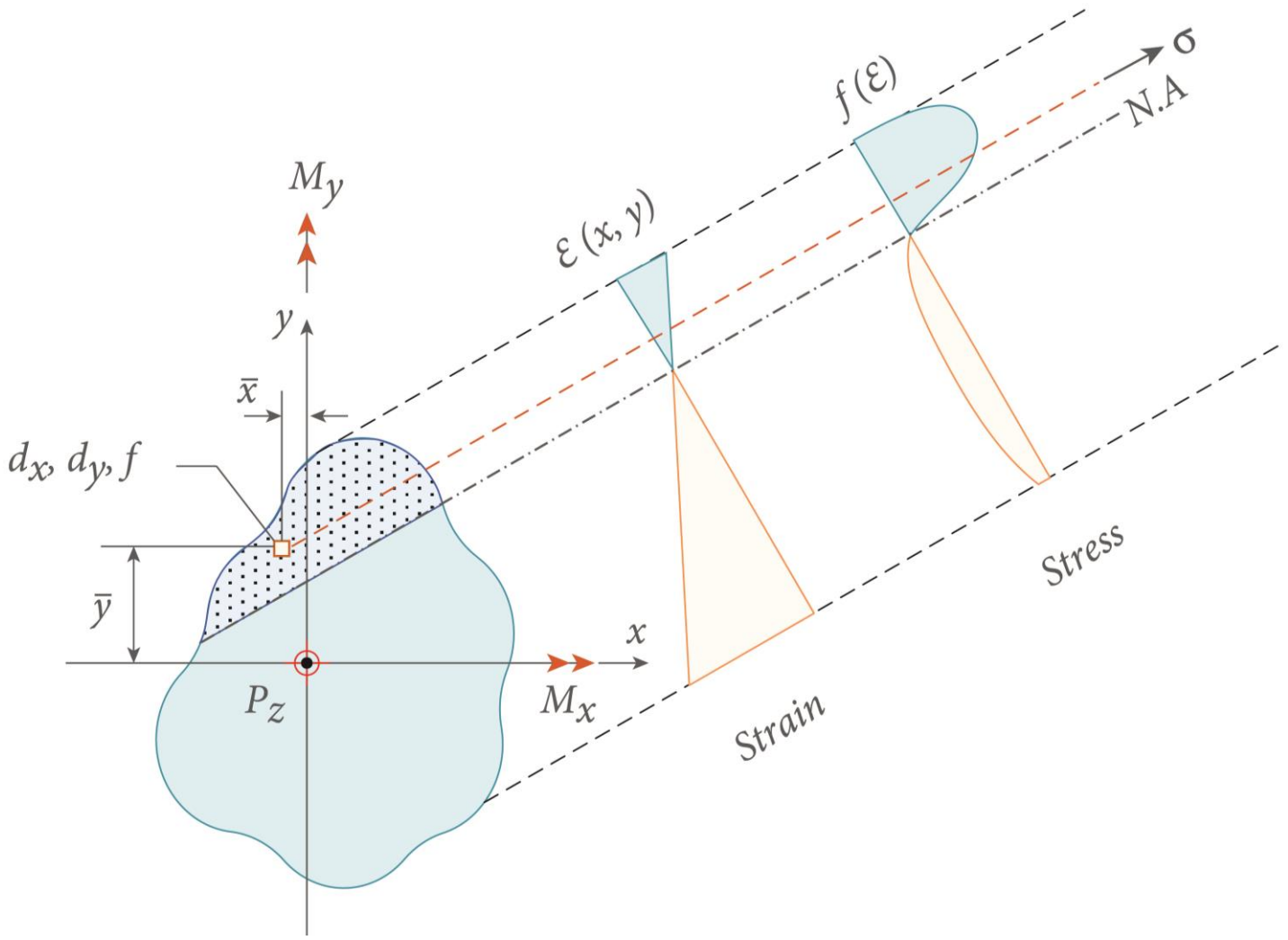
- Service State Calculations
    - Neutral axis depth controlled by limit on concrete (or steel) stresses directly

- Ultimate State Calculations
    - Neutral axis depth controlled by limit on strain in concrete (or in steel) and indirect control on material stresses
- General
    - Section Capacity based on location of neutral axis, strain compatibility and equilibrium of stress resultants and actions

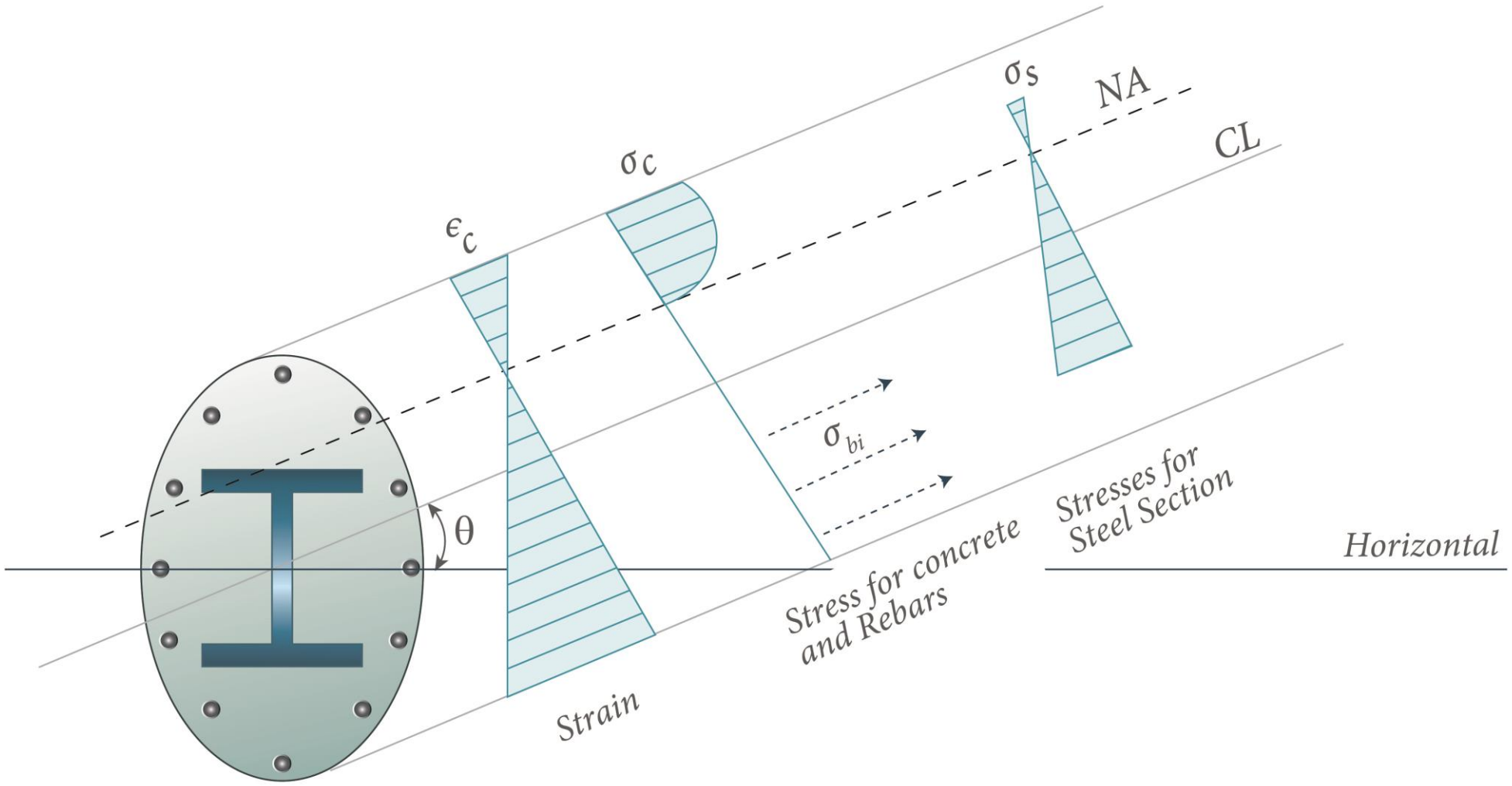
# General Procedure for Computing Capacity

- Assume Strain Profile
  - Assume a specific angle of neutral axis
  - Assume a specific depth of neutral axis
  - Assume maximum strain and determine the strain in concrete, re-bars, strands, and steel from the strain diagram
- Determine the stress in each component from the corresponding stress-strain Relationship
- Calculate stress-resultant of each component
- Calculate the total stress resultant of the section by summation of stress resultant of individual components

# A General Cross-section with General Strain and Stress Profiles



# The General Cross-section



## The General Stress Resultants

$$N_z = \phi_1 \left[ \frac{1}{\gamma_1} \int_x \int_y \sigma(x,y)\,dx\,dy \ \ldots + \frac{1}{\gamma_2} \sum_{i=1}^{n} A_i \sigma_i(x,y) \ \ldots \right]$$

$$M_x = \phi_2 \left[ \frac{1}{\gamma_1} \int_x \int_y \sigma(x,y)\,dx\,dy.y \ \ldots + \frac{1}{\gamma_2} \sum_{i=1}^{n} A_i \sigma_i(x,y)\ y_i \ \ldots \right]$$

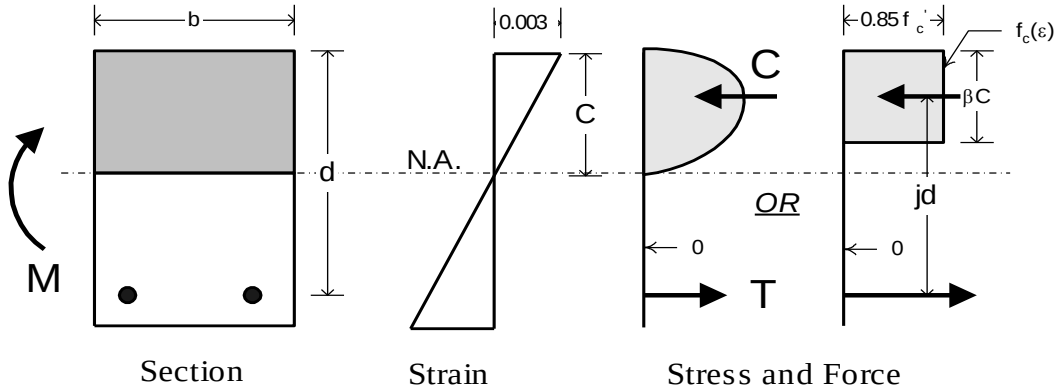
$$M_y = \phi_3 \left[ \frac{1}{\gamma_1} \int_x \int_y \sigma(x,y)\,dx\,dy.x \ \ldots + \frac{1}{\gamma_2} \sum_{i=1}^{n} A_i \sigma_i(x,y)\ x_i \ \ldots \right]$$

The Comprehensive Case

# Flexural Theory: Stress Resultants

## The Most Simple Case

$$M_n = \phi \quad f_y A_{st}\left(d - \frac{a}{2}\right)$$

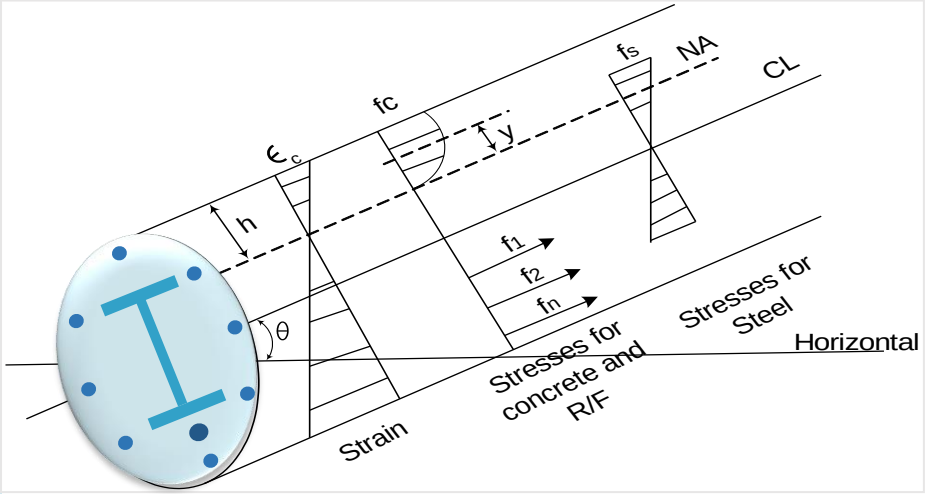



## The Most Comprehensive Case

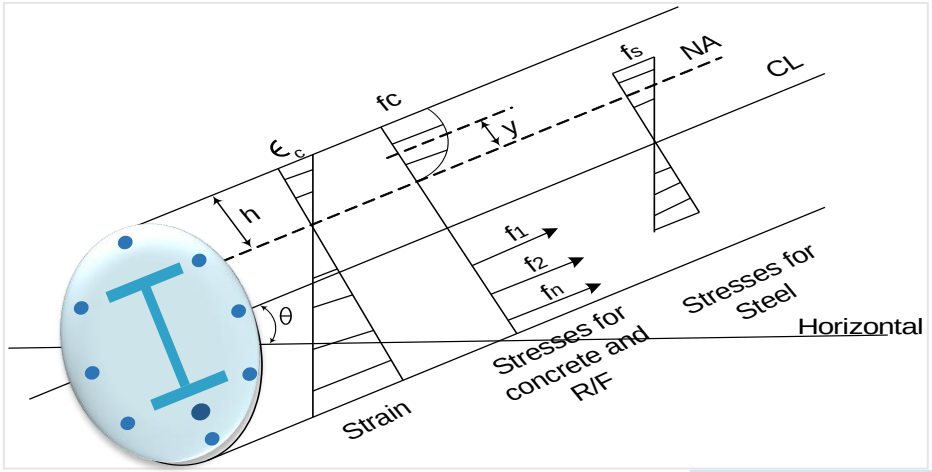
$$N_z = \phi_1\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)dx\,dy \ldots + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y) \ldots\right]$$

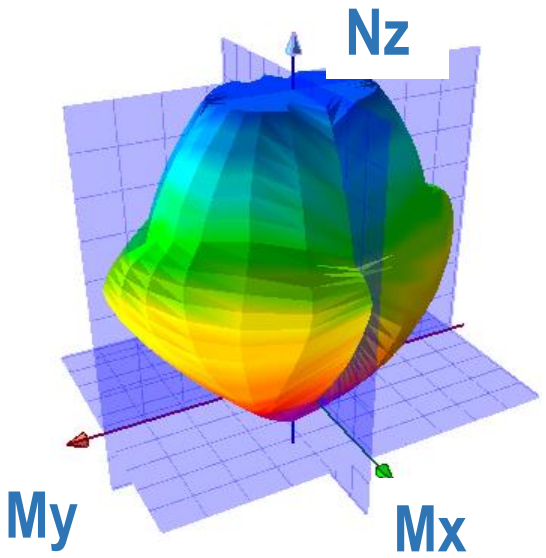
$$M_x = \phi_2\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)dx\,dy.y \ldots + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y)\ y_i \ldots\right]$$

$$M_y = \phi_3\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)dx\,dy.x \ldots + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y)\ x_i \ldots\right]$$



# The Governing Equations



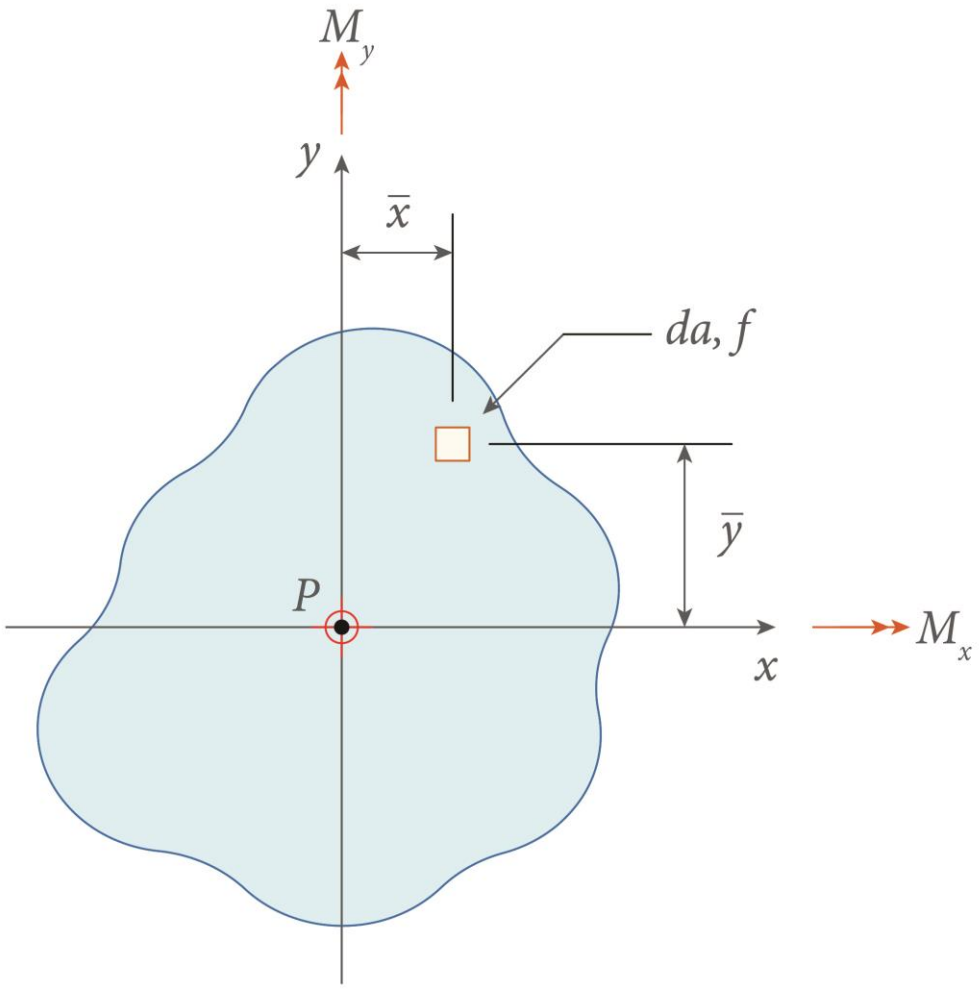



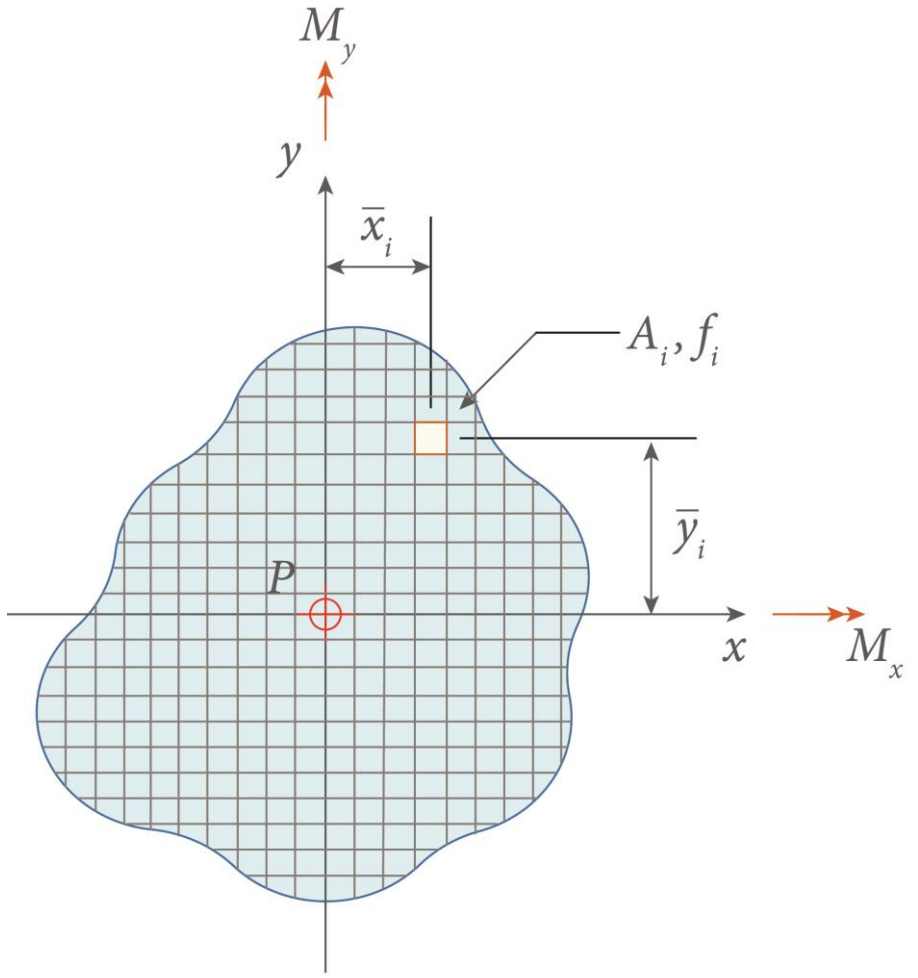

$$N_z = \phi_1\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)dx\,dy \ldots + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y) \ldots\right]$$

$$M_x = \phi_2\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)dx\,dy.y \ldots + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y)\ y_i \ldots\right]$$

$$M_y = \phi_3\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)dx\,dy.x \ldots + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y)\ x_i \ldots\right]$$

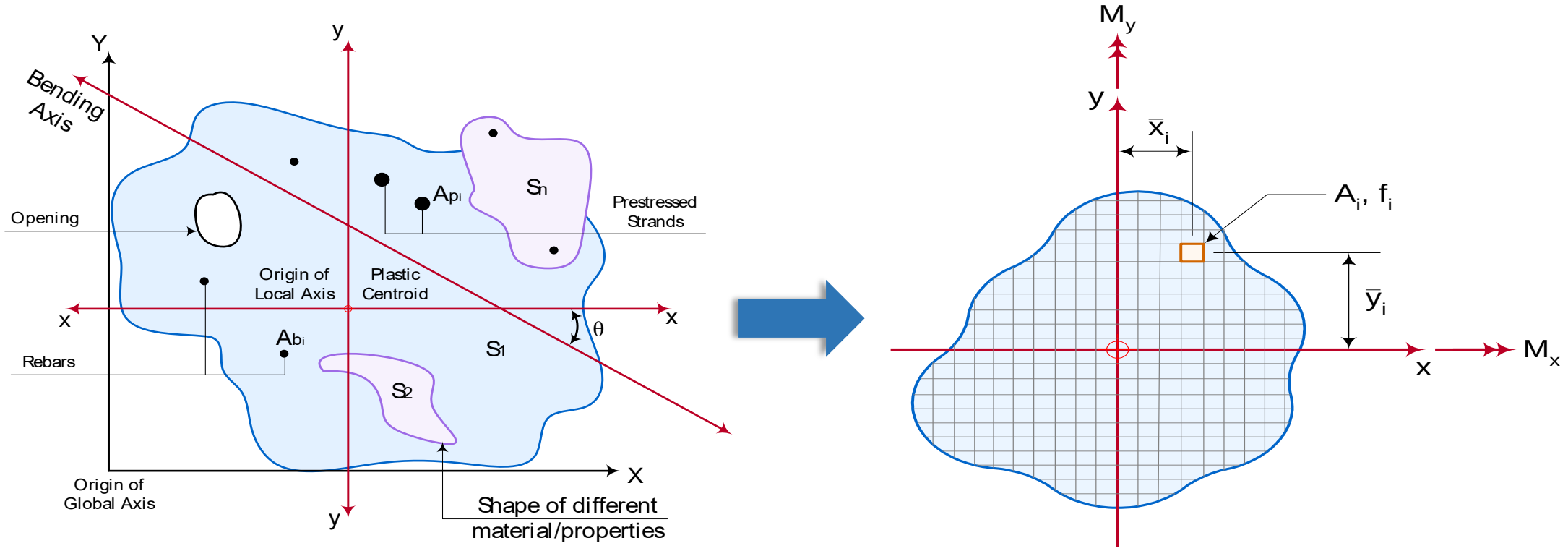
# Fiber Modeling

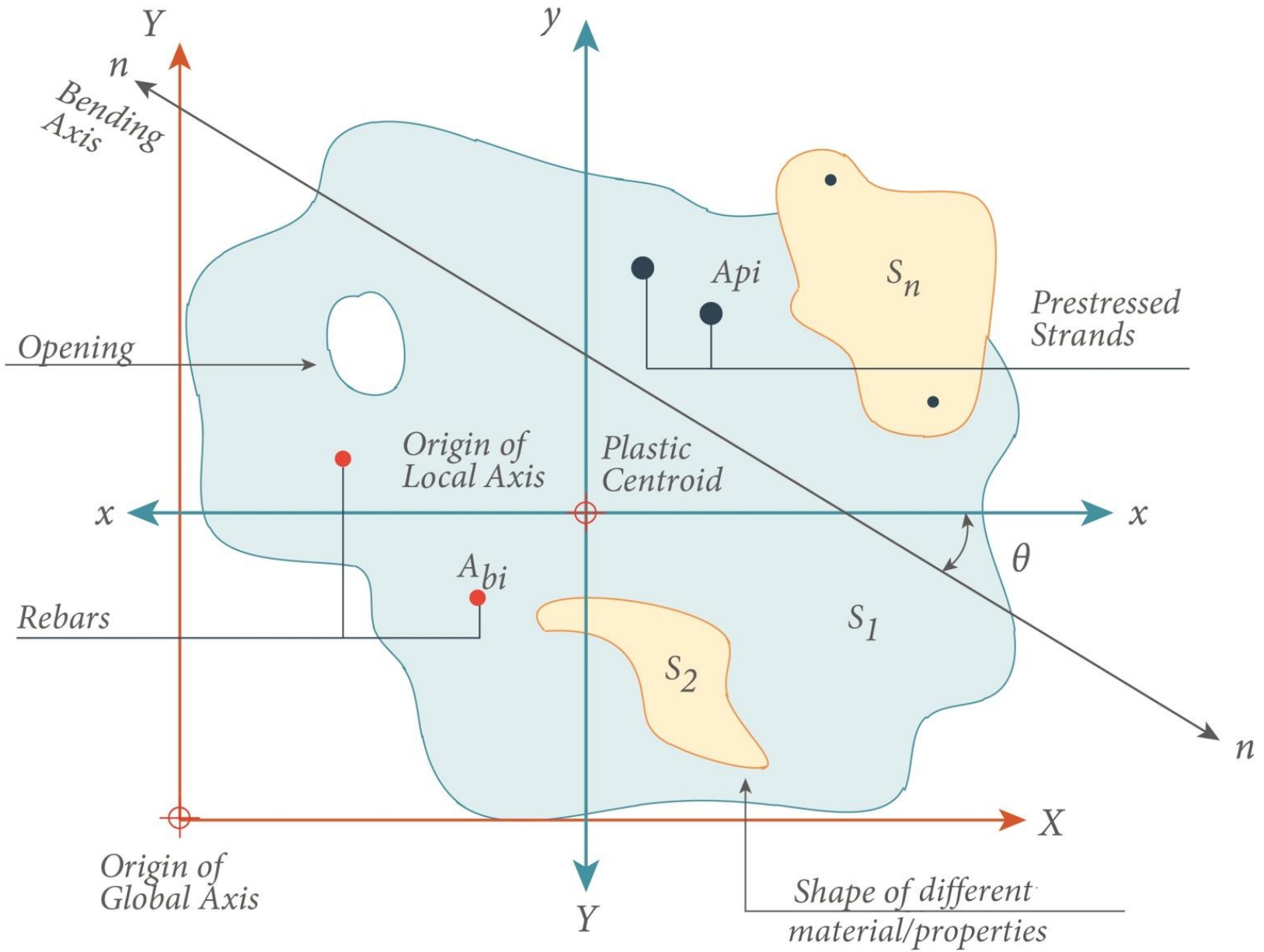


# The Fiber Model and Implementation

- In this approach, the section is sub-divided into a mesh, each element called a Fiber. A particular material model is attached to each Fiber and then solved to compute the response.




Y
n
Bending Axis
y
Api
$S_n$
Prestressed Strands
Opening
Origin of Local Axis
Plastic Centroid
x
x
θ
$A_{bi}$
Rebars
$S_1$
$S_2$
n
X
Origin of Global Axis
Y
Shape of different material/properties

## Fiber Model - Equations

Equilibrium equation based on Integration

$$N_z = \int_A f_i \, dA; \quad M_x = \int_A f_i \, \bar{y}_i \, dA; \quad M_y = \int_A f_i \, \bar{x}_i \, dA$$

Equilibrium equation based on Summation

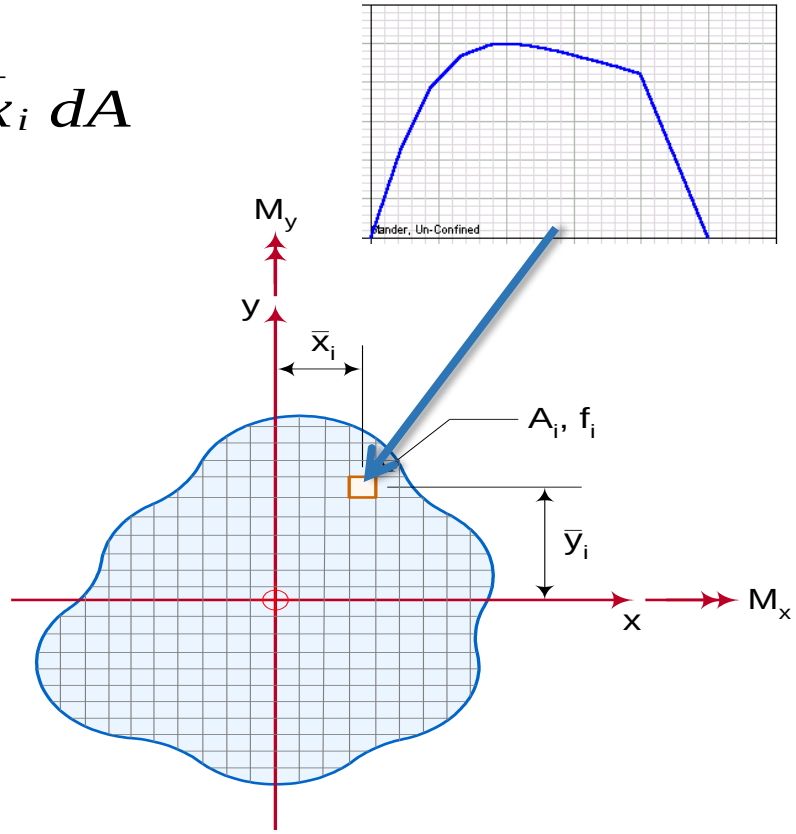
$$N_z = \sum_{i=1}^{n} f_i A_i \; ; \quad M_x = \sum_{i=1}^{n} f_i A_i \, \bar{y}; \quad M_y = \sum_{i=1}^{n} f_i$$

Expanded Summation for Complex Models

$$N_z = \phi_1 \left[ \sum_{p=1}^{q} \frac{1}{\gamma_p} \left( \sum_{k=1}^{l} \left( \sum_{i=1}^{m} N_{zi} + \sum_{j=1}^{n} f_j A_j \right) \right) \right]$$

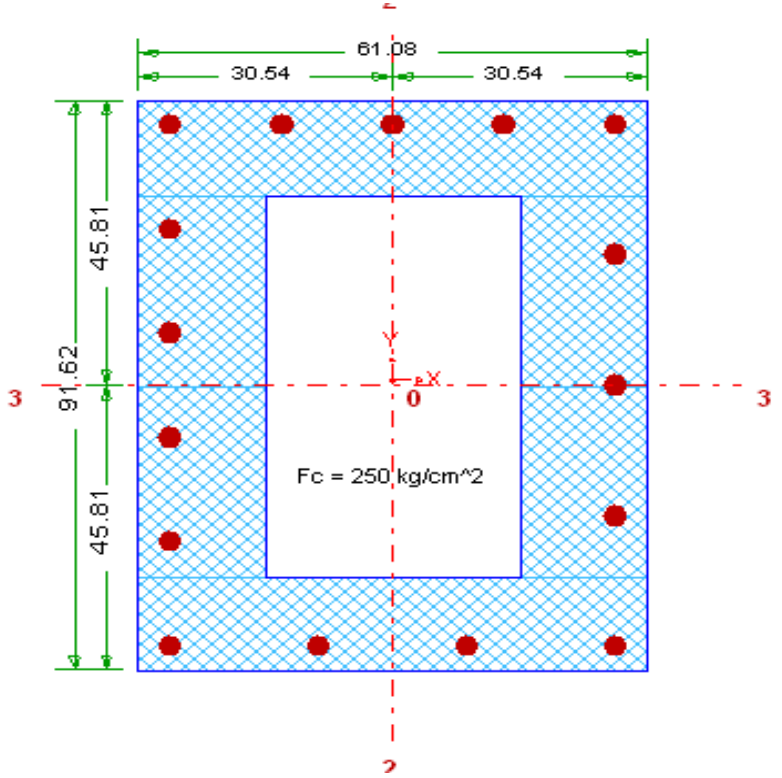
$$M_x = \phi_2 \left[ \sum_{p=1}^{q} \frac{1}{\gamma_p} \left( \sum_{k=1}^{l} \left( \sum_{i=1}^{m} M_{xi} + \sum_{j=1}^{n} f_j A_j y_j \right) \right) \right]$$

$$M_y = \phi_3 \left[ \sum_{p=1}^{q} \frac{1}{\gamma_p} \left( \sum_{k=1}^{l} \left( \sum_{i=1}^{m} M_{yi} + \sum_{j=1}^{n} f_j A_j x_j \right) \right) \right]$$

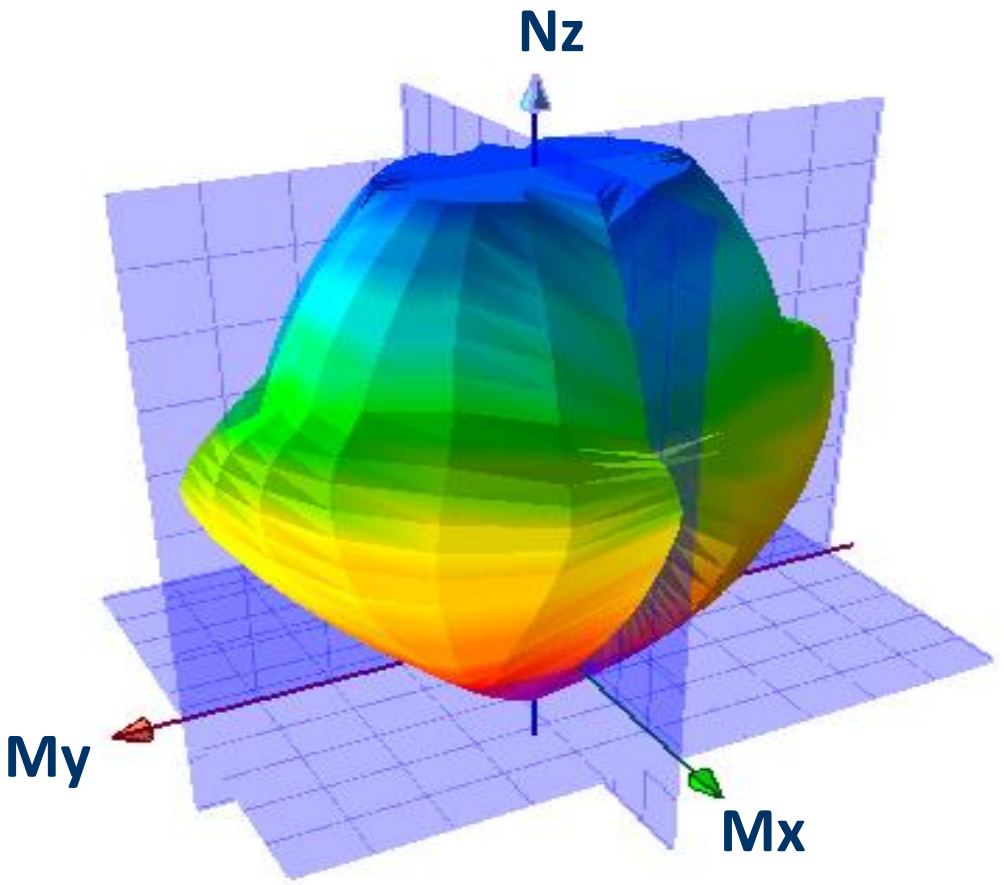


# Example: Cross-Section Response

- The Section Geometry
- Elastic Stresses
- Load Point
- Neutral Axis
- Ultimate Stresses
- Cracked Section Stresses
- Section Capacity
- Moment Curvature Curve

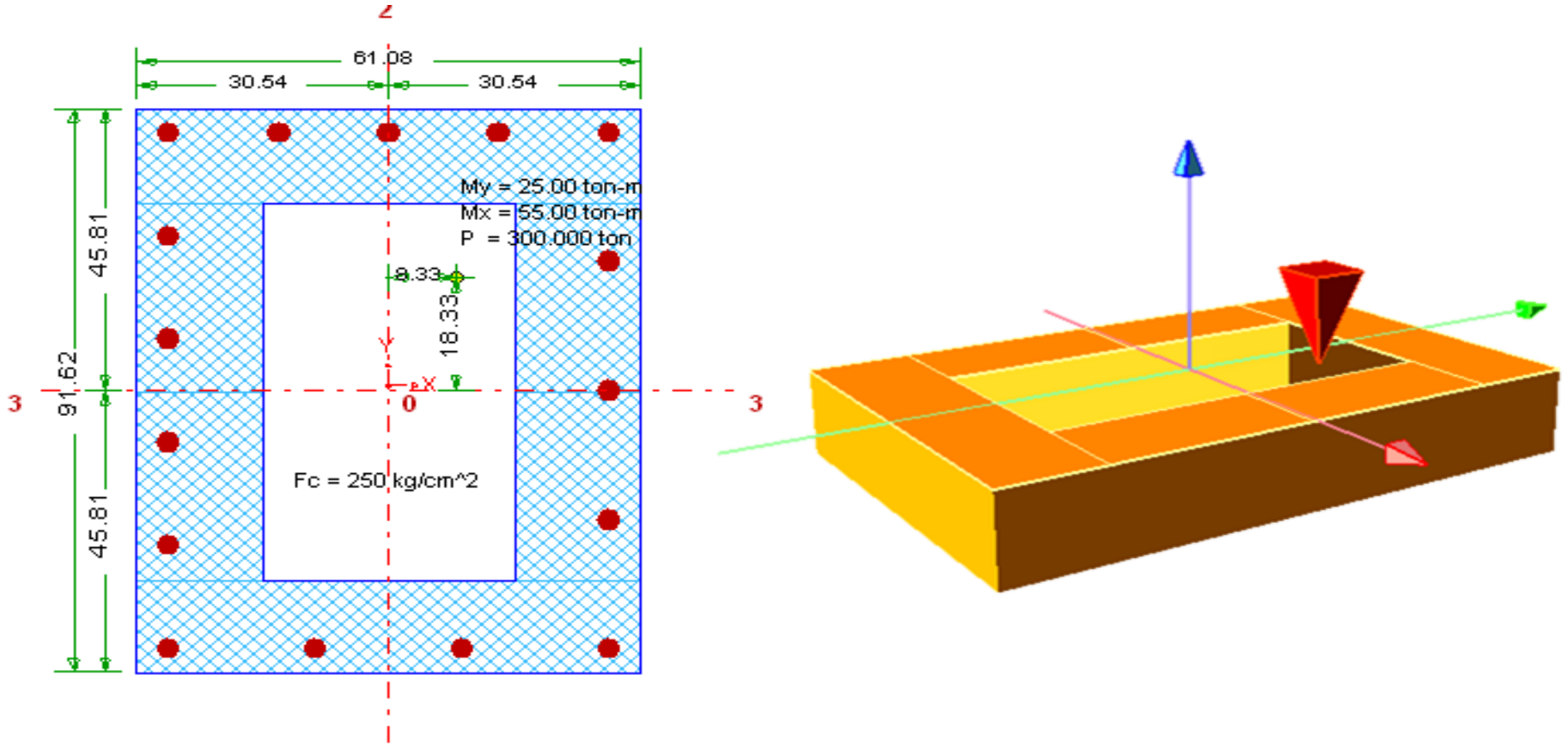


# Axial-Flexural Capacity

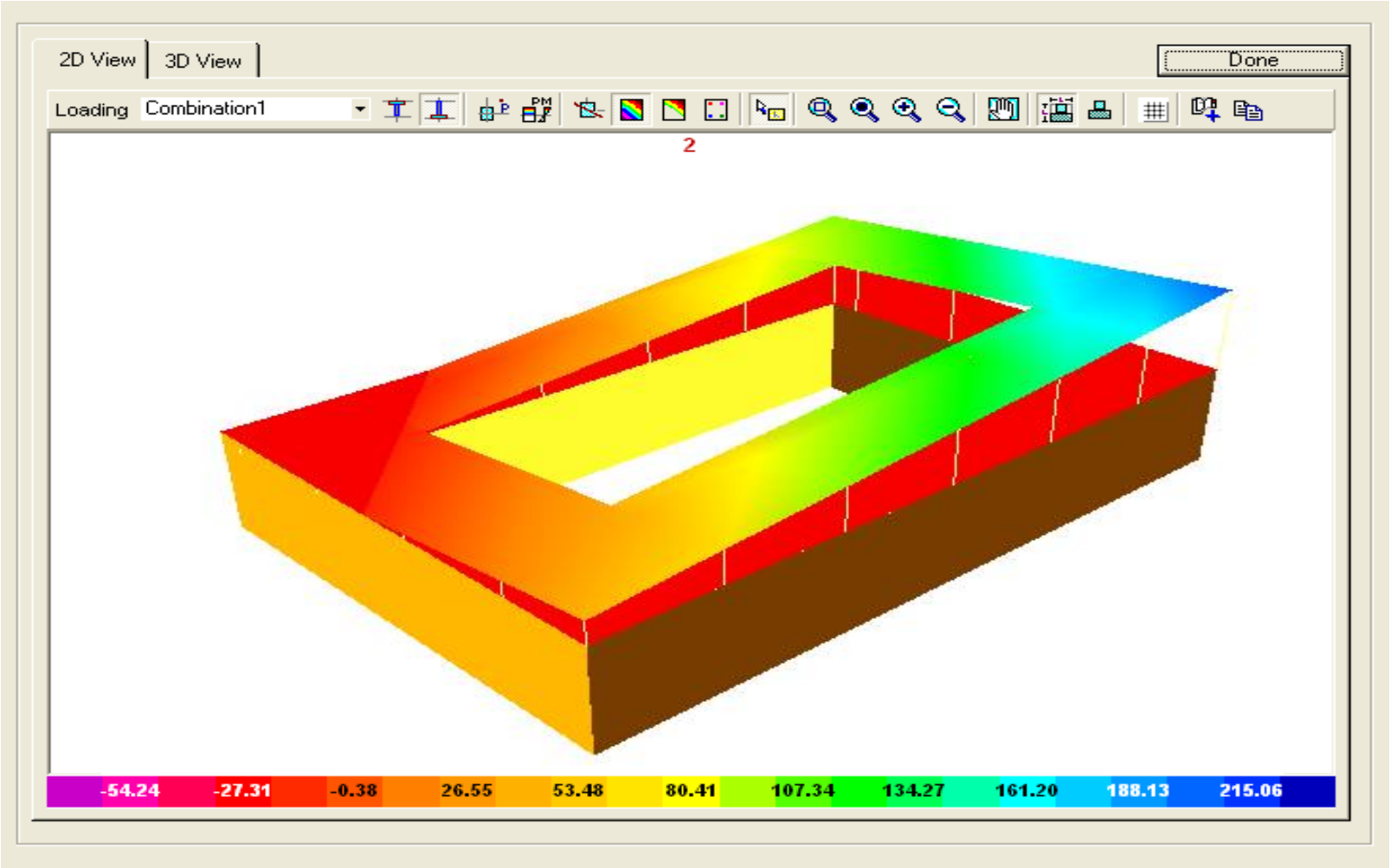



The Stress-Resultants for Bi-Axial Bending

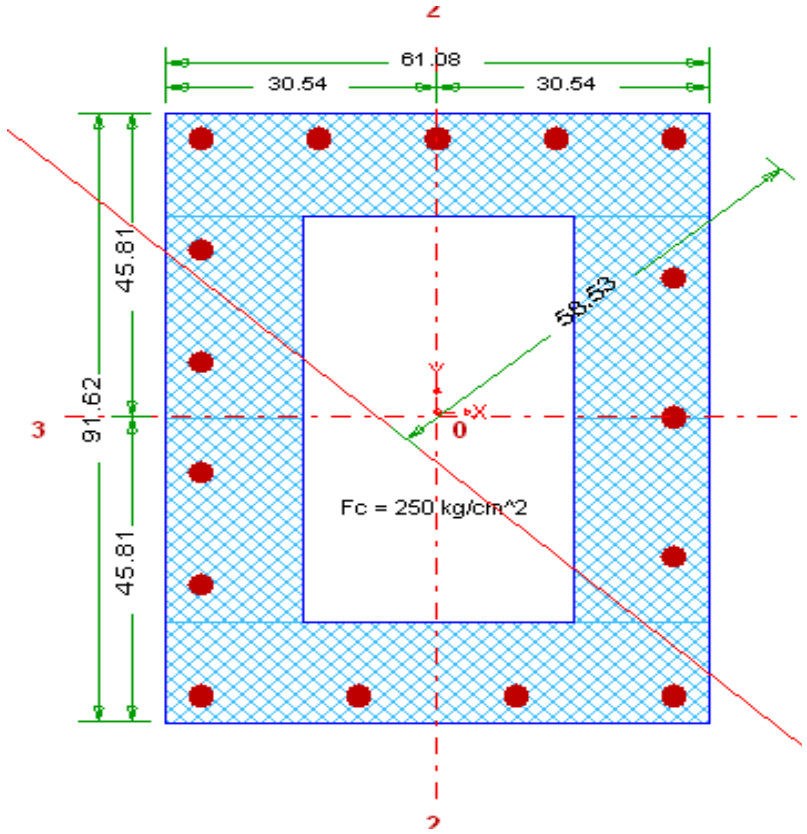
# Load Point and Eccentricity

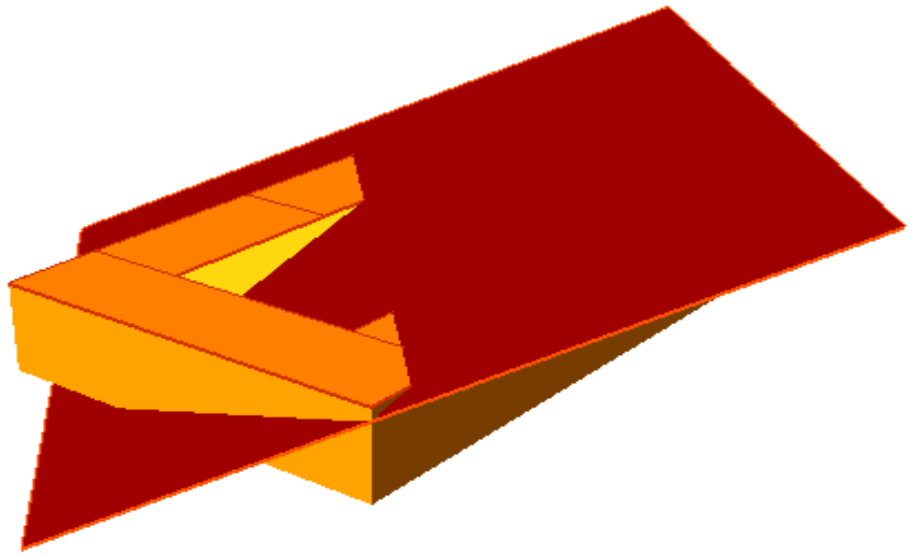


# Biaxial Elastic Stress Distribution

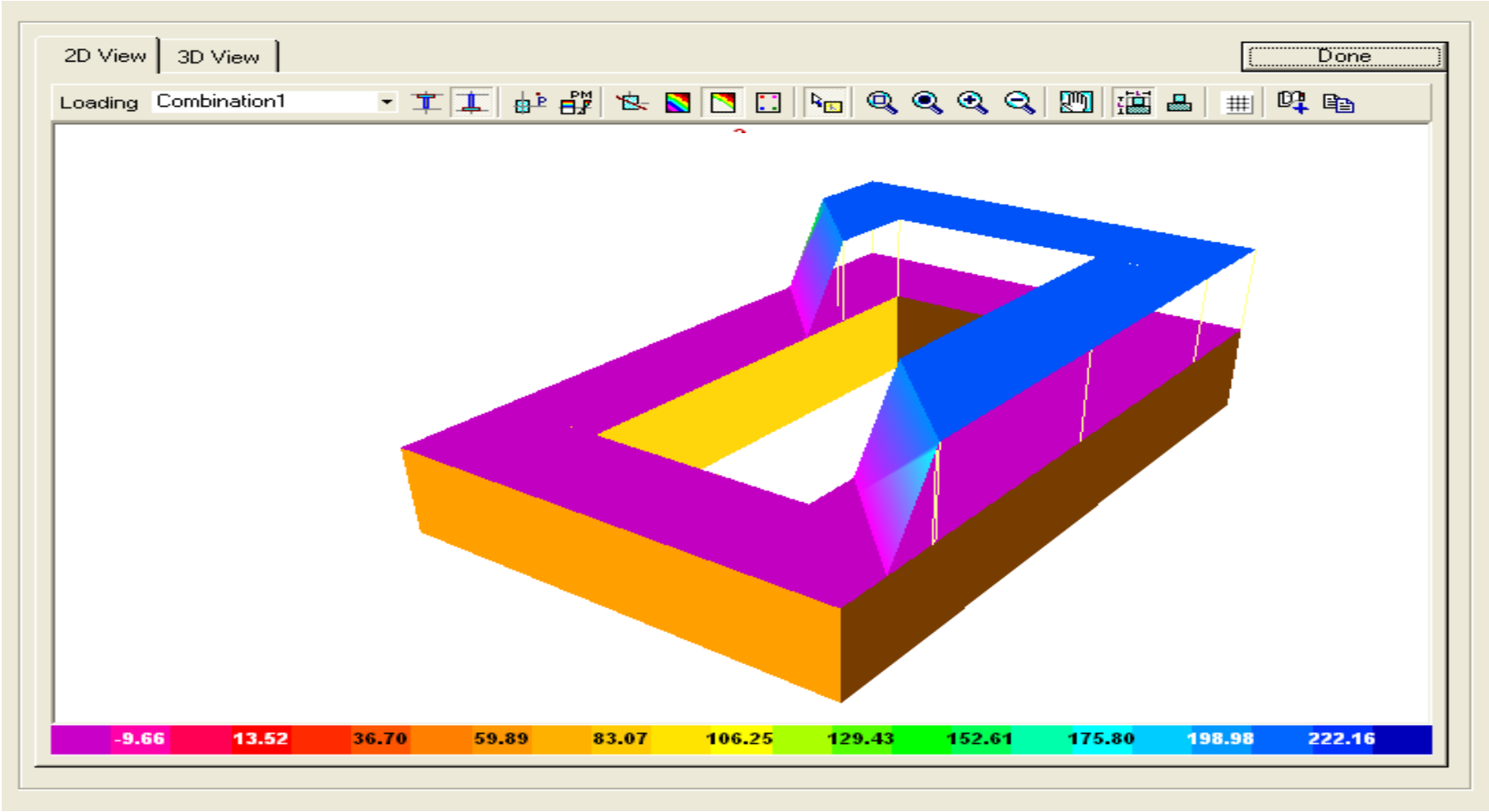


# Neutral Axis and Strain Plane

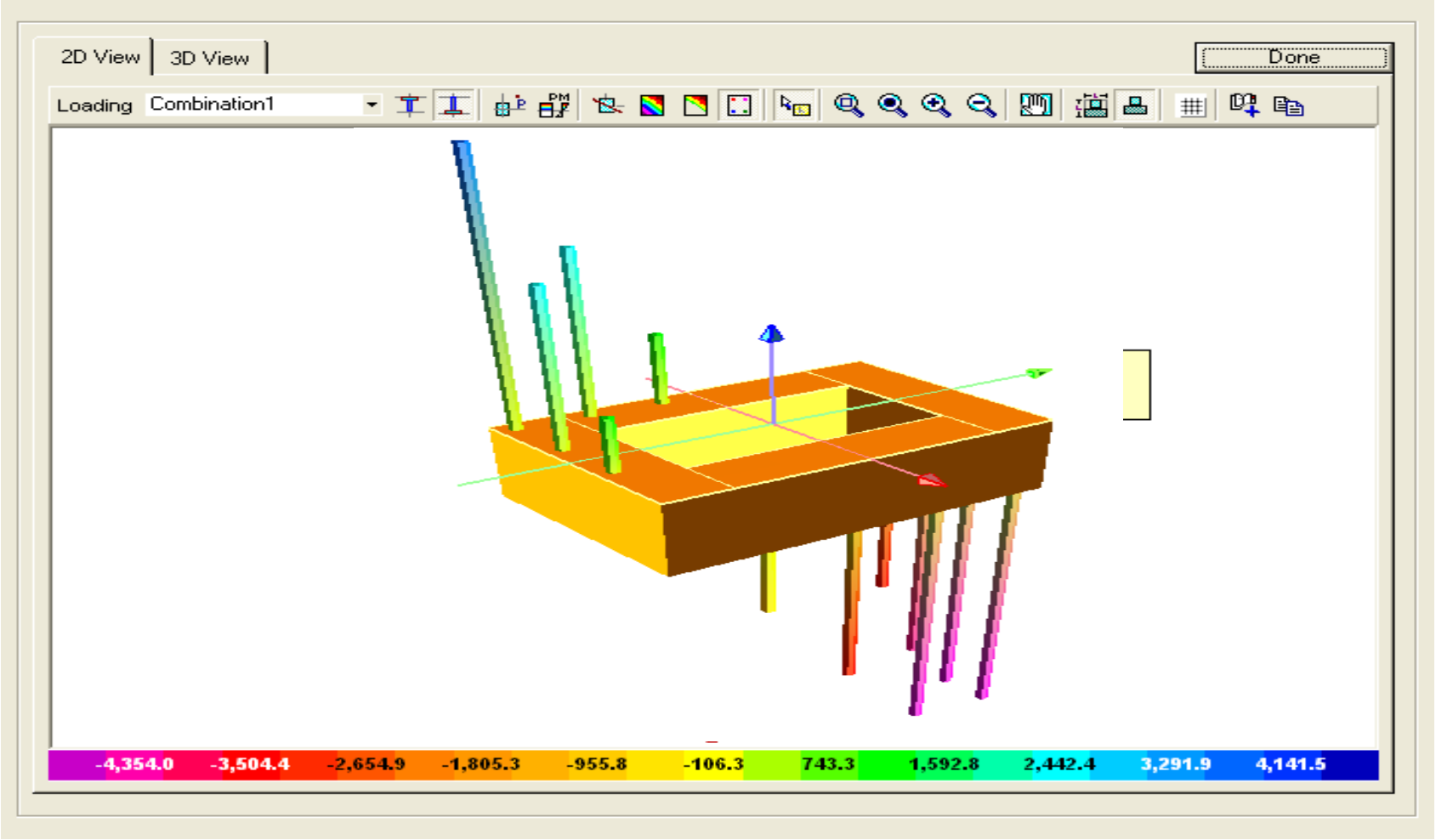


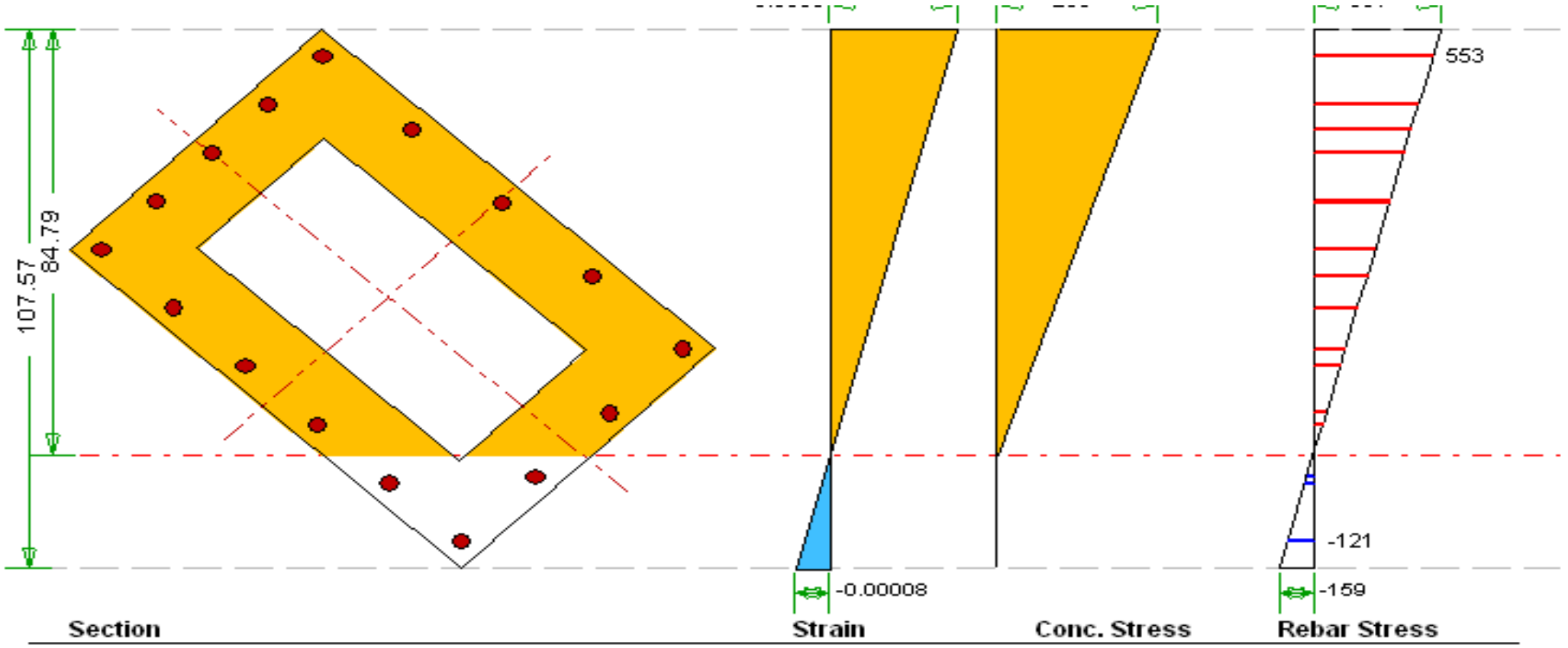
# Ultimate Stress – Rectangular Block



# Stresses in Rebars



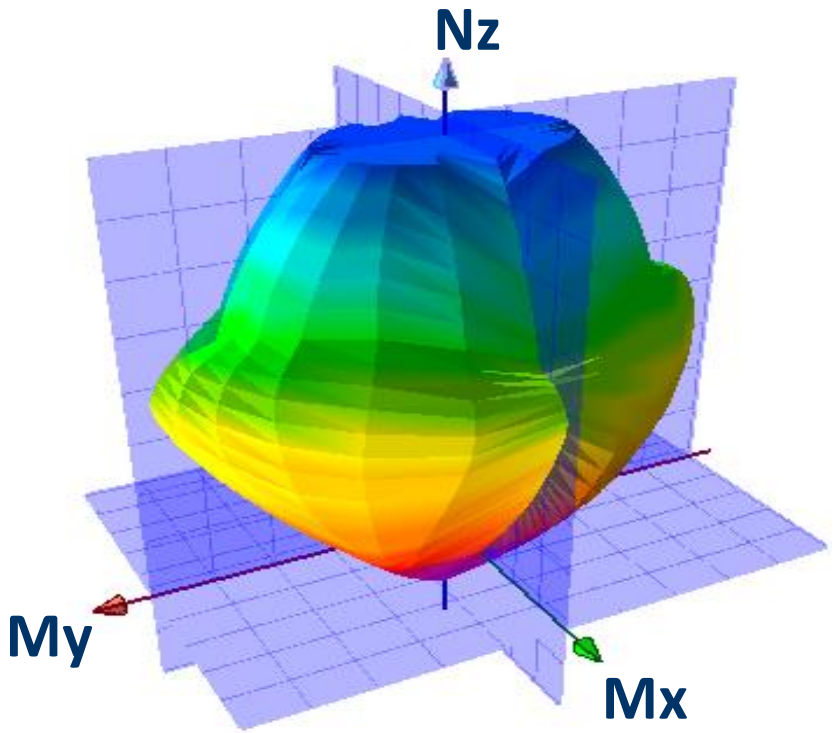
# Cracked Section Stresses



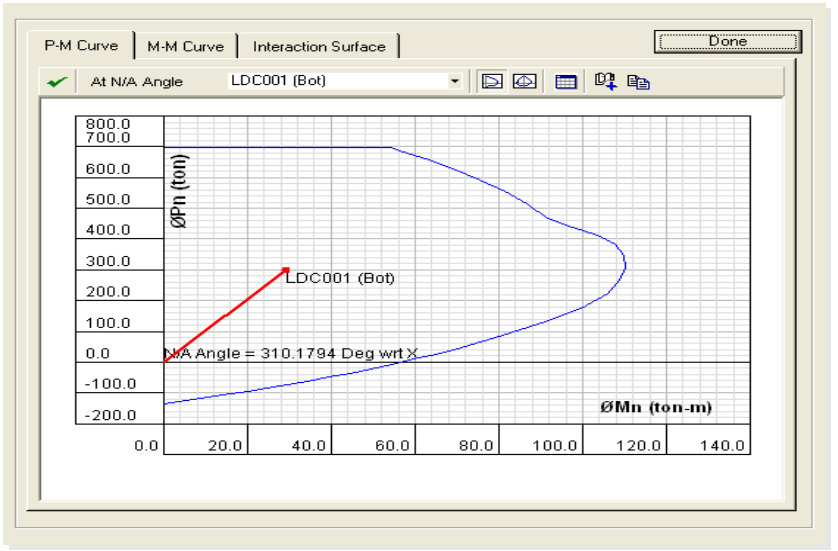

**Loading - 1**

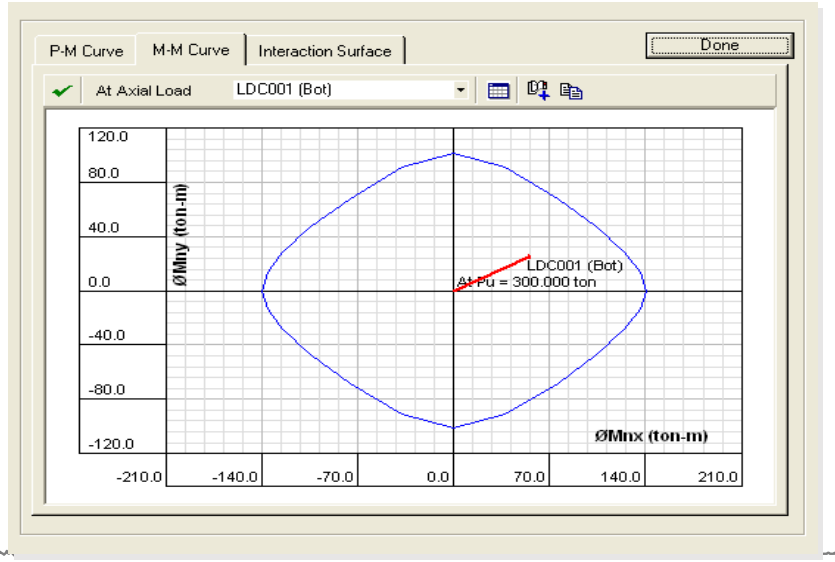
| | | | | | |
|---|---|---|---|---|---|
| Load P | = 300.000 ton | Moment Angle | = 24.4 Deg | Load Pn | = 301.694 ton |
| Moment Mx | = 55.00 ton-m | N.A. Angle | = 314.0 Deg | Moment Mn | = 53.87 ton-m |
| Moment My | = 25.00 ton-m | Curvature | = 0.349 1/1000 | Moment Mn | = 27.38 ton-m |

# Axial-Flexural Capacity




+



## Procedure for Computing Stress Resultants

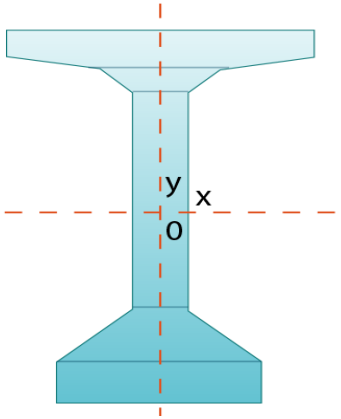
- Define the material models in terms of basic stress-strain functions. Convert these functions to discretized curves in their respective local axes;
- Model the geometry of the cross-section using polygon shapes and points, called “fibers”
- Assign the material models to various fibers
- Locate the reference strain plane based on the failure criterion. The failure criterion is a strain in concrete defined in corresponding material model and design code;

## Procedure for Computing Stress Resultants

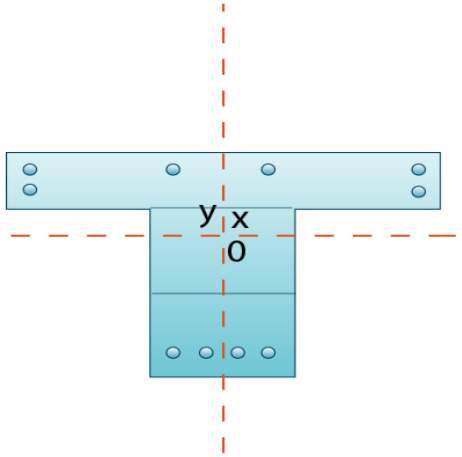
- Compute the basic stress profiles for all materials, using the reference strain profile;
- Modify the stress profiles for each material based on appropriate material functions, and special factors;
- For each material stress profile compute the corresponding stress resultant for the resulting triangles and points in the descretized cross-section.

## Procedure for Computing Stress Resultants

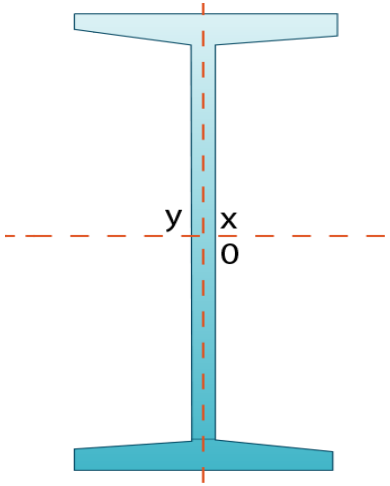
- Modify the stress resultants using the appropriate material specific and strain-dependent capacity reduction factors as defined in design codes; and,
- Compute the total stress resultants for all material stress profiles.
- Steps 5 to 9 are repeated for other locations of the reference strain plane. The computed sets of Nz, Mx, and My are used to define the capacity surface.



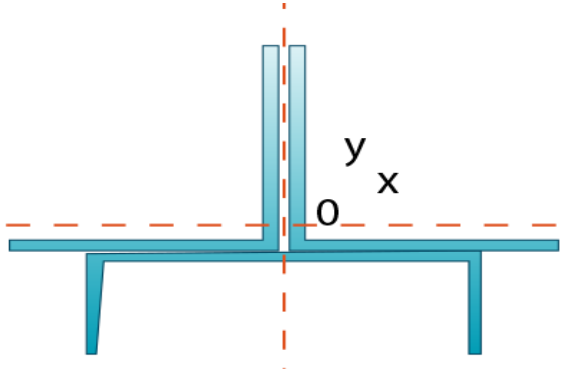

Plain concrete shape



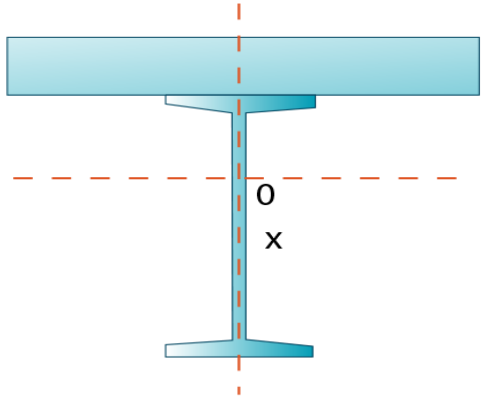

Reinforced concrete section



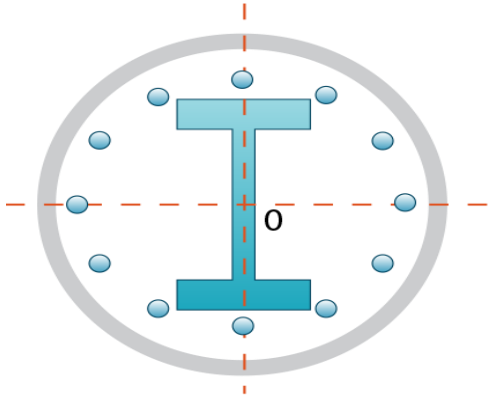

*Compact Hot-rolled steel shape*




Compact Built-up steel section




Composite section




Reinforced concrete, composite section

## Application of General Equations

# Thank You